

The Geopolitical Determinants of Economic Growth, 1960–2024

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This paper shows that geopolitical relations are a first-order determinant of economic growth. We introduce a new event-based measure of bilateral geopolitical relations, using large language models to compile and classify 373,020 geopolitical events across 193 countries from 1960 to 2024. Exploiting within-country temporal variation with local projections, we find that a one-standard-deviation permanent improvement in geopolitical relations increases GDP per capita by approximately 10 percent over 25 years. These effects operate through domestic stability, investment, trade, and productivity. In accounting exercises, geopolitical factors account for GDP variations ranging from –30 to +30 percent across countries and time periods.

Keywords. Geopolitical relations, Economic growth, Large language models, Local projections

JEL Classification. F50, F51, O11, O40, C82, P16

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1. Introduction

What determines the wealth of nations has long been a foundational question in economics and the social sciences. Early empirical contributions by Barro (1991) and Mankiw, Romer, and Weil (1992) identified physical and human capital accumulation as key drivers of cross-country income differences. Subsequent research established technology, institutions, geography, culture, and financial systems as fundamental determinants, with Acemoglu et al. (2019) providing causal evidence that democratization increases GDP per capita by approximately 20 percent in the long run. Yet despite decades of research on the determinants of economic growth, large cross-country growth and income differences remain unexplained (Kremer, Willis, and You 2022).

This paper shows that geopolitical alignment—the state of a country’s diplomatic and strategic relationships with major nations—constitutes a first-order yet underexplored determinant of economic growth. While the growth literature has extensively studied domestic factors such as institutions and human capital, the role of international political relationships has received surprisingly little attention despite the evident importance of global integration for economic development. We find that geopolitical factors account for a sizable portion of growth and income variation, as international relations shape stability and access to markets, technology, and investment. The accelerating fragmentation of international relations makes quantifying these dynamics especially pressing.

To systematically investigate this geopolitical growth determinant, we introduce a novel event-based measure of bilateral geopolitical relations that captures both the timing and intensity of diplomatic dynamics. Using large language models with web search capabilities, we compile 373,020 geopolitical events involving bilateral interactions between 193 United Nations member states and 24 major nations from 1960 to 2024. For each bilateral relationship and year, we identify significant geopolitical events—ranging from trade agreements and formal alliances to sanctions and wars—classify them using the Conflict and Mediation Event Observations (CAMEO) framework,¹ and assign Goldstein scores quantifying their cooperative or conflictual intensity.² This approach overcomes limitations of measures based on UN voting patterns or binary indicators, enabling us to exploit within-country variation in both the timing and magnitude of geopolitical changes. We aggregate bilateral scores using GDP-weighted averages to construct country-year indices that reflect each country’s overall position in the global geopolitical landscape. The resulting measure spans the full arc of contemporary international relations—from Cold War bipolarity through post-1990 globalization to the recent era of geopolitical fragmentation.

Our main empirical analysis employs local projection methods with country fixed effects to trace the dynamic effects of geopolitical relations on economic growth. Growth effects build gradually and persist for decades, with even transitory improvements generating lasting benefits. A one-

¹CAMEO (Conflict and Mediation Event Observations) is a framework for systematically coding international political events into cooperation-conflict categories. See Schrodtt and Yilmaz (2012) and Appendix C for implementation details.

²The Goldstein Scale assigns numerical values from −10 (maximum conflict) to +10 (maximum cooperation) to quantify the intensity of international political actions. See Goldstein (1992) and Appendix C for scoring guidelines.

standard-deviation permanent improvement in geopolitical relations increases GDP per capita by approximately 10 log points (10.5 percent) over 25 years. Moving from the most hostile to the most cooperative relations in our sample—a near-unit change—raises long-run GDP by 70 log points. South Africa’s post-apartheid transformation, one of the largest observed shifts at 0.78 units, implies gains of approximately 55 log points.

We support the causal interpretation of these results through multiple complementary approaches. Decomposing the index into economic, diplomatic, and security components shows that all three contribute independently—the effects are not driven by economic content alone. Results are stable across alternative fixed effects specifications, lag structures, and time-varying controls including trade openness, domestic unrest, and war exposure measures. Estimates are also symmetric across improvements and deteriorations in geopolitical relations, across six decades, and whether alignment is with the United States, Russia, China, Western democracies, or non-Western powers—consistent with a general cooperation channel rather than country-specific confounds.

Placebo exercises that randomly reassign geopolitical shocks within region-year cells or shift them to future periods produce near-zero responses, providing evidence against mechanical artifacts of the estimator. We also implement an instrumental variables strategy exploiting variation from non-economic verbal conflicts—diplomatic disputes, border tensions, and cultural disagreements that move bilateral relations but are that lack direct economic content. The convergence of IV and OLS estimates supports the view that the baseline relationship is not driven solely by reverse causality or omitted economic confounders.

The growth effects operate through multiple reinforcing channels—domestic stability, physical capital accumulation, productivity gains, and trade expansion—with distinct temporal patterns ranging from immediate stabilization to gradual human capital gains. Decomposing the democracy-growth nexus reveals that democracy and geopolitical relations are complementary but distinct: democracy’s short-run growth effects are substantially mediated by improved international relations, while its long-run benefits—strengthened property rights, reduced expropriation risk—persist independently of geopolitical channels.

We apply the estimated impulse responses to observed geopolitical changes in growth accounting exercises. The 1980s–2000s period of improving international relations generated median long-run GDP gains of 2–5% per decade, while the post-2010 fragmentation reversed this pattern, with the median country experiencing negative geopolitical growth contributions for the first time in our sample. Extending the analysis to cross-country differences under maintained assumptions, these estimates suggest that persistent differences in geopolitical positioning can account for GDP variations ranging from –30% to +30%, with countries like Singapore benefiting from broad international engagement and others, like Russia, bearing substantial costs from geopolitical isolation.

Beyond the baseline identification, our findings are robust to alternative estimation methods and measurement approaches. Dynamic panel estimates exploiting the autoregressive structure yield similar dynamic effects, with permanent improvements generating long-run GDP gains

close to our baseline. Alternative measures based on unsmoothed geopolitical events converge to similar long-run impacts once we account for their transitory nature. In contrast, measures based on UN voting patterns fail to generate any growth effects when subjected to the same empirical framework, highlighting the importance of our bilateral event-based approach. Even when compared to standard bilateral proxies—economic sanctions and formal military alliances—our comprehensive geopolitical index absorbs much of their explanatory power while capturing additional growth-relevant variation.

Recent shifts in major-power relations—including U.S.-China trade tensions, the Russia-Ukraine war, and escalating conflict in the Middle East—underscore the importance of quantifying these dynamics. Our estimates suggest that geopolitical misalignment can reduce GDP by 20–30%, undermining gains from domestic reforms.

Literature Review. This paper contributes to three strands of literature: the measurement of geopolitical relations, the determinants of economic growth, and the emerging field of geoeconomics.

Measuring Bilateral Geopolitical Relations. Our primary contribution is to provide a continuous, time-varying representation of bilateral geopolitical relations that addresses key limitations in existing approaches. The predominant method relies on UN General Assembly (UNGA) voting similarity (Bailey, Strezhnev, and Voeten 2017; Signorino and Ritter 1999), which captures multilateral positions rather than bilateral dynamics. Alternative categorical measures—including strategic rivalries (Aghion et al. 2019; Thompson 2001), sanctions (Ahn and Ludema 2020; Felbermayr et al. 2021), military alliances (Gibler 2008), and bilateral treaties (Broner et al. 2025)—provide discrete classifications but do not track the continuous evolution of international relationships. Our event-based approach tracks both the timing and intensity of bilateral geopolitical dynamics at annual frequency. Drawing on textual analysis methods for measuring economic and political risks (Baker, Bloom, and Davis 2016; Caldara and Iacoviello 2022; Hassan et al. 2019, 2024) and the growing application of large language models for data construction in economics (Dell 2025; Lagakos, Michalopoulos, and Voth 2025; Asirvatham, Mokski, and Shleifer 2026; Clayton et al. 2025), we extract, classify, and score 373,020 bilateral political events across six decades using standardized frameworks (CAMEO/Goldstein).

Economic Growth Determinants. We contribute to the literature on cross-country growth determinants (Barro 1996; Levine and Renelt 1992; Durlauf, Johnson, and Temple 2005; Johnson and Papageorgiou 2020). While existing research has identified fundamental drivers including institutions (Acemoglu, Johnson, and Robinson 2001; Dell 2010), human capital (Barro 1991; Mankiw, Romer, and Weil 1992), culture (Nunn 2008; Tabellini 2010), geography (Dell, Jones, and Olken 2012; Sachs and Warner 1995), and trade (Alcala and Ciccone 2004; Frankel and Romer 1999), the role of geopolitical relationships remains underexplored. Building on Acemoglu et al. (2019)’s empirical framework, we employ local projection methods (Jordà 2005; Jordà and Taylor 2025) to estimate the dynamic growth effects of geopolitical shocks.

Geoeconomics. Our research bridges classical economic statecraft (Hirschman 1945) with contem-

porary geoeconomic analysis (Clayton, Maggiori, and Schreger 2024, 2026; Fernández-Villaverde, Mineyama, and Song 2024; Flynn et al. 2025; Gopinath et al. 2025; Kleinman, Liu, and Redding 2024; Liu and Yang 2025). While recent theoretical and empirical work has formalized the mechanisms, evidence on the growth consequences of geopolitical relations remains limited. In a companion paper, Fan, Wo, and Xiang (2025) extend the measurement approach introduced here to study how geopolitical relations shape trade globalization; this paper focuses on the aggregate growth consequences. Our framework situates aid (Alesina and Dollar 2000; Nunn and Qian 2014), sanctions (Morgan, Syropoulos, and Yotov 2023), and conflicts (Federle et al. 2026; Martin, Mayer, and Thoenig 2008) within a unified spectrum of geopolitical relations.

Road Map. The remainder of the paper proceeds as follows. Section 2 introduces our event-based measure of geopolitical relations. Section 3 presents our main empirical results, estimating the dynamic effects of geopolitical relations on long-run growth. Section 4 unpacks the channels through which geopolitics affects growth and disentangles the interplay between democracy and international alignment. Section 5 applies these estimates to conduct growth accounting exercises. Section 6 provides additional robustness tests. Section 7 concludes.

2. Event-based Measure of Geopolitical Relations

Investigating the growth effects of geopolitical relations requires a measure that captures bilateral relationship intensity—not merely multilateral voting positions or discrete institutional thresholds—at annual frequency and over long historical horizons. Existing approaches fall short on at least one of these dimensions: UN voting similarity reflects multilateral preferences rather than bilateral dynamics, while categorical indicators of sanctions, alliances, or strategic rivalries mark specific relationship thresholds but not the continuous evolution between them.

To construct such a measure, we leverage a large language model augmented with search capabilities to compile and analyze 373,020 geopolitical events worldwide from 1960 to 2024. Our dataset covers bilateral interactions between all 193 UN member states, with particular focus on relationships involving 24 major economic and geopolitical nations (denoted as $\mathcal{N}$)³. Using these events, we construct a novel measure of bilateral geopolitical relations that varies by country pair and year, which we then aggregate to create country-year measures of average geopolitical relations. This section describes the construction, validates the measure against historical episodes and alternative pipelines, and documents the evolution of global geopolitical relations over six decades.

³The 24 major nations are: Argentina, Australia, Belgium, Brazil, Canada, Switzerland, People’s Republic of China, Germany, Denmark, Spain, France, United Kingdom, Indonesia, India, Italy, Japan, Republic of Korea, Mexico, Netherlands, Poland, Russian Federation (Soviet Union), Saudi Arabia, Turkey, and United States. Appendix A.1 discusses these choices. Collectively, these 24 nations account for 83–90 percent of global GDP over the study period, underscoring their dominant role in the world economy.

2.1. LLM: Compile and Analyze Geopolitical Events

Bilateral geopolitical events—diplomatic agreements, economic sanctions, military actions, and territorial disputes—are extensively documented across news archives, government publications, and scholarly databases, making them well-suited for systematic extraction by large language models (LLMs). We employ Gemini 2.5 Pro, an LLM equipped with web search capabilities, to collect, verify, and analyze bilateral geopolitical events through a structured prompt engineering framework. Figure 1 illustrates our LLM-based analysis procedure, with complete analytical framework and prompt specifications provided in Appendix A.2.

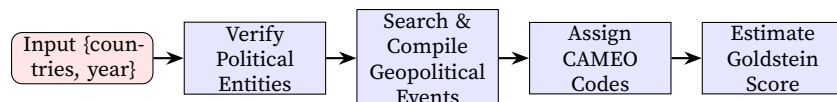


FIGURE 1. LLM Geopolitical Event Analysis Procedure

Our methodology instructs the LLM to perform five sequential tasks: (i) verify the historical political entities for each country pair and year, accounting for state succession (e.g., Soviet Union to Russian Federation); (ii) conduct systematic searches across its knowledge base and internet sources to identify bilateral geopolitical events from authoritative sources—spanning six dimensions: economic relations, diplomatic and political relations, security and defense, legal and territorial issues, multilateral governance, and other significant interactions (Table A1 in Appendix A.2; see Appendix C.2 for complete taxonomy); (iii) classify each event using the Conflict and Mediation Event Observations (CAMEO) framework into cooperation-conflict categories; (iv) assign Goldstein Scale scores ranging from -10 (maximum conflict) to $+10$ (maximum cooperation) based on event intensity;⁴ and (v) categorize economic content when applicable.

Table 1 demonstrates our methodology using U.S.-Russia bilateral events from 2022, the year Russia launched its full-scale war on Ukraine. Five of the six events represent conflict, with Goldstein scores ranging from -4.5 (UN General Assembly condemnation) to -8.0 (U.S. military assistance to Ukraine), spanning material and verbal dimensions. The sole cooperative event—the Griner-Bout prisoner exchange—scores $+7.0$, illustrating how the framework captures rare bilateral cooperation even amid otherwise hostile relations. Table A2 in Appendix A.2 provides a contrasting example from the 1972 détente period, where all nine U.S.-Soviet events represent cooperation with scores ranging from $+5.0$ to $+9.0$.

Table 2 synthesizes patterns across our complete dataset of 373,020 events spanning 1960–2024. The data reveal three distinct phases in international relations: Cold War tensions (1960–1989) with cooperation comprising 75–79% of interactions and mean Goldstein scores ranging from 2.86 to 3.28; the globalization era (1990–2010) with cooperative events expanding to 84–86% with scores

⁴Goldstein scores are assigned following the standard CAMEO-to-Goldstein mapping used by the Global Database of Events, Language, and Tone (GDELT) and Integrated Crisis Early Warning System (ICEWS) databases. While we maintain consistency with these established mappings, our LLM implementation allows for limited contextual adjustments based on the bilateral relationship’s historical context and event severity. For the reference mapping between CAMEO codes and Goldstein scores, see <https://eventdata.parusanalytics.com/cameo.dir/CAMEO.SCALE.txt>.

TABLE 1. U.S.-Russia Bilateral Geopolitical Events in 2022: LLM Analysis Results

Event Name	Event Description	CAMEO Class.	Econ. Type	Goldstein Score
US Military Assistance to Ukraine	US committed billions in security assistance through Presidential Drawdown Authority and Ukraine Security Assistance Initiative	Material Conflict (17-170)	Not econ.	−8.0
Sweeping Sanctions on Russia	Following Russia’s invasion of Ukraine, US coordinated with G7 and EU to impose extensive sanctions targeting major banks, state enterprises, and officials	Material Conflict (16-163)	Sanctions	−7.5
US Leads Russia’s Suspension from UN Human Rights Council	Following reports of civilian deaths in Bucha, US led successful campaign to suspend Russia from UN Human Rights Council by 93–24 vote	Material Conflict (16-166)	Not econ.	−6.5
Biden Labels Putin a “War Criminal”	President Biden publicly called President Putin a “war criminal” for actions during the Ukraine invasion, a label the Kremlin called “unacceptable and unforgivable”	Verbal Conflict (11-112)	Not econ.	−5.5
US Leads Diplomatic Condemnation at UN	US led effort resulting in UN General Assembly Resolution ES-11/1, which passed 141–5 deploring Russia’s aggression and demanding immediate withdrawal from Ukraine	Verbal Conflict (11-113)	Not econ.	−4.5
Griner-Bout Prisoner Exchange	US and Russia conducted prisoner swap in Abu Dhabi, exchanging Russian arms dealer Viktor Bout for American basketball player Brittney Griner after months of negotiations	Material Coop. (08-084)	Not econ.	+7.0

peaking at 4.09; and the contemporary fragmentation period (2010–2024) where conflict events increase relatively faster than cooperation, with mean scores declining from 3.88 in the 2010s to 3.73 in the 2020s amid major geopolitical ruptures including the Russia-Ukraine war. Appendix A.2.2 provides detailed statistics on event classification and temporal patterns.

Comparison with Existing Event Databases. Our geopolitical events compilation differs from existing global event databases such as GDELT (Leetaru and Schrodtt 2013) and ICEWS (Boschec et al. 2015) in two key dimensions. First, leveraging the LLM’s contextual capabilities, we focus exclusively on events that shape bilateral relations between country pairs, enabling more precise measurement of relationship intensity.⁵ This targeted approach prioritizes events with clear bilateral significance rather than attempting to synthesize the full spectrum of international interactions. Second, our compilation provides extended temporal coverage from 1960, aligning with the availability of

⁵GDELT and ICEWS collect comprehensive global events across all actors and issue areas, making it challenging to aggregate these events into meaningful measures of bilateral relationship intensity between specific country pairs.

TABLE 2. Geopolitical Events Summary by Decade, 1960–2024

	1960s	1970s	1980s	1990s	2000s	2010s	2020s	Total
Cooperation Events	22,010	26,872	29,657	40,538	59,232	82,879	48,546	309,734
Conflict Events	7,279	7,353	9,390	7,988	9,647	13,296	8,333	63,286
Mean Goldstein Score	2.856	3.278	2.920	3.976	4.086	3.879	3.727	3.671

Notes: Cooperation events include verbal and material cooperation (CAMEO classes 1-2). Conflict events include verbal and material conflict (CAMEO classes 3-4). Goldstein Scale: -10 (most conflictual) to +10 (most cooperative).

economic data necessary for comprehensive panel analysis.⁶

2.2. Measuring Geopolitical Relations

Dynamic Bilateral Geopolitical Relations. We construct a measure of bilateral geopolitical relations based on geopolitical events between country pairs. Let $\{s_{ij,t}^n\}_{n=1}^{\tilde{N}_{ij,t}}$ denote the Goldstein scores for the set of events between countries i and j in year t , where $\tilde{N}_{ij,t}$ is the total number of events. The average event score, normalized to the $[-1, +1]$ interval, is:

$$\tilde{S}_{ij,t} = \frac{1}{\tilde{N}_{ij,t}} \sum_{n=1}^{\tilde{N}_{ij,t}} s_{ij,t}^n / 10$$

To capture both the immediate impact of political events and the institutional memory that characterizes international relationships, we then construct a dynamic geopolitical score for each country pair as a weighted moving average:

$$(1) \quad \begin{aligned} S_{ij,t} &= (1 - \phi_{ij,t}) \cdot S_{ij,t-1} + \phi_{ij,t} \cdot \tilde{S}_{ij,t} \\ \phi_{ij,t} &= \tilde{N}_{ij,t} / N_{ij,t}, \quad N_{ij,t} = (1 - \delta) N_{ij,t-1} + \tilde{N}_{ij,t} \end{aligned}$$

where $\phi_{ij,t}$ represents the updating weight, $N_{ij,t}$ is the effective cumulative number of events, and δ is the depreciation rate for past observations. We set $\delta = 0.3$, which yields a half-life of approximately two years and captures dynamics at the frequency of a typical four-year electoral cycle.⁷ The resulting dynamic score exhibits a half-life of approximately five years—versus under one year for raw event scores—confirming that it reflects the persistent bilateral relationship stock rather than transitory event noise (Appendix A.5). The dynamic geopolitical score is normalized to

⁶GDELT provides daily event data only from 1979 onward, while ICEWS covers 1995 to present. Both databases have limited historical depth compared to the timespan required for comprehensive economic analysis of geopolitical relationships.

⁷In years with no observed events ($\tilde{N}_{ij,t} = 0$), both the effective event count and the score decay by δ : $N_{ij,t} = (1 - \delta) N_{ij,t-1}$ and $S_{ij,t} = (1 - \delta) S_{ij,t-1}$. For the initial year of each bilateral pair, $S_{ij,t_0} = \tilde{S}_{ij,t_0}$ and $N_{ij,t_0} = \tilde{N}_{ij,t_0}$. Our results are robust to alternative depreciation rates. Section 6.2 examines the case $\delta = 1$ (unsmoothed event scores): while raw event scores show rapid mean reversion and smaller immediate GDP effects, the cumulative long-run impact of permanent changes converges to our baseline estimates, because the local projection controls for lags of geopolitical relations and thus identifies off innovations rather than the accumulated stock.

range from -1 (maximum conflict) to $+1$ (maximum cooperation). Throughout the paper, we refer to $S_{ij,t}$ as the *bilateral geopolitical score* and to the GDP-weighted aggregate p_{ct} (defined below) as the *geopolitical relations index*.

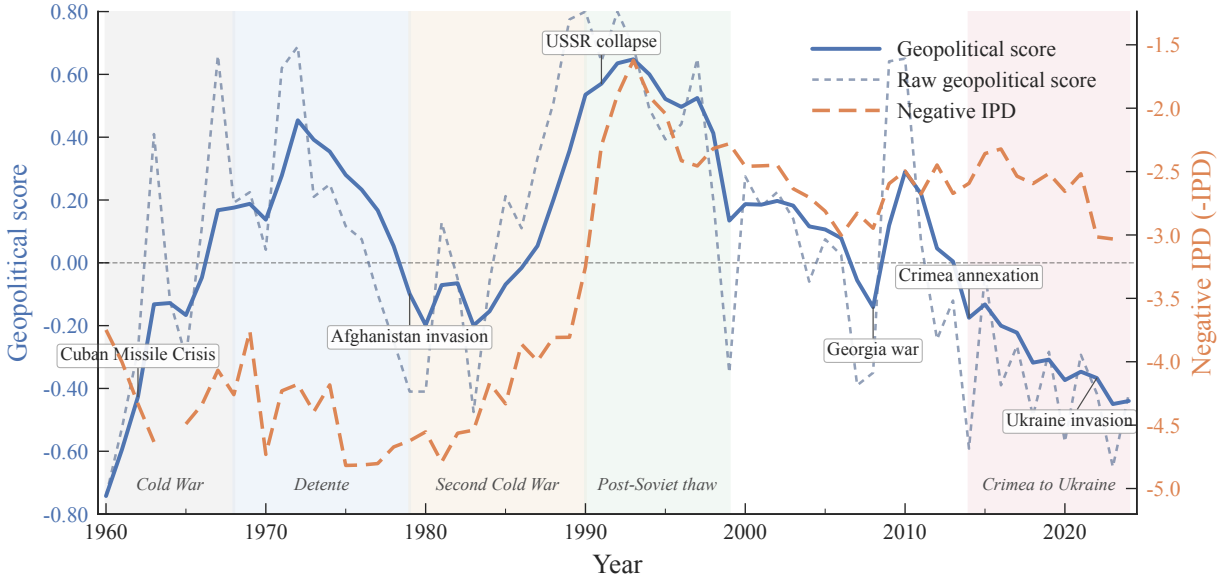


FIGURE 2. Geopolitical Scores Between United States and Russia (Soviet Union)

Time series comparison of geopolitical relationship measures between the United States and Russia, 1960–2024. The blue solid line shows our dynamic geopolitical score (left axis), the gray dashed line shows the raw (unsmoothed) geopolitical score, and the orange dashed line displays the negative Ideal Point Distance (–IPD) from UN voting data (right axis). Shaded regions indicate five geopolitical eras: Cold War (1960–1968), Détente (1968–1979), Second Cold War (1979–1989), Post-Soviet thaw (1989–2000), and Crimea to Ukraine (2014–2024). Key geopolitical events are annotated.

We validate the resulting measure against known bilateral dynamics and compare it with existing alternatives. Figure 2 plots our dynamic geopolitical score (blue line) between the United States and Russia from 1960 to 2024. The measure tracks major historical episodes: the Cuban Missile Crisis marking peak Cold War tensions, the Détente period (1968–1979), deterioration following the Soviet invasion of Afghanistan, improvement beginning with Gorbachev’s reforms in the mid-1980s, the post-Soviet peak in 1992, and subsequent decline culminating in the Crimean Crisis. The 2014–2024 period represents a historical low, exceeded only by the peak Cold War years. In contrast, the Ideal Point Distance measure from UN voting data (orange dashed line) fails to capture the Détente period and shows little movement after the Crimean Crisis despite the sustained bilateral deterioration. Figure 3 extends this validation to two non-superpower relationships that expose a critical limitation of UN voting-based measures.

Panel (a) of Figure 3 traces India-Pakistan relations through six decades of recurrent conflict. Our measure tracks the wars of 1965 and 1971, the Mumbai terrorist attacks (2008), and the Uri attack and subsequent surgical strikes (2016), with scores ranging from $+0.70$ in the early 1960s to -0.45 at the 1971 war nadir. The measure captures repeated cycles of conflict and cautious engagement: scores recover to $+0.40$ in the late 1970s, rise again during the Composite Dialogue

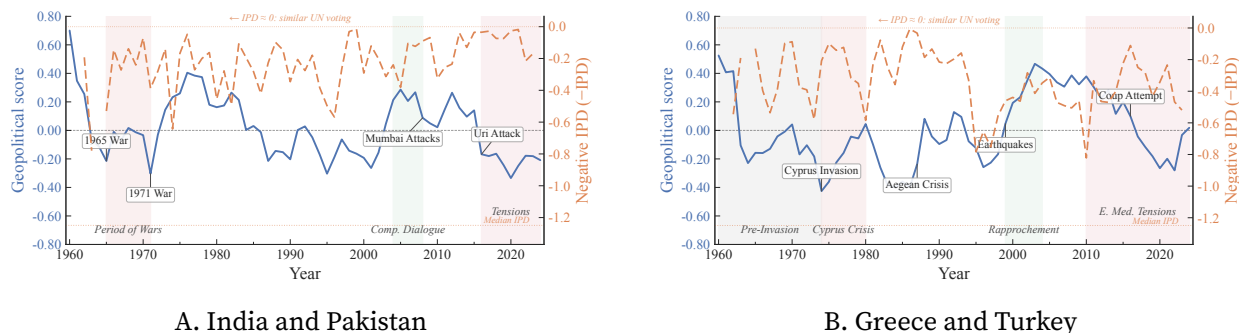


FIGURE 3. Geopolitical Scores: India-Pakistan and Greece-Turkey

Time series comparison of geopolitical relationship measures for India-Pakistan (panel a) and Greece-Turkey (panel b), 1960–2024. Blue solid lines show our dynamic geopolitical score (left axis); orange dashed lines display the negative Ideal Point Distance (–IPD) from UN voting data (right axis). The dotted horizontal line marks the cross-country median IPD. Shaded regions highlight key periods of conflict and cooperation; boxed labels mark major bilateral events.

period (2004–2008) to approximately +0.30, and collapse after the Mumbai attacks. The IPD measure, by contrast, shows these bitter rivals as increasingly aligned—because both countries vote similarly on postcolonial issues at the UN—completely inverting the actual bilateral relationship.

Panel (b) reveals a similar pattern in Greece-Turkey relations: our measure registers the sharp deterioration following the 1974 Cyprus invasion, the 1987 Aegean crisis, the brief rapprochement following the 1999 earthquakes, and the sustained decline during Eastern Mediterranean tensions (2010–2024). The IPD measure shows these NATO allies as consistently close throughout—because both vote with the Western bloc at the UN—despite decades of bilateral hostility over Cyprus, the Aegean, and maritime boundaries. In both cases, the entire IPD series remains well below the cross-country median (dotted line), confirming that UN voting similarity reflects bloc affiliation rather than bilateral dynamics.⁸ Appendix A.3 provides additional case validation across eight bilateral relationships spanning major power dynamics (Figure A3) and regional rivalries (Figure A4); geopolitical maps appear in Appendix A.6.

Robustness and Replicability. Beyond historical face validity, our measure demonstrates strong reliability across multiple dimensions. Cross-model agreement tests using three frontier LLMs—Gemini 2.5 Pro, GPT 5.4, and Claude (Opus 4.6 and Sonnet 4.6)—on 300 randomly sampled dyad-years yield a pooled correlation of 0.88, with within-model replicability reaching 0.93. The residual cross-model variation reflects differences in training data and scoring conventions rather than measurement instability (Appendix A.4).

A more demanding test varies the entire measurement pipeline—both the LLM and the prompt structure—simultaneously. An independent pipeline using Gemini 2.5 Flash with an earlier prompt design produces a country-year index correlated at 0.88 with our baseline, and nearly identical

⁸Fan, Wo, and Xiang (2025) show formally that UNGA voting similarity correlates with the low-dimensional bloc structure of our bilateral scores—the distance over the first three principal components—but misses the higher-dimensional bilateral variation that drives economic outcomes.

growth impulse responses with a horizon-by-horizon correlation of 0.97 (Figure A6). Because bilateral geopolitical events are extensively documented in public sources, different models and prompt architectures converge on similar assessments.

Advantages over Existing Measures. As Figures 2 and 3 demonstrate, UN voting similarity can miss or invert bilateral dynamics because it reflects multilateral bloc positions rather than bilateral interactions. Our measure complements these and other categorical approaches by situating discrete events—rivalries, sanctions, alliances, treaties—within a continuous bilateral spectrum.⁹ Section 6.2 presents empirical comparisons.

Country-Level Geopolitical Relations. To capture a country’s overall geopolitical stance, we aggregate bilateral geopolitical scores $S_{ij,t}$ into a country-level measure using a GDP-weighted average:

$$(2) \quad p_{ct} = \sum_{j \in \mathcal{N}} S_{cj,t} \times \text{GDP share}_{jt}$$

Here, GDP share_{jt} denotes country j ’s share of world nominal GDP in current U.S. dollars.¹⁰ We interpret nominal GDP as a proxy for geopolitical influence, as it reflects both economic strength and military capacity. The resulting index p_{ct} serves as our primary measure of a country’s geopolitical position in subsequent analyses of growth outcomes. Values of p_{ct} approaching -1 indicate strong global conflict, while values near $+1$ indicate strong alignment with major nations.

2.3. Landscape of Geopolitical Relations: 1960–2024

The country-level index reveals three macro-historical transformations in the global geopolitical landscape—Cold War bipolarity, post-Cold War convergence, and contemporary fragmentation—that also establish the within-country temporal variation exploited in Section 3.

Figure 4 illustrates the transformation from Cold War bipolarity to contemporary multipolarity. In 1980, the United States and Soviet Union anchor two distinct clusters, with extensive red dashed lines between blocs indicating widespread hostility. Notably, China appears positioned between the two superpowers but closer to the Western bloc, reflecting the Sino-American rapprochement following Nixon’s opening and shared opposition to Soviet expansion.¹¹

⁹UN voting-based measures include Signorino and Ritter (1999), Bailey, Strezhnev, and Voeten (2017), and Kleinman, Liu, and Redding (2024); see Broner et al. (2025) for limitations. Categorical alternatives include Strategic Rivalry (Thompson 2001; Aghion et al. 2019), Sanctions (Ahn and Ludema 2020; Felbermayr et al. 2021), Formal Alliance (Gibler 2008), and Treaties (Broner et al. 2025).

¹⁰The identifying variation in p_{ct} comes from changes in bilateral scores $S_{cj,t}$ rather than from movements in GDP shares. Figure B1(c) in Appendix B.3 confirms that replacing time-varying weights with fixed average or regime-fixed GDP shares yields nearly identical impulse responses.

¹¹The U.S.-China alignment in 1980 represented a dramatic reversal from the 1960s, demonstrating how quickly geopolitical relations can shift when strategic interests align. Non-aligned states like India, Brazil, and Mexico occupy intermediate positions, though their placement reveals varying degrees of practical tilt beyond stated non-alignment policies.

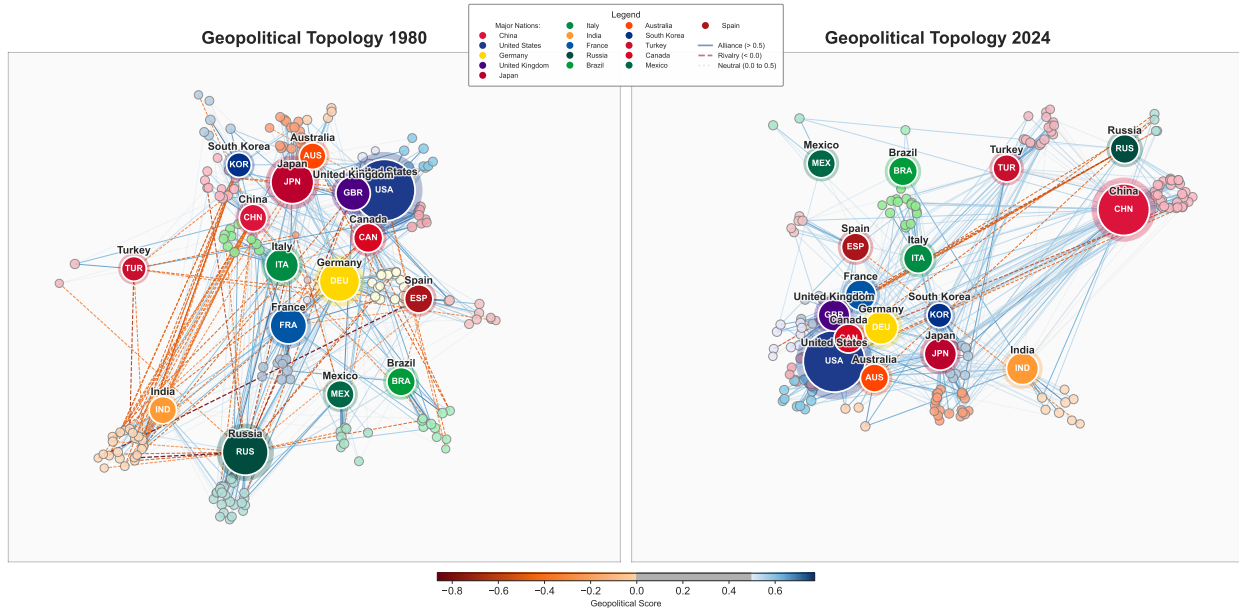


FIGURE 4. Geopolitical Relations Topology, 1980 and 2024

Geopolitical topology constructed using multidimensional scaling based on pairwise geopolitical scores between major nations. Distances between major nations reflect dissimilarity in their geopolitical relations with the rest of the world. Circle sizes indicate GDP shares. Smaller circles represent non-major countries, colored by strongest alignment and positioned near primary patron. Blue lines indicate alliances (scores > 0.5); red dashed lines indicate hostile relations (scores < 0.0).

By 2024, the bipolar structure has transformed into a fragmented system shaped by Russia's invasion of Ukraine. The Western alliance—comprising the United States, United Kingdom, Germany, France, Japan, Australia, and South Korea—forms a tightly integrated cluster, while China and Russia occupy a separate pole connected by alliance ties and jointly isolated from the West. India maintains a strategically ambiguous position between these blocs, and “connector” states like Turkey, Brazil, and Mexico continue bridging different clusters amid intensifying competition.¹²

Figure 5 documents shifts in the global distribution of geopolitical relations across six decades. During the Cold War era (1960–1990), the 5th percentile fluctuated between -0.30 and -0.15 , reflecting the volatile position of countries caught in superpower competition, while upper percentiles remained relatively stable. The 1970 and 1980 distributions are left-skewed—with a primary peak around 0.2 and considerable mass extending into negative territory—capturing the division into opposing geopolitical camps.

The post-Cold War transformation (1990–2010) brought dramatic convergence: the 5th percentile improved sharply from -0.15 to nearly 0.0 and the median rose from 0.20 to above 0.30 . By 2010, the distribution concentrated tightly around 0.25 with minimal mass below 0.0 , reflecting the globalization era's widespread cooperation. The 2010–2024 period shows a notable reversal—the 5th percentile declined back toward 0.0 , variance increased, and the 2020 distribution shows renewed

¹²These nonaligned “connector” countries are rapidly gaining importance and serving as bridges between blocs, potentially undermining the effectiveness of policies aimed at economic decoupling.

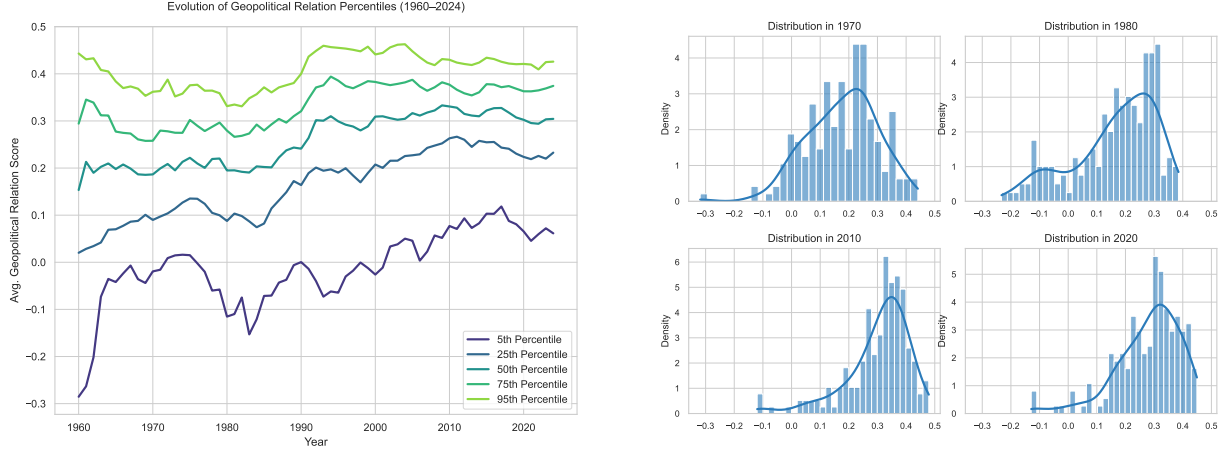


FIGURE 5. Evolution of the Distribution of Geopolitical Relations

Left panel shows the evolution of geopolitical relation percentiles from 1960–2024, with lines representing the 5th, 25th, 50th, 75th, and 95th percentiles. Right panels display kernel density estimates of the distribution of country-level average geopolitical relations for selected years (1970, 1980, 2010, 2020), highlighting the shift from Cold War bipolarity through post-Cold War convergence to contemporary re-polarization.

dispersion with a flattened peak and re-emerging left tail extending below 0.0.¹³

These distributional dynamics emerge from purely data-driven patterns, without imposing any historical periodization on the estimation.¹⁴ Table A4 in Appendix A.7 provides additional summary statistics by decade.

3. Dynamic Growth Effects of Geopolitics

This section estimates the dynamic effects of geopolitical relations on economic growth using local projections with country fixed effects. We present baseline impulse responses, extensive robustness analysis, and an instrumental variables strategy.

3.1. Economic Data

Our analysis employs a comprehensive panel dataset covering 193 countries from 1960 to 2024. The primary outcome variable is log GDP per capita in constant US dollars from the World Development Indicators, which provides broad country coverage and facilitates cross-country comparisons.¹⁵ We build upon the dataset constructed by Acemoglu et al. (2019), expanding coverage and incorporating additional variables from the Penn World Tables (Feenstra, Inklaar, and Timmer 2015) and

¹³The percentile spread narrowed from roughly 0.60 (ranging from -0.30 to 0.30) during peak Cold War to 0.35 (0.0 to 0.35) by 2010, before widening again to 0.40 by 2024. This compression and re-expansion of the distribution aligns with the rise and partial retreat of economic globalization.

¹⁴Mean geopolitical relations improved from 0.177 in the 1960s to 0.296 in the 2010s before declining to 0.281 in the 2020s; standard deviation fell from 0.141 to 0.108 over the same period before rising to 0.114 in the 2020s.

¹⁵Section 4.1 presents robustness checks using alternative output measures from the Penn World Tables, confirming that our results are not sensitive to the choice of GDP measure.

Acemoglu et al. (2025).¹⁶

Beyond our main outcome, we examine key growth determinants spanning multiple categories: enhanced Solow fundamentals (physical capital, investment rates, population growth, education), institutional measures (democracy indices, governance quality), trade openness, and additional correlates identified in the growth literature. These variables enable us to explore the channels through which geopolitical relations affect economic development.

Distinctiveness from Economic Fundamentals. Table B2 in Appendix B.2 reports within-year cross-sectional correlations between the geopolitical index and standard economic fundamentals. These correlations are generally small, unstable across decades, and typically below 0.2 in absolute value. In contrast, the same fundamentals exhibit strong and stable correlations with GDP per capita.

3.2. Empirical Specification

We denote our measure of country-level geopolitical relations as p_{ct} . Rather than examining the sources of geopolitical dynamics, we take geopolitical relations as given and identify their dynamic causal effects on economic outcomes. Specifically, we examine how a change in geopolitical relations at time t influences economic outcomes at future horizons relative to a baseline of no change. Formally, following Jordà and Taylor (2025), we define the impulse response function (IRF) as:

$$(3) \quad \mathcal{R}_{p \rightarrow y}(h) \equiv E[y_{c,t+h} | p_{ct} = p_{c0} + 1; \mathbf{x}_{ct}] - E[y_{c,t+h} | p_{ct} = p_{c0}; \mathbf{x}_{ct}], \quad h = 0, 1, \dots, H$$

where $y_{c,t+h}$ represents the economic outcome (log GDP per capita $\times 100$) at horizon h , and $\mathbf{x}_{ct}$ is a vector of control variables.¹⁷ The control vector includes:

$$\mathbf{x}_{ct} = \left\{ \{y_{c,t-\ell}, p_{c,t-\ell}\}_{\ell=1}^J, \delta_c, \delta_{r(c)t}, \mathbf{m}_{ct} \right\}$$

where $y_{c,t-\ell}$ and $p_{c,t-\ell}$ are lagged values of GDP and geopolitical relations, respectively, δ_c denotes country fixed effects, $\delta_{r(c)t}$ represents region-time fixed effects, and $\mathbf{m}_{ct}$ includes time-varying country-specific controls employed in robustness checks.

To estimate the impulse responses, we employ the local projection (LP) method proposed by Jordà (2005) and extended to panel settings by Jordà, Schularick, and Taylor (2020) and Bilal and Känzig (2024):

$$(4) \quad y_{c,t+h} = \alpha_h^{\text{LP}} p_{ct} + \gamma_h' \mathbf{x}_{ct} + \mu_{c,t+h}, \quad h = 0, 1, \dots, H$$

where each horizon h is estimated via a separate regression, providing a semiparametric approxi-

¹⁶The panel is unbalanced, with coverage varying across countries and time periods. Table B1 in Appendix B.1 provides detailed information on country coverage and data availability for each variable.

¹⁷In our linear specification, this impulse response equals the marginal effect of geopolitical relations on GDP. The multiplication by 100 means coefficients are expressed in log points.

mation of the conditional expectation in equation (3). In our baseline specification, we include four lags ($J = 4$) of both GDP and geopolitical relations, consistent with Acemoglu et al. (2019), to capture growth dynamics while ensuring robust inference in the presence of serial correlation (Montiel Olea and Plagborg-Møller 2021).

Country fixed effects δ_c account for time-invariant heterogeneity across countries, enabling identification from within-country variation—a stringent requirement in growth empirics (Durlauf, Johnson, and Temple 2005). Region-time fixed effects $\delta_{r(c)t}$ control for regional and global shocks, including shared temporal dynamics such as regional conflicts, financial crises, and globalization waves.¹⁸ The accurate measurement of the timing and intensity of geopolitical relations is critical for empirical relevance in this demanding panel setting.

ASSUMPTION 1. *The structural shock to GDP satisfies $\mathbb{E} \left[\mu_{c,t+h} \mid p_{ct}, \{\mathbf{x}_{c\tau}\}_{t_0 \leq \tau \leq t} \right] = 0$ for all $\{\mathbf{x}_{c\tau}\}_{t_0 \leq \tau \leq t}$, all countries c , and all $t \geq t_0$.*

Assumption 1 ensures that, conditional on lagged GDP, lagged geopolitical relations, and other controls, geopolitical relations are orthogonal to contemporaneous and future shocks to GDP. This identification condition implies that $\mathcal{R}_{p \rightarrow y}(h) = \alpha_h^{\text{LP}}$ (Jordà 2005), allowing consistent estimation via ordinary least squares (OLS). Compared to vector autoregressions (VARs), local projections offer greater robustness to model misspecification, as they do not require a fully specified dynamic system to extrapolate responses across horizons (Montiel Olea and Plagborg-Møller 2021).

Economically, Assumption 1 requires that countries with different geopolitical relations exhibit similar potential GDP growth trends after conditioning on our controls. This is a strong assumption that warrants careful examination. Our rich set of controls—country fixed effects, lagged GDP, and lagged geopolitical relations—captures key determinants of both future growth and geopolitical dynamics, including geographical endowments, historical legacies, institutional quality, and economic fundamentals. Section 3.4 first establishes that the estimated effects are not driven by economic content embedded in the measure. We then probe the credibility of this identification through four complementary exercises.

First, Section 3.5.1 tests sensitivity to alternative fixed effects structures and additional time-varying controls, including lags of trade openness, domestic political stability, and casualty-based war exposure measures. Second, Section 3.5.2 examines the identifying variation by decomposing our geopolitical measure into relations with the United States versus other major nations and inspecting the distribution of effects across the shock distribution. Third, Section 3.5.3 conducts placebo exercises that test whether the estimated GDP response depends on the realized country-specific assignment and timing of geopolitical shocks. Fourth, Section 3.6 implements an instrumental variables strategy restricting identifying variation to non-economic verbal conflicts—diplomatic disputes, border tensions, and cultural disagreements that move bilateral relations but lack direct economic content. The exclusion restriction is not beyond dispute, and we discuss its limitations.

¹⁸Our estimates are robust to using only time fixed effects, as shown in Section 3.5.

3.3. Baseline Results

Figure 6 displays two central results: the persistence of geopolitical shocks and their dynamic impact on GDP per capita.

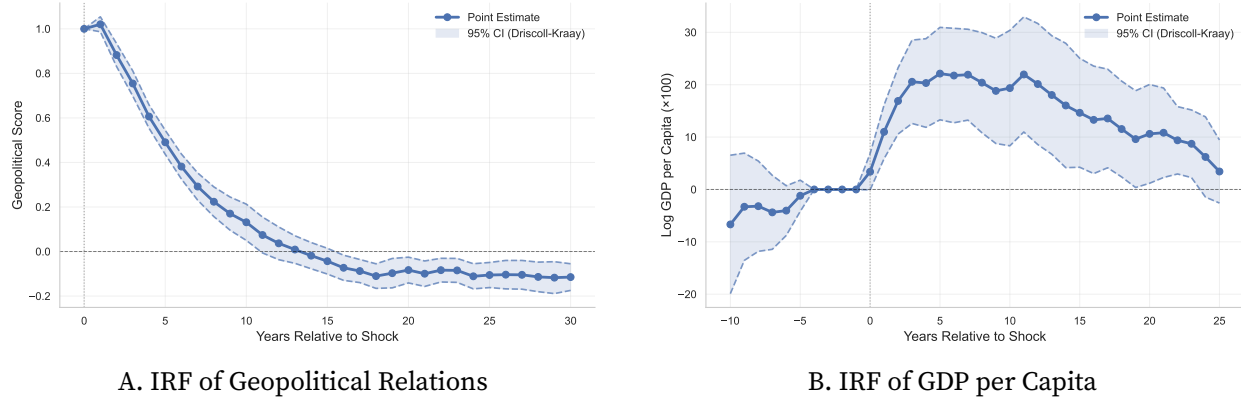


FIGURE 6. Dynamic Responses to Geopolitical Relations Shock

Panel (a) shows the impulse response of geopolitical relations to a unit shock. Panel (b) displays the impulse response of log GDP per capita ($\times 100$) to a unit improvement in geopolitical relations. Both panels estimate equation (4) with four lags of the dependent variable and geopolitical relations, country fixed effects, and region-year fixed effects. Shaded areas represent 95% confidence intervals based on Driscoll-Kraay standard errors. Horizons range from 0 to 30 years in panel (a) and -10 to 25 years in panel (b), where negative horizons test for pre-trends.

Dynamics of Geopolitical Relations and Economic Growth. Panel (a) of Figure 6 shows that geopolitical relations exhibit substantial persistence following a shock. A unit improvement decays gradually—approximately 50% of the initial effect persists after five years, and 15% remains after ten years. Full dissipation occurs after approximately 15 years. This persistence drives the cumulative economic impact: the impulse response in panel (b) captures both the direct effect of the initial shock and the compounding influence of sustained improvements in geopolitical relations.¹⁹

Panel (b) shows three key patterns in the GDP response. First, the absence of pre-trends for horizons -10 to -1 is consistent with our identification assumption, indicating no systematic relationship between future geopolitical changes and current economic outcomes.²⁰ Second, GDP rises on impact, peaking at approximately 20 log points around year 5. Third, the response follows a hump-shaped path, gradually declining to roughly 10 log points by year 20 as the underlying geopolitical impulse dissipates. Our inference employs Driscoll-Kraay standard errors to account for serial correlation and cross-sectional dependence (Driscoll and Kraay 1998; Montiel Olea and Plagborg-Møller 2021).²¹

¹⁹The dynamics of geopolitical relations can be approximated by a mean-reverting AR(2) process with overshooting. Appendix A.5 provides additional analysis.

²⁰Panel (b) of Figure 6 includes all 184 countries with GDP data available in any year. In Sections 3.5.1 and 4.1, estimates using countries with comprehensive data coverage show point estimates close to zero for horizons -10 to -5 .

²¹Appendix B.3 presents four robustness checks: (i) restricting to 146 countries with complete data yields virtually identical impulse responses; (ii) bootstrap inference accounting for estimation uncertainty in the geopolitical measure

Transitory versus Permanent Shocks. The impulse responses in Figure 6 capture the combined effect of an initial geopolitical shock and its subsequent dynamics. To isolate the impact of purely transitory changes, we follow Sims (1986) and Bilal and Känzig (2024) in constructing counterfactual impulse responses to shocks that increase by one unit on impact and immediately return to zero.²²

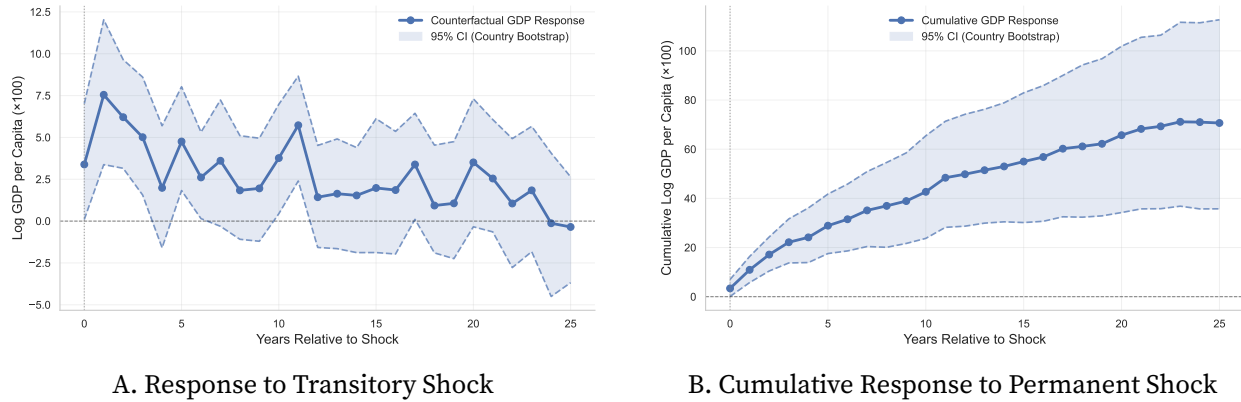


FIGURE 7. GDP Responses to Transitory and Permanent Geopolitical Shocks

Panel (a) shows the response to a purely transitory unit shock (shock equals 1 at $h = 0$ and 0 thereafter). Panel (b) displays the cumulative response to a permanent unit shock. Baseline specification as in Figure 6. Shaded areas represent 95% confidence intervals from 1,000 country-block bootstrap iterations.

Figure 7 decomposes the effects into responses to transitory and permanent shocks. Panel (a) shows that even purely transitory improvements in geopolitical relations generate persistent economic gains. GDP per capita increases by approximately 3.5 log points on impact, rises to a peak of 7.5 log points after one year, and gradually fades over 20 years. The wide confidence intervals reflect substantial uncertainty in medium-term dynamics, consistent with the challenges of estimating long-horizon effects in cross-country panels.

Panel (b) demonstrates the cumulative gains from permanent improvements: GDP per capita rises steadily, reaching approximately 70 log points after 25 years. The trajectory stabilizes after year 20, suggesting convergence to a new steady state. While local projections provide robust estimates, they sacrifice statistical efficiency at long horizons. Section 6.1 presents a complementary analysis using dynamic panel methods that exploit the autoregressive structure for more precise long-run estimates.

Economic Magnitude. The economic significance of our findings depends on the plausible range of geopolitical variation. The geopolitical relations index has a standard deviation of 0.143 and ranges from -0.43 (hostile relations) to 0.55 (strong cooperation), with a median of 0.26 .²³ A one-standard-

produces only marginally wider confidence intervals; (iii) replacing time-varying GDP weights with fixed or regime-fixed weights yields nearly identical responses; (iv) alternative lag specifications confirm the stability of our results.

²²While local projections semiparametrically estimate the empirical impulse response, constructing counterfactual responses requires the structural assumption that effects of a series of unanticipated shocks equal those of an anticipated path announced at time zero. Appendix B.4 details our methodology.

²³The 25th percentile is 0.155 and the 75th percentile is 0.342.

deviation improvement in geopolitical relations generates a long-run GDP gain of approximately 10 log points (0.143×70). Moving from the 25th to the 75th percentile—representing a shift from limited to moderate cooperation—increases GDP by 13 log points over 25 years.

More substantial improvements yield proportionally larger effects. Progressing from verbal tensions (index value of 0) to the cooperation level observed among allied nations (0.42) generates a 29 log point increase in GDP per capita. Distributed over 25 years, this represents an average annual growth contribution of 1.2 log points. For a country at the median geopolitical relations level (0.26), achieving the highest observed cooperation (0.55) results in a cumulative GDP increase of 20 log points. These gradual, compounding effects underscore the importance of long-horizon analysis.

While the absence of pre-trends is consistent with our identification assumption, pre-trend tests have limited power against confounders that accumulate gradually. A separate concern is whether the growth effects are driven by economic content embedded in the index—trade agreements, sanctions, or investment treaties that mechanically correlate with growth conditions. We address this next by decomposing the index into its constituent dimensions.

3.4. Decomposing the Geopolitical Index

Our event classification system (Section 2.1) assigns each bilateral event to one of six major categories. We construct three components of the geopolitical index from these categories: *economic relations* (category A), *diplomatic and political relations* (category B), and *security, defense, and territorial relations* (categories C and D).²⁴ This classification is defined ex ante in the LLM prompt and held fixed across the entire sample. Each component is aggregated to the country-year level using the same GDP-weighted procedure as the baseline index (equation (2)). The three components correlate with the full index at 0.74, 0.78, and 0.67, with pairwise cross-correlations between 0.36 and 0.53—sufficiently distinct to support a meaningful decomposition.

Horse Race. Figure 8 reports local projection estimates from two specifications that assess the independent contribution of each component. Panel (a) includes all three components simultaneously, each with four lags, in a multivariate extension of equation (4). All three generate positive and statistically significant growth responses. Diplomatic relations exhibit the largest and most persistent effect, rising to approximately 11 log points at horizon 5 and remaining significant through horizon 20. Security and territorial relations produce a sustained response, reaching 12 log points at horizon 10. Economic relations peak early—approximately 11 log points at horizon 5—before gradually attenuating. The joint significance of all three components shows that the growth effects of geopolitical relations are not reducible to economic content alone.

²⁴Category A covers trade policy, financial relations, economic coercion, strategic sectors, and economic integration. Category B covers formal diplomacy, high-level interactions, public diplomacy, and cultural exchanges. Categories C and D cover military cooperation, security incidents, intelligence, territorial disputes, and maritime issues. The remaining categories—multilateral governance (E) and other events (F)—account for 20% of events and are omitted; including them as a fourth component does not affect the results. See Table A1 for subcategory details.

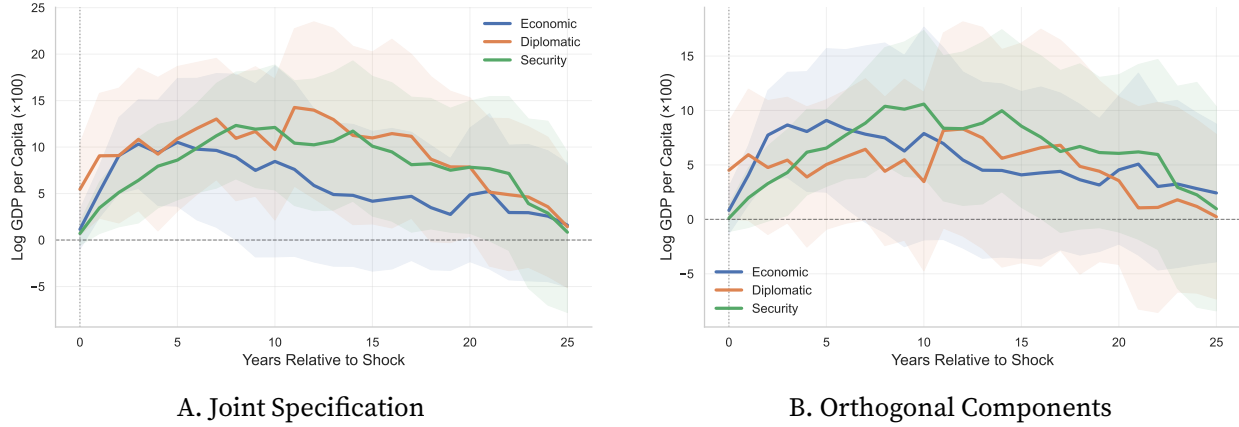


FIGURE 8. Decomposing Geopolitical Relations: Component-Level Growth Effects

Panel (a) estimates a multivariate extension of equation (4) with all three geopolitical components entered simultaneously, each with four lags. Panel (b) uses orthogonal components obtained by regressing each component on the other two with country and year fixed effects and using the residuals as shocks. Both panels show the impulse response of log GDP per capita ($\times 100$) with baseline fixed effects and Driscoll-Kraay standard errors as in Figure 6. Shaded areas represent 95% confidence intervals. Components: Economic (category A), Diplomatic (category B), Security (categories C and D).

Orthogonal Variation. Panel (b) isolates variation unique to each component by regressing each on the other two with country and year fixed effects, then using the residuals as shocks in the baseline local projection. Security and territorial relations show the strongest independent signal, significant from horizon 5 through 15 and peaking at approximately 11 log points. Economic relations retain significance at short horizons, while diplomatic relations show a positive but less precisely estimated response. This contrast with panel (a)—where diplomacy exhibits the largest joint effect—is informative: much of diplomatic variation is shared with economic and security conditions, so its incremental predictive content is high but its orthogonal content is more limited.

The security component—military cooperation, defense agreements, territorial arrangements, and conflict dynamics—generates the most robust independent growth response, consistent with security alignment playing a distinct role beyond economic channels.²⁵ Having shown that the growth effects are not driven by economic content alone, we now present complementary evidence strengthening the causal interpretation of these estimates.

3.5. Robustness

We evaluate the credibility of our baseline estimates through three sets of exercises: robustness to alternative specifications and controls, examination of the identifying variation, and placebo tests that probe the assignment and timing of geopolitical shocks.

²⁵Additional results in the appendix include individual component impulse responses, the baseline index overlaid with component estimates, and a timing analysis showing mutual reinforcement between diplomatic and economic components.

3.5.1. Additional Controls and Fixed Effects

We first verify that results are robust to alternative fixed effects specifications and additional time-varying controls, using a balanced panel of countries with at least 30 years of GDP data.²⁶

Alternative Fixed Effects Specifications. We examine sensitivity to different assumptions about unobserved heterogeneity. For these tests, we restrict our analysis to 146 countries with complete GDP coverage across all horizons, ensuring that compositional changes do not drive differences across specifications.

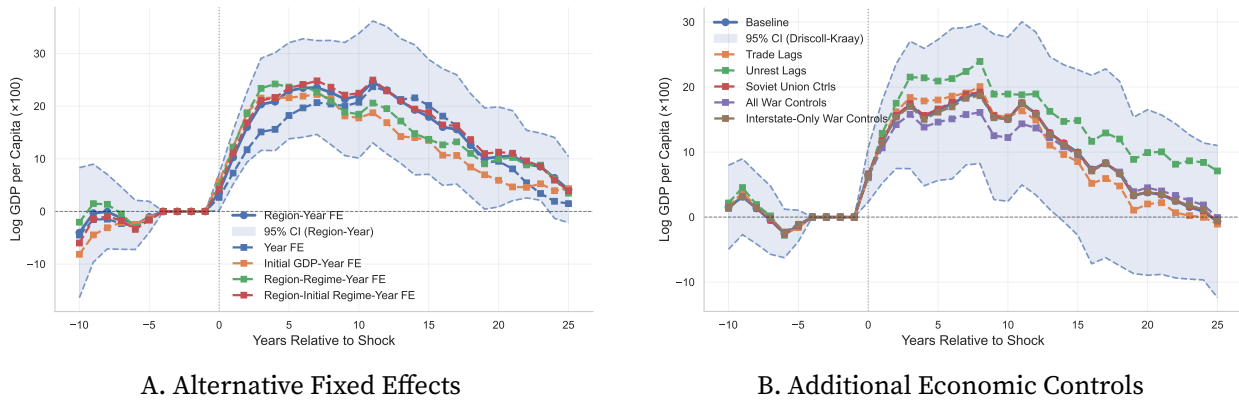


FIGURE 9. Robustness to Alternative Specifications

Panel (a) compares impulse responses under alternative fixed effects specifications. The baseline specification (solid blue line with 95% confidence interval) includes region-year fixed effects. Alternative specifications shown as dashed lines include: year fixed effects only, initial GDP quintile-year fixed effects, region-regime-year fixed effects, and region-initial regime-year fixed effects. Panel (b) presents robustness to additional economic and political controls, with the baseline specification compared to models including four lags of trade openness, domestic unrest, Soviet transition controls, and casualty-based war exposure measures (all wars and interstate wars only) from Federle et al. (2026). All specifications use the baseline lag structure and Driscoll-Kraay standard errors as in Figure 6.

Panel (a) of Figure 9 confirms stability across four alternative fixed effects structures: year fixed effects only, initial GDP quintile-year effects,²⁷ region-initial regime-year effects, and region-current regime-year effects. All specifications yield impulse responses closely tracking the baseline, with peaks between 20 and 23 log points. The modest attenuation under region-current regime-year effects—which absorb contemporaneous democratization—suggests that geopolitical relations and democratic transitions represent complementary but distinct growth channels, a finding we explore in Section 4.3.

Time-Varying Economic and Political Controls. We next examine robustness to additional time-varying controls that address specific historical events and potential confounding economic factors.

²⁶Following our approach in Appendix B.3, we restrict the sample to ensure consistent composition across specifications.

²⁷Following Acemoglu et al. (2019), we rank countries using Angus Maddison's GDP estimates for 1960, which are available for 149 countries, to maximize sample coverage.

For these specifications, we restrict our sample to 108 countries with complete data coverage for all control variables, ensuring comparability across estimates.

Panel (b) of Figure 9 presents these results. Including four lags of trade openness slightly attenuates long-run effects—consistent with trade being one channel through which geopolitical relations affect growth—but the impulse response remains significant, peaking at approximately 20 log points. Controls for the Soviet Union’s collapse (interactions between Soviet/satellite indicators and year dummies for 1989–1992) have minimal impact. Results are also robust to controlling for four lags of domestic unrest,²⁸ alleviating the concern that geopolitical deterioration proxies for domestic political instability. Finally, we include contemporaneous and lagged casualty-based war exposure measures²⁹ to absorb the direct economic effects of armed conflict. The impulse response remains stable, indicating that geopolitical relations affect growth through channels beyond the destruction of war.

Appendix B.5 shows that results are also robust to controlling for a broad set of economic fundamentals from the Penn World Tables, including investment, capital, human capital, and labor market variables.

3.5.2. Identifying Variation and Symmetry

We investigate whether our results are driven by outliers or depend critically on specific components of our geopolitical relations measure. The Frisch-Waugh-Lovell (FWL) theorem allows us to visualize the identifying variation by plotting the relationship between GDP and geopolitical relations after partialling out all controls and fixed effects.

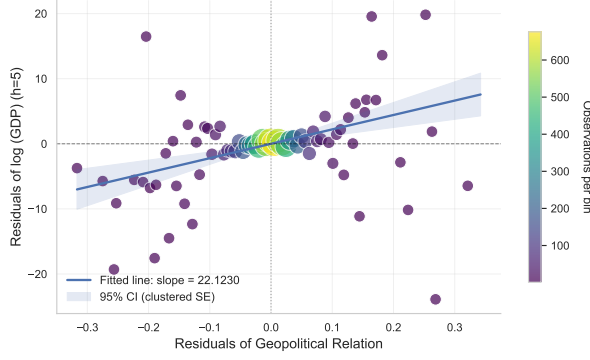
Figure 10 shows a robust positive relationship between geopolitical relations and future GDP growth. Panel (a) confirms that the relationship remains stable across the entire distribution of geopolitical shocks—from large negative changes (deteriorating relations) to large positive changes (improving relations). The approximately linear relationship indicates that both positive and negative changes in geopolitical relations generate symmetric economic responses, with no evidence of threshold effects or nonlinearities.

Panel (b) provides complementary evidence by examining GDP at $t + 15$ against cumulative geopolitical changes measured at $t + 5$. Using these forward geopolitical relations—which capture the accumulation of diplomatic changes over 5-year windows—yields a slope of approximately 17, consistent with the baseline impulse response at comparable horizons. The stability of the relationship at this extended horizon, combined with consistency across different measures (contemporaneous versus cumulative changes), reinforces that our findings reflect a robust economic relationship rather than the influence of outliers or extreme events.³⁰

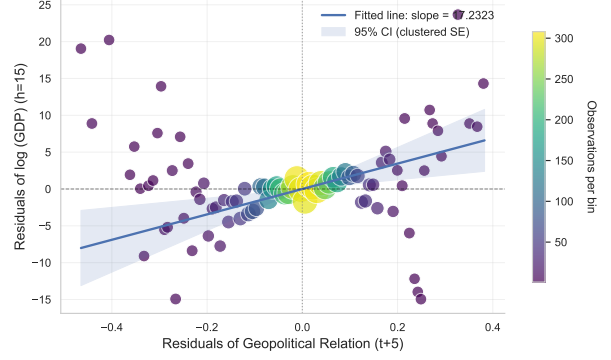
²⁸The unrest measure captures strikes, demonstrations, and political violence from the Cross-National Time-Series Data Archive.

²⁹We include contemporaneous values and four lags of five war variables: domestic war-site onset, domestic casualty intensity (scaled by prewar population), trade-weighted foreign war exposure, proximity-weighted foreign war exposure, and the count of active foreign war sites. We report specifications using both all wars and interstate wars only.

³⁰Appendix B.6 presents additional binscatter plots for the 10-year horizon, as well as raw scatter plots. The conditional



A. GDP at $t + 5$ vs. Geo Relations at t



B. GDP at $t + 15$ vs. Geo Relations at $t + 5$

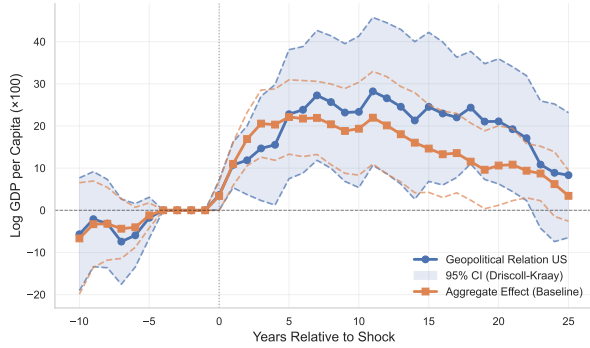
FIGURE 10. Binscatter Plots of Residualized GDP and Geopolitical Relations

Binscatter of residualized log GDP per capita against residualized geopolitical relations after partialling out all baseline controls (as in Figure 6). Panel (a): GDP at $t + 5$ vs. geopolitical relations at t . Panel (b): GDP at $t + 15$ vs. forward geopolitical relations at $t + 5$. Each dot represents approximately 100 observations; fitted lines use Driscoll-Kraay standard errors.

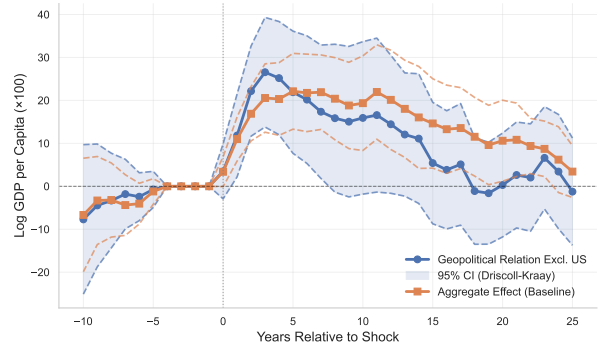
Next, we examine whether our results depend on geopolitical relations with the United States specifically or reflect broader patterns of international alignment. We decompose our geopolitical relations measure into relations with the US (p_{ct}^{US}) and relations with all other major nations ($p_{ct}^{Excl.US}$), then jointly estimate their effects:

$$y_{c,t+h} = \alpha_h^{US} p_{ct}^{US} + \alpha_h^{Excl.US} p_{ct}^{Excl.US} + \gamma_h' \mathbf{x}_{ct} + \delta_c + \delta_{r(c)}t + \varepsilon_{c,t+h}, \quad h = -10, \dots, 25$$

where $\mathbf{x}_{ct}$ includes four lags of GDP, geopolitical relations with the US, and geopolitical relations excluding the US.



A. Geopolitical Relations with US



B. Geopolitical Relations Excluding US

FIGURE 11. Decomposing Geopolitical Relations: US versus Other Major Nations

Impulse responses from joint estimation of US and non-US geopolitical relations. Blue: decomposed effect; orange: baseline aggregate for comparison. Baseline specification as in Figure 6.

Figure 11 shows that both components generate nearly identical dynamic responses. The covariance between GDP and geopolitical relations remains stable across all horizons.

impulse response to improved relations with the US (panel a) closely mirrors that from improved relations with other major nations (panel b), with both peaking at approximately 25–28 log points within the first decade.³¹ The overlapping confidence intervals and similar trajectory to our baseline aggregate measure (shown in orange) indicate that our results reflect a general phenomenon rather than US-specific dynamics. This finding also supports our use of nominal GDP shares to aggregate bilateral geopolitical relations, suggesting that GDP weights capture the economic importance of different bilateral relationships. Appendix B.7 shows that decomposing relations into Western versus non-Western countries yields similarly robust results.

The stability of our estimates across distinct sources of variation strengthens the credibility of the baseline design. Both ideologically and economically different partners generate similar growth responses, which is inconsistent with ideology-specific or country-specific confounds and consistent with a general cooperation channel.³² Combined with the linearity of effects across the distribution, these results support the interpretation that the estimated effects are not driven by a narrow source of geopolitical variation. Appendix B.8 confirms that these effects are present in both the 1960–1989 and 1990–2019 sub-periods, alleviating the concern that our results are driven exclusively by Cold War realignments.

3.5.3. Placebo and Timing Validation

We conduct two placebo exercises that test whether the estimated GDP response depends on the realized country-specific assignment and timing of geopolitical shocks. Both exercises re-estimate the full baseline local projection from scratch in each placebo draw, preserving the specification's dynamics and fixed effects structure.

Within-Region-Year Reassignment. The first exercise randomly reassigns geopolitical shocks across countries within each region-year cell, preserving common regional shocks and the cross-sectional distribution of the shock variable while breaking the link between each country's outcome path and its own realized geopolitical shock. Since our baseline already includes region-year fixed effects, this placebo does not test for common regional dynamics—those are absorbed in the main specification. Instead, it asks whether, conditional on the regional-time environment, the estimated GDP response depends on the realized country-specific assignment of shocks within a region-year cell.

Panel (a) of Figure 12 shows that the placebo median remains flat near zero across all horizons and the placebo band stays tight, while the realized response lies well outside the placebo envelope throughout. This rules out a specific class of threats: mechanical artifacts from GDP persistence,

³¹The conditional correlation between US and non-US relations—after partialling out country fixed effects, region-year fixed effects, and lagged controls—is only 0.28, so that over 90% of the identifying variation in each component is independent of the other.

³²As Section 4.3 documents, democratization improves relations with Western countries but not with Russia—and only modestly with China and India—yet both components generate similar growth responses.

incidental within-cell covariance structure, and estimator-specific biases that could generate spurious positive responses to any persistent regressor.

However, this exercise does not rule out time-varying country-specific confounders that are contemporaneously correlated with geopolitical changes. Such confounders remain attached to each country-year observation, but the random reassignment breaks their correlation with the geopolitical measure, just as it breaks the true treatment assignment. A null placebo result is therefore consistent both with a genuine treatment effect and with omitted-variable bias in the original specification.

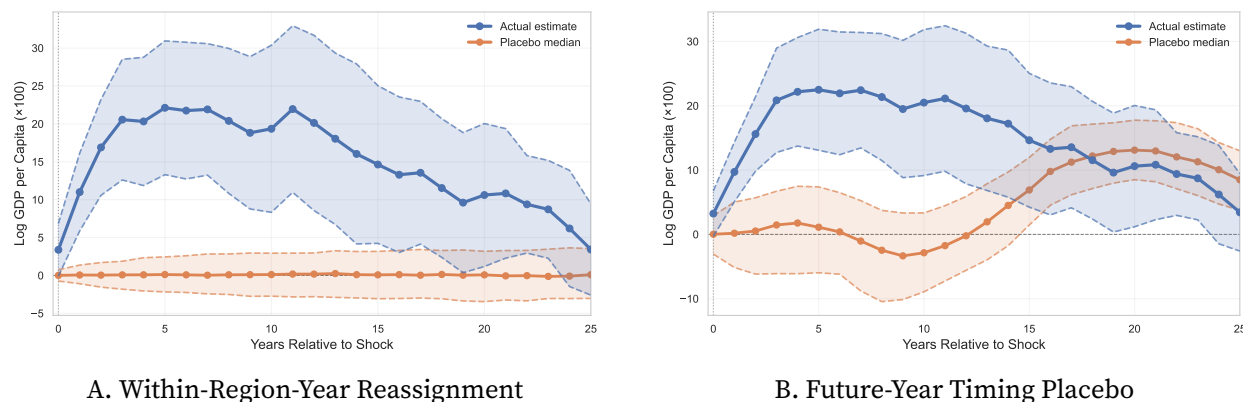


FIGURE 12. Placebo Tests for the Dynamic GDP Response to Geopolitical Shocks

Blue: actual estimates with 95% Driscoll-Kraay CIs. Orange: placebo median and 5th–95th percentile range across 500 draws. Panel (a): within-region-year reassignment of geopolitical shocks. Panel (b): replacement with future shocks drawn 8–15 years ahead. Both re-estimate the full baseline specification (Figure 6) in each draw.

Future-Year Timing Placebo. The second exercise replaces each country’s contemporaneous shock with a randomly drawn future shock (8 to 15 years ahead), preserving each country’s own shock distribution while breaking the contemporaneous timing. Unlike the cross-country reassignment, this placebo preserves country identity and therefore tests whether the baseline result depends on the actual temporal alignment between geopolitical changes and subsequent GDP dynamics.

Panel (b) of Figure 12 reveals a pattern consistent with the timing structure of the design. The placebo median remains near zero through approximately horizon 8, then begins rising as the local projection window reaches horizons where the shifted shocks have actually occurred. By horizons 15–18, the placebo response reaches magnitudes comparable to the actual response over its peak horizons 0–10. This delayed emergence serves as a positive control for timing: the LP recovers the expected response once the outcome window overlaps with the actual occurrence of the reassigned shocks, but produces null results at horizons where the shifted shocks have not yet materialized.

This pattern supports the interpretation that the baseline estimates depend on the temporal location of geopolitical shocks rather than on slow-moving country trajectories. At the same time, because the shift is long, the exercise is less informative about more proximate time-varying confounders that may jointly influence current growth and contemporaneous geopolitical changes.

Appendix B.9 reports complementary randomization-inference results based on the average response over horizons 0 to 10, which place the realized estimate in the extreme right tail of the placebo distribution (randomization $p < 0.002$ in both exercises).

3.6. Instrumental Variables Estimates

The preceding analyses show that geopolitical relations predict economic growth in a demanding panel setting with rich controls and robust inference. A remaining concern is that the economic components of our geopolitical index—trade agreements, sanctions, investment treaties—may themselves respond to growth prospects. To assess the sensitivity of our results to this specific form of reverse causality, we implement an instrumental variables strategy that restricts the identifying variation to non-economic verbal conflicts: diplomatic disputes, border tensions, and cultural disagreements that move bilateral relations but lack direct economic content. This exercise tests whether the baseline relationship survives when we exclude the most endogeneity-prone components of the index from the identifying variation.

Instrument description. Our instrument isolates variation from a specific subset of geopolitical events: non-economic verbal conflicts. These events—including diplomatic protests, public criticisms, formal demands, and symbolic political gestures—move bilateral relations but are plausibly less directly tied to contemporaneous macroeconomic conditions than economic agreements, sanctions, or material conflicts. They arise from political and institutional disputes, security and strategic concerns, social and cultural tensions, and legal disagreements over non-economic matters.³³ These conflicts generate considerable variation in bilateral relations through escalation patterns—from formal protests to public condemnations and threatened retaliation—that deteriorate relations without involving material actions.

Exclusion restriction. The identifying assumption is that these verbal conflicts affect GDP only through their impact on overall geopolitical relations, conditional on our controls (stated formally as Assumption 2 below). Several features of our design support this restriction. We exclude material and severe conflicts (CAMEO codes 15–20) that may carry direct economic consequences, and we control for four lags of the instrument, absorbing persistent confounders correlated with escalation dynamics. The baseline specification already conditions on lagged GDP, lagged geopolitical relations, and region-year fixed effects, so the exclusion restriction requires only that verbal conflicts carry no *additional* information about future growth beyond what these controls and the bilateral relations channel capture.

We do not claim this restriction is beyond dispute. Verbal diplomatic conflicts may reflect underlying shocks—leadership changes, strategic realignments, regime deterioration—that inde-

³³The economic/non-economic classification is detailed in Appendix C.2. We focus on CAMEO root codes 9–14, which represent verbal conflicts ranging from diplomatic investigations to formal protests, excluding material conflicts (codes 15–16) that may have direct economic consequences through disrupted diplomatic channels or investment signaling, and severe conflicts (codes 17–20) that may directly interact with economic activity. Appendix A.8 provides a comprehensive analysis of these 37,519 events spanning 1960–2024.

pendently affect growth through channels not fully mediated by our bilateral relations index. The IV estimates should therefore be interpreted as a check on whether the baseline results are driven by reverse causality through economic channels, not as a resolution of all identification concerns.

Empirical implementation. We construct our instrument by isolating the component of geopolitical relations driven by non-economic verbal conflicts:

$$z_{ct} = \sum_{j \in \mathcal{N}} \left(\frac{1}{N_{cj,t}^Q} \sum_{n \in Q} s_{cj,t}^n / 10 \right) \times \text{GDP share}_{jt}$$

where Q denotes the set of non-economic verbal conflict events, $N_{cj,t}^Q$ is the count of such events between countries c and j in year t , and $s_{cj,t}^n$ represents the Goldstein score of event n . This construction parallels our main geopolitical relations measure but restricts attention to verbal conflict events.

We estimate impulse responses using the local projection instrumental variables (LP-IV) method introduced by Jordà, Schularick, and Taylor (2015). Let $\tilde{\mathbf{x}}_{ct} = \left\{ \{z_{c,t-\ell}\}_{\ell=1}^4, \mathbf{x}_{ct} \right\}$ denote our extended control vector including four lags of the instrument. The LP-IV estimator proceeds in two stages. First, we estimate the reduced-form projection:

$$y_{c,t+h} = \alpha_h^{\text{RF}} z_{ct} + \gamma_h' \tilde{\mathbf{x}}_{ct} + \mu_{c,t+h}^{\text{RF}}$$

and the first-stage relationship:

$$p_{ct} = \alpha^{\text{FS}} z_{ct} + \gamma' \tilde{\mathbf{x}}_{ct} + \mu_{ct}^{\text{FS}}$$

The LP-IV estimate of the impulse response is then $\hat{\alpha}_h^{\text{LP-IV}} = \hat{\alpha}_h^{\text{RF}} / \hat{\alpha}^{\text{FS}}$. Following Plagborg-Møller and Wolf (2021), this approach requires the following exclusion restriction:

ASSUMPTION 2 (LP-IV Exclusion Restriction). $\mathbb{E} \left[\mu_{c,t+h} z_{ct} \mid \{\tilde{\mathbf{x}}_{c\tau}\}_{t_0 \leq \tau \leq t} \right] = 0$ for all countries c , all $t \geq t_0$, and all horizons h .

This assumption requires that non-economic verbal conflicts affect future GDP only through their impact on overall geopolitical relations, conditional on our controls.³⁴ The decomposition in Section 3.4 reinforces this restriction: the instrument operates through diplomatic and security channels distinct from the economic content most susceptible to reverse causality.

³⁴The assumption requires that, once we control for all lagged data, the instrument is not contaminated by other structural shocks or by lags of the shock of interest. It is equivalent to estimating a VAR with the instrument ordered first (Plagborg-Møller and Wolf 2021).

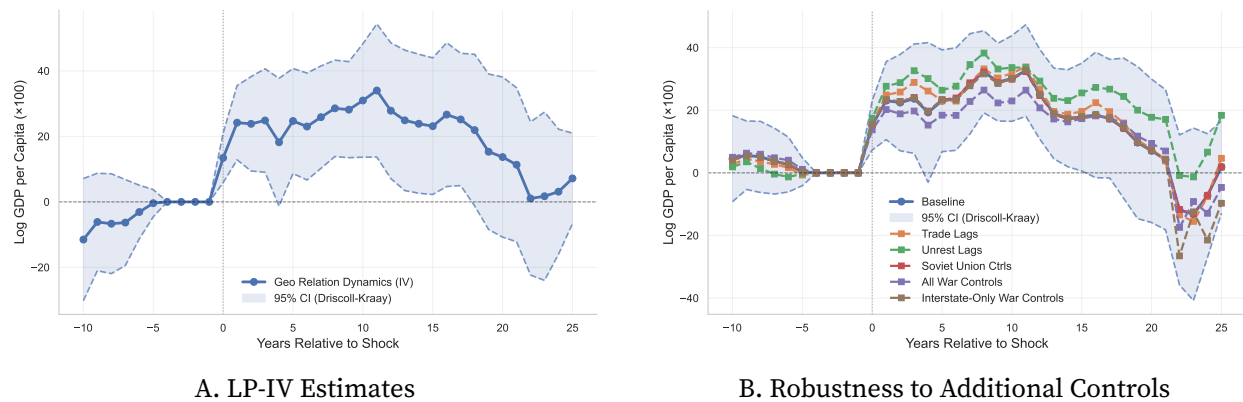


FIGURE 13. Instrumental Variables Estimates of Geopolitical Effects on Growth

Panel (a) shows the LP-IV impulse response of log GDP per capita ($\times 100$) to a unit improvement in geopolitical relations, instrumenting with non-economic verbal conflicts. The specification includes four lags of GDP, geopolitical relations, and the instrument, plus country and region-year fixed effects. Shaded areas represent 95% confidence intervals based on Driscoll-Kraay standard errors. Panel (b) demonstrates robustness across specifications with additional controls: trade openness (4 lags), domestic unrest (4 lags), Soviet transition indicators (contemporaneous), and casualty-based war exposure measures (all wars and interstate wars only) from Federle et al. (2026). All alternative specifications maintain the same fixed effects structure and lag orders for core variables. The baseline specification is shown with confidence intervals; alternative specifications are presented as point estimates only for clarity.

Results and interpretation. Figure 13 presents our LP-IV estimates.³⁵ Panel (a) shows that the IV impulse responses closely track the baseline OLS results, with GDP per capita increasing by approximately 20–30 log points within ten years. The IV confidence intervals are wider, reflecting the efficiency loss from instrumentation, but maintain statistical significance through horizon 15. The absence of significant pre-trends is consistent with the exclusion restriction, though pre-trend tests are a necessary rather than sufficient condition for instrument validity. Panel (b) confirms that this convergence is stable across specifications with additional time-varying controls.

The convergence of IV and OLS estimates is informative under the maintained assumption that the instrument isolates variation distinct from the economic components most susceptible to reverse causality. If the baseline relationship were driven primarily by feedback from economic growth to trade agreements, sanctions, or investment treaties, we would expect the IV—which excludes these components—to yield attenuated estimates. It does not. However, this convergence would also obtain if the confounders driving the OLS bias operated through non-economic channels (e.g., leadership quality affecting both diplomatic rhetoric and growth), so the similarity of estimates is consistent with a causal interpretation but not sufficient to establish it. Appendix B.10 provides detailed first-stage dynamics and demonstrates that our IV results remain stable across alternative fixed effects specifications.

³⁵The Kleibergen-Paap F -statistic for the first stage at $h = 0$ is 672. This high value reflects the mechanical correlation between non-economic verbal conflicts—a component of the aggregate index—and overall geopolitical relations. Standard critical values for weak-instrument inference are not directly applicable when the instrument is a component of the endogenous variable, so this statistic should be interpreted as confirming mechanical relevance rather than instrument strength in the conventional sense.

4. Channels and Mechanisms

This section examines how geopolitical relations influence the fundamental determinants of economic growth. We first verify robustness to alternative output measures, then trace the dynamic effects through key growth channels—stability, investment, productivity, trade, and human capital. We also revisit the democracy-growth nexus, showing that geopolitical relations mediate a substantial share of democracy’s short-run growth effects while democracy retains independent long-run benefits.

4.1. Alternative Output Measures

Our main analysis employs GDP per capita from the World Bank to maximize country and temporal coverage. To verify that our results are not sensitive to this choice, we first examine the response using real GDP per capita from the Penn World Table. Panel (a) of Figure 14 shows that the impulse response closely matches our baseline estimates, with output increasing by approximately 20 log points within 15 years. This convergence across data sources reinforces the robustness of our core finding.

4.2. Mechanisms and Growth Fundamentals

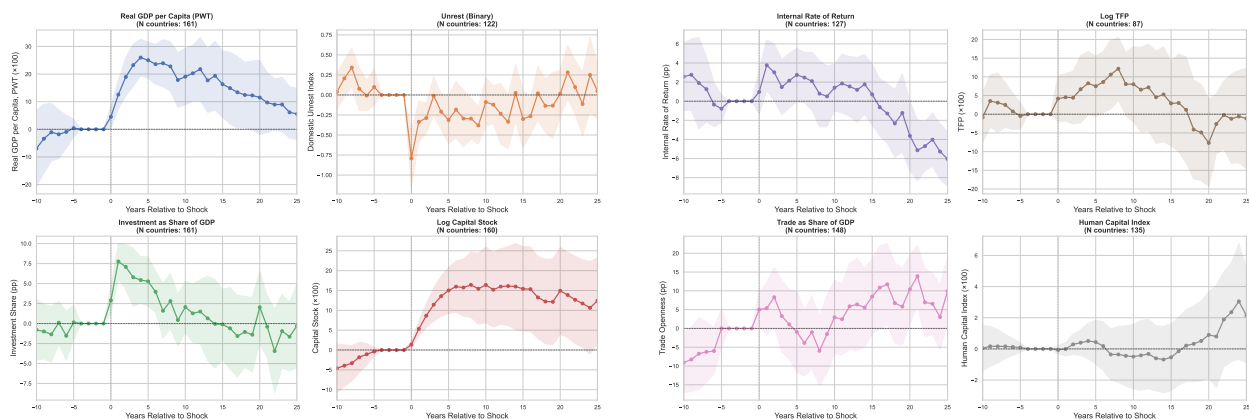
To trace the channels through which geopolitical relations may affect growth, we estimate reduced-form responses of key growth determinants to the same geopolitical shock:

$$(5) \quad m_{c,t+h} = \alpha_h^m p_{ct} + \sum_{\ell=1}^4 \beta_{\ell} y_{c,t-\ell} + \sum_{\ell=1}^4 \gamma_{\ell} p_{c,t-\ell} + \sum_{\ell=1}^4 \lambda_{\ell} m_{c,t-\ell} + \delta_c + \delta_{r(c)t} + \mu_{c,t+h}^m$$

where $m_{c,t+h}$ represents various growth correlates at horizon h . We include four lags of the outcome variable alongside our standard controls.

Figure 14 traces how geopolitical relations generate reinforcing effects across growth fundamentals. The responses exhibit distinct temporal patterns that illuminate the underlying mechanisms. Domestic political stability responds immediately and substantially—the unrest indicator drops sharply on impact before recovering toward baseline over the following years. This rapid stabilization likely reflects both enhanced regime legitimacy from international recognition and concrete external support that helps manage internal tensions, consistent with Rodrik (1998)’s evidence on openness as a buffer against political volatility.

Physical capital also responds quickly. Investment rises by 7–8 percentage points within the first few years, driving capital stock accumulation that peaks at 15–16 log points around year 5–10 and persists thereafter. The internal rate of return initially increases on impact—signaling improved investment opportunities—before declining as capital deepening proceeds, eventually turning negative after year 15, matching neoclassical predictions of diminishing returns. Total factor productivity shows sizable gains, rising to 10–12 log points around year 5–10, suggesting that



A. Output and Stability

B. Growth Fundamentals

FIGURE 14. Dynamic Effects of Geopolitical Relations on Growth Correlates

This figure displays impulse responses of various economic outcomes to a unit improvement in geopolitical relations. Panel (a) shows responses for real GDP per capita (Penn World Tables), domestic unrest, investment share, and capital stock. Panel (b) presents internal rate of return, total factor productivity, trade openness, and human capital. All specifications follow equation (5) with four lags of the dependent variable, GDP, and geopolitical relations, plus country and region-year fixed effects. The sample is restricted to countries with complete data for each variable across all horizons to ensure compositional stability. Numbers in parentheses indicate the country count for each balanced panel. Shaded areas represent 95% confidence intervals based on Driscoll-Kraay standard errors.

geopolitical alignment enables productivity improvements through technology transfer, better resource allocation, or institutional spillovers—though these gains attenuate at longer horizons.

The responses of trade and human capital unfold more gradually, reflecting inherent adjustment costs. Trade openness exhibits a volatile but generally positive trajectory, expanding to about 5 percentage points of GDP in the initial years and reaching peaks of 10–13 percentage points at longer horizons as diplomatic alignment reduces both formal and informal barriers to commerce. This pattern corroborates Frankel and Romer (1999)’s findings on the growth impact of trade integration. Human capital accumulation proceeds even more slowly, remaining near zero for most of the horizon before rising at the very end of our 25-year window—a trajectory reflecting the substantial time required for educational investments to mature and the sustained conditions for human development that international stability provides.

Appendix B.11 extends these findings across additional dimensions of economic development. Government expenditure responds positively, rising by 20–30 log points over the first decade as improved international relations enhance fiscal capacity. Primary school enrollment improves gradually by 10–15 log points over 25 years, consistent with the slow accumulation of educational gains under sustained stability. In contrast, market reforms, secondary school enrollment, employment rates, and labor share show no significant responses—consistent with geopolitical alignment operating primarily through existing economic structures, though limited statistical power may partly explain the null results. Both consumption measures—real consumption and domestic absorption per capita—closely track GDP responses, confirming that growth translates into broad household welfare improvements.

This multi-channel propagation explains why geopolitical relations exert large and persistent growth effects. Alignment simultaneously relaxes constraints across multiple margins: reducing political risk to unlock investment, facilitating technology transfer to boost productivity, expanding market access through trade, and enabling human capital investments that compound over generations. The temporal sequencing—from immediate stabilization effects through medium-term capital deepening to long-run human development—explains both the magnitude and persistence of our baseline results. Conversely, geopolitical misalignment likely triggers corresponding deterioration across the same channels—destabilizing politics, deterring investment, blocking technology diffusion, and restricting trade.³⁶

4.3. Democracy and Geopolitical Alignment

The relationship between political institutions and economic growth has been extensively studied, with Acemoglu, Johnson, and Robinson (2001) and Acemoglu et al. (2019) providing evidence for the causal effect of democratic institutions on development. Parallel literature examines how regime type shapes international relations, with democratic peace theory suggesting that democracies form more stable alliances (Maoz and Russett 1993; Leeds 2003). Democracy’s growth effects may operate partly through international economic channels such as preferential foreign aid (Alesina and Dollar 2000) and reduced sanctions (Park 2024)—both manifestations of geopolitical relations.³⁷ This section examines how democracy and geopolitical alignment jointly influence economic growth, disentangling their relative contributions.

4.3.1. Democracy and Differential Geopolitical Associations

We first examine the conditional correlation between democratization and bilateral relations with different major nations. Using the democracy measure from Acemoglu et al. (2019) (henceforth ANRR), we estimate local projections of country-specific geopolitical scores following democratization episodes:

$$(6) \quad S_{cj,t+h} = \beta_h^j D_{ct}^{\text{ANRR}} + \sum_{\ell=1}^4 \gamma_{\ell} S_{cj,t-\ell} + \delta_c + \delta_t + \varepsilon_{cjt+h}$$

where $S_{cj,t}$ represents the bilateral geopolitical score between country c and major nation j , and D_{ct}^{ANRR} is the democracy indicator. These estimates quantify the dynamic association between democratic transitions and subsequent bilateral relations, conditional on past relationship dynamics and time-invariant country characteristics.

Figure 15 reveals substantial heterogeneity in how bilateral relations with major nations evolve following democratization. Panel (a) shows that relations with Western democracies—the United

³⁶Section 3.5.2 confirms symmetric effects for both positive and negative geopolitical changes.

³⁷In our framework, sanctions constitute negative geopolitical events that lower bilateral Goldstein scores, directly contributing to our measure of geopolitical relations.

States, United Kingdom, Germany, and France—improve substantially following democratic transitions. The association is strongest and most persistent for the United States, where bilateral scores increase by approximately 0.07 within five years of democratization and remain elevated through year 15. Germany and France display similar patterns, with bilateral scores rising by 0.05–0.06 points within the first few years and persisting for over a decade. The United Kingdom shows comparable initial gains, peaking at approximately 0.06 around year 5 before gradually declining. These improvements suggest sustained alignment between countries that share similar political institutions, consistent with democratic peace theory.

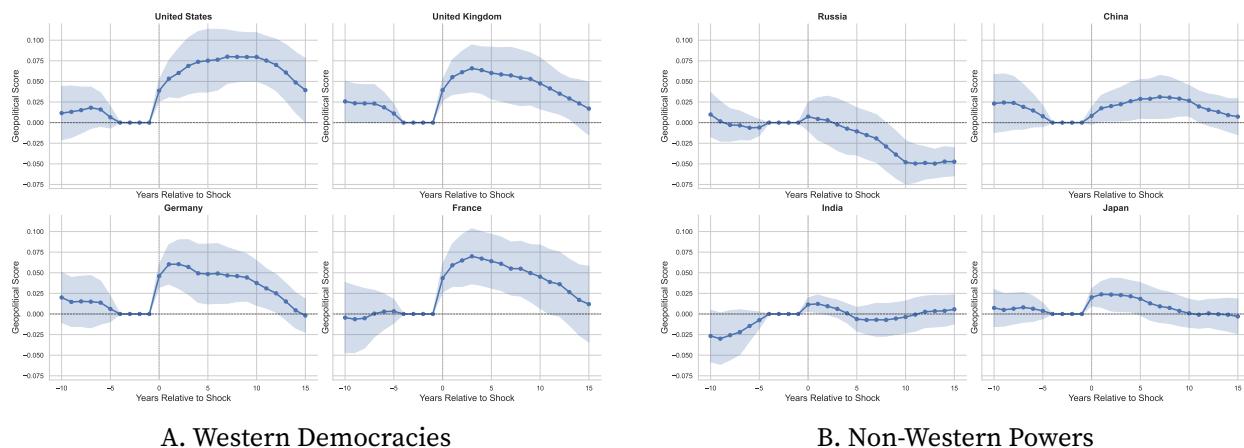


FIGURE 15. Bilateral Geopolitical Responses to Democratization

This figure shows impulse responses of bilateral geopolitical scores with major nations to a democratization shock. Panel (a) displays responses for Western democracies (USA, GBR, DEU, FRA), while panel (b) shows non-Western powers (RUS, CHN, IND, JPN). Specifications follow equation (6) with four lags of the bilateral score, country fixed effects, and year fixed effects. The horizons span -10 to $+15$ years, with negative values testing for pre-trends. Shaded areas represent 95% confidence intervals based on Driscoll-Kraay standard errors.

In contrast, panel (b) shows markedly different patterns for non-Western powers. Democratization is associated with significantly deteriorating relations with Russia, with bilateral scores declining by approximately 0.04 points over 10–15 years—consistent with geopolitical tensions that often accompany political realignment away from Russian influence. China shows a modest positive association, with bilateral scores rising by approximately 0.03 points over the medium term, possibly reflecting China’s continued economic engagement with democratizing countries. India and Japan display smaller and more transitory responses: both show modest initial improvements of approximately 0.02 points on impact, but these gains dissipate over the following decade. These differential patterns indicate that democratization is primarily associated with a reorientation of international relations toward Western democracies, while generating divergent responses among non-Western powers—deterioration with Russia, modest improvement with China, and minimal lasting effects with India and Japan.

4.3.2. Disentangling Democracy and Geopolitical Channels

To assess the relative importance of democracy versus geopolitical alignment for growth, we estimate a horse-race specification:

$$y_{c,t+h} = \alpha_h^{\text{Geo}} p_{ct} + \alpha_h^{\text{Dem}} D_{ct}^{\text{ANRR}} + \gamma'_h \mathbf{x}_{ct} + \delta_c + \delta_t + \mu_{c,t+h}$$

where both geopolitical relations (p_{ct}) and democracy enter jointly. Following ANRR, we use year fixed effects rather than region-year effects to avoid absorbing variation from regional democratization waves.³⁸

Figure 16 presents the results. Panel (a) shows that geopolitical relations maintain strong growth effects even after controlling for democracy. The joint specification (dashed line) yields an impulse response only modestly attenuated relative to the univariate model (solid line), with GDP increasing by approximately 20 log points after 7–8 years—compared to approximately 23 log points without democracy controls. This modest reduction of about 15% suggests that while some of the benefits of geopolitical relations operate through associated democratic transitions, the vast majority of geopolitical effects on growth are independent of regime type.

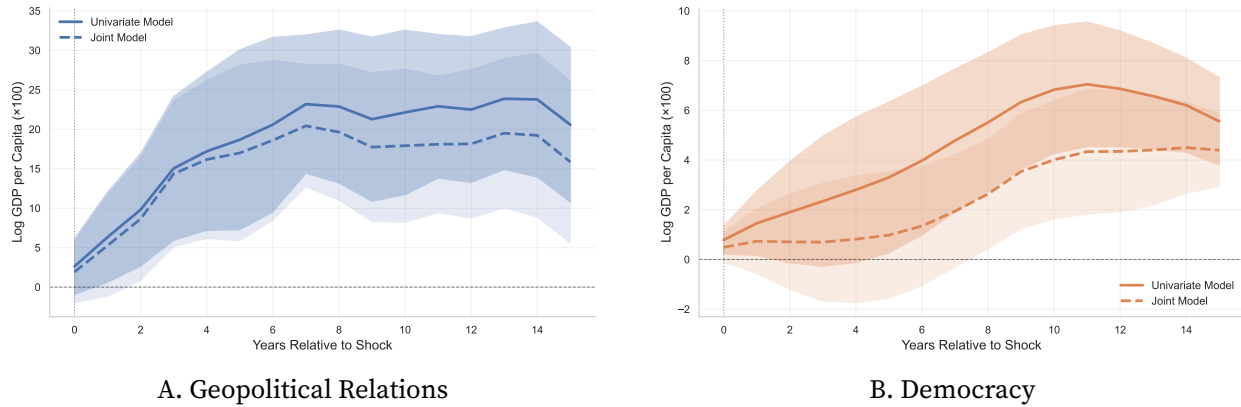


FIGURE 16. Horse-Race: Democracy versus Geopolitical Channels

This figure compares univariate and joint specifications for geopolitical relations and democracy effects on GDP. Panel (a) shows the impulse response of geopolitical relations, comparing the univariate model (solid line) with the joint specification controlling for democracy (dashed line). Panel (b) presents democracy's effects, contrasting the univariate specification without geopolitical controls (solid line) against the joint model (dashed line). Both panels use year fixed effects following ANRR, with four lags of GDP. The geopolitical specifications include four lags of geopolitical relations; the democracy univariate model excludes these lags to match ANRR's approach. Coefficients represent log point changes in GDP per capita ($\times 100$).

Panel (b) shows how geopolitical relations mediate democracy's growth effects across the entire horizon. In the univariate specification, democracy generates steadily increasing GDP gains, rising from about 1 log point on impact to a peak of roughly 7 log points around year 10–11

³⁸Regional democratization waves—such as those in Latin America (1980s), Eastern Europe (1990s), and the Arab Spring (2010s)—create strong regional correlation in democratic transitions. Region-year fixed effects would absorb this variation, potentially understating democracy's effects.

before declining modestly. When controlling for geopolitical relations, democracy's effects are substantially attenuated throughout: the joint specification shows effects rising more gradually from near zero on impact to 4–4.5 log points at year 10–12. This attenuation—reducing the peak effect by 35–40%—indicates that a substantial portion of democracy's growth benefits operates through improved international relations, particularly enhanced access to Western markets and reduced economic restrictions, consistent with Park (2024)'s findings on sanctions.

However, the majority of democracy's long-run growth impact persists even after accounting for geopolitical channels. By horizon 15, democracy continues to generate about 4.5 log points of additional GDP growth in the joint specification, representing roughly two-thirds of the univariate effect. The distinction between transitory and permanent democratization sharpens these findings: as shown in Appendix B.12, transitory democratic episodes generate growth almost exclusively through temporary geopolitical improvements. In contrast, permanent democratic transitions yield cumulative GDP gains of roughly 20 log points over 25 years, with geopolitical channels explaining only 30–40% of this effect. The remaining 60–70% operates through sustained domestic institutional improvements—enhanced property rights, political stability, reduced expropriation risk, and human capital accumulation—that materialize gradually but persist independently of international alignment.³⁹

These results provide a nuanced decomposition of how political institutions and international relations jointly determine economic development. Democracy's growth effects operate partially through the geopolitical channel, as democratic transitions trigger improved relations with Western powers, reduced sanctions, and enhanced market access. Yet the persistence of substantial democracy effects even after controlling for geopolitics confirms that domestic institutional improvements provide independent growth benefits. This decomposition helps reconcile conflicting findings in the literature—studies emphasizing international relations and those highlighting institutional quality both capture real phenomena operating through distinct channels.

Having established the dynamic effects, channels, and institutional complementarities, we now ask what these estimates imply for observed growth and income differences across countries and time periods.

5. Geopolitical Growth Accounting

The estimates in Section 3 identify the dynamic GDP response to within-country changes in geopolitical relations. In this section, we use those estimated impulse responses as an accounting tool to quantify the implied contributions of observed geopolitical changes over time and to illustrate how persistent cross-country differences in geopolitical positioning may translate into growth and income differences. The first exercise applies the estimated IRFs directly to observed within-country

³⁹Our estimate of 20 log points for permanent democratization aligns closely with Acemoglu et al. (2019)'s finding that democracy increases GDP per capita by approximately 20% in the long run. The decomposition reveals that while geopolitical improvements account for a substantial share of democracy's effects, domestic institutional channels remain quantitatively important. Appendix B.12 provides detailed impulse responses to transitory versus permanent democracy shocks using the methodology described in Appendix B.4.

changes; the second extrapolates these estimates to cross-country differences under a maintained homogeneity assumption. These cross-country results should be interpreted as illustrative rather than separately identified.

5.1. Growth Effects of Changes in Geopolitical Relations

We quantify the economic implications of the geopolitical transformations documented in Section 2.3.

For each country-decade pair, we calculate the change in geopolitical relations from the beginning to the end of the decade and apply the relevant impulse response function to obtain growth effects.⁴⁰ This approach captures both the immediate economic impact within each decade (contemporaneous effect) and the projected long-term consequences (long-run effect).

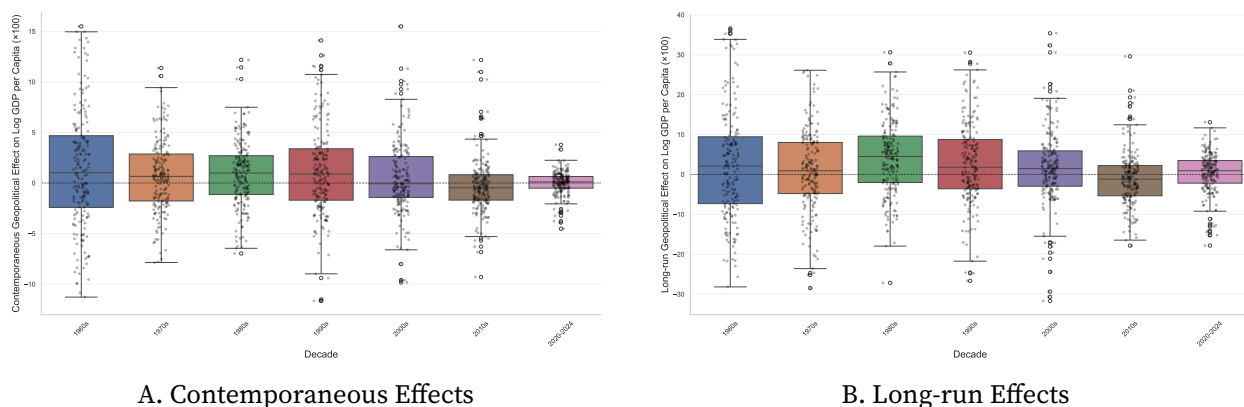


FIGURE 17. Distribution of Growth Effects from Changes in Geopolitical Relations by Decade

This figure displays the distribution of GDP effects from within-decade changes in geopolitical relations. Panel (a) shows contemporaneous effects—the cumulative GDP impact realized within each decade. Panel (b) presents long-run effects—the projected 25-year GDP impact of geopolitical changes. Each boxplot represents the distribution across countries, with boxes indicating interquartile ranges, whiskers extending to the 5th and 95th percentiles, and individual country observations shown as gray points. The horizontal dashed line marks zero effect. The sample includes all countries with complete geopolitical and GDP data for the respective decade.

Figure 17 presents the distribution of growth effects across countries for each decade from 1960 to 2024. Panel (a) displays contemporaneous effects—the cumulative GDP impact realized within each decade. The distributions reveal predominantly positive median effects from the 1970s through the 2000s, with median values ranging from 0.5% to 1%. The 1960s show a median near zero, reflecting the volatile Cold War environment. The wide dispersion, with interquartile ranges spanning 3–5 percentage points, reflects heterogeneous country experiences. Individual countries experienced contemporaneous effects ranging from –15% to +15% across the sample period. The

⁴⁰Specifically, for contemporaneous effects, we calculate $\sum_{t=0}^9 \alpha_t^{\text{transitory}} \Delta p_{c,\tau+t}$, where $\alpha_t^{\text{transitory}}$ is the transitory IRF at horizon t and $\Delta p_{c,\tau+t}$ represents the year-on-year change in geopolitical relations, and τ denotes the first year of the decade. For long-run effects, we use $\alpha_{25}^{\text{permanent}} \times (p_{c,\tau+9} - p_{c,\tau})$, where $\alpha_{25}^{\text{permanent}}$ is the permanent IRF at the 25-year horizon.

2010s mark a notable shift, with the median turning negative for the first time, while the 2020–2024 period shows a return toward zero with a notably compressed distribution.

Panel (b) presents long-run effects—the projected 25-year GDP impact of within-decade geopolitical changes. The patterns reveal important temporal variation: the 1980s exhibit the highest median long-run gains at 4–5%, followed by the 1990s and 2000s with median effects around 2%. The 1960s and 1970s show medians closer to zero, reflecting the constraints of Cold War bipolarity. Individual countries experienced effects ranging from –30% to +30% across the sample period. The 2010s stand out as the first decade with a negative median effect, consistent with the geopolitical fragmentation documented in Section 2.3. The 2020–2024 period shows a modest recovery in median effects, though with substantial dispersion reflecting the heterogeneous impacts of recent geopolitical upheavals including the Russia-Ukraine war.

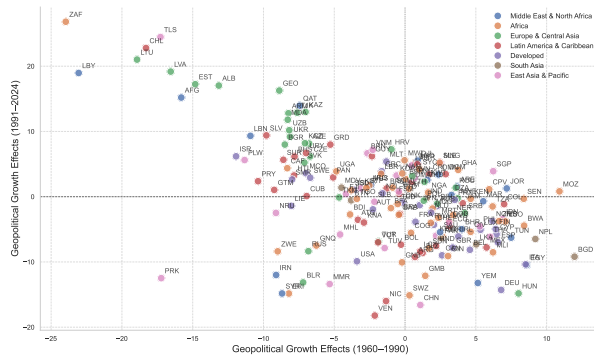
These results quantify the economic stakes of geopolitical alignment. The 1980s–2000s period of improving international relations generated median long-run GDP gains of 2–5% per decade across countries. Conversely, the deterioration observed in the 2010s threatened to reverse decades of geopolitically driven growth, with particular implications for countries navigating between competing powers. The amplification from contemporaneous to long-run impacts underscores that geopolitical choices carry lasting economic consequences.

5.2. Cross-Country Growth and Income Differences

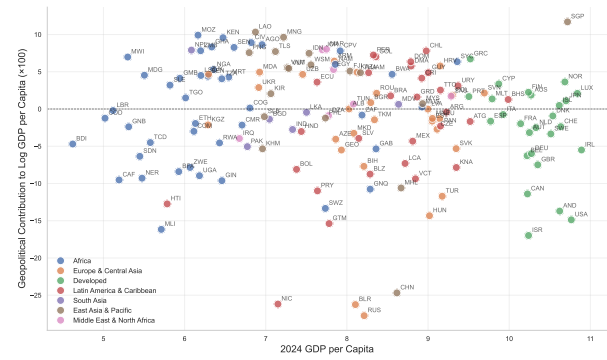
While the empirical design in Section 3 identifies responses to within-country changes in geopolitical relations, it is useful to ask what these estimates imply for broader cross-country differences. We apply the estimated impulse responses to persistent differences in countries’ geopolitical relations relative to a common benchmark—the median level of geopolitical relations across countries in each year.⁴¹

Figure 18 presents the cross-country implications. Panel (a) compares implied geopolitical growth effects during the Cold War (1960–1990) and post-Cold War (1991–2024) periods. Both periods show wide dispersion, with implied effects spanning approximately –25% to +30%, but the composition of beneficiaries and adversely affected countries shifted substantially. Countries in the upper-left quadrant—including South Africa, the Baltic states, Chile, and Georgia—experienced the largest reversals, moving from negative implied effects during the Cold War to positive effects afterward. South Africa’s trajectory is the most pronounced, shifting from approximately –28% under apartheid-era isolation to +25% following democratic transition. Conversely, Venezuela, Nicaragua, and China show deteriorating implied effects in the post-Cold War period. The wide dispersion of countries across the plot—rather than clustering along the diagonal—indicates that geopolitical fortunes are not persistent across eras.

⁴¹Formally, the geopolitical contribution to country c ’s GDP in year t is: $\Delta y_{c,t}^{\text{geo}} = \sum_{s=\max(1960,t-25)}^t \alpha_{t-s}^{\text{transitory}} \times (p_{c,s} - p_s^{\text{median}})$, where $\alpha_h^{\text{transitory}}$ is the transitory IRF at horizon h . This extrapolation assumes that the within-country dynamic response generalizes across countries—an assumption supported by the stability of our estimates across diverse specifications and country subsamples, but one that should be kept in mind when interpreting the cross-country results.



A. Implied Growth Differences



B. Implied Income Differences, 2024

FIGURE 18. Cross-Country Growth and Income Implications of Geopolitical Relations

This figure applies the estimated impulse responses from Section 3 to persistent cross-country differences in geopolitical relations relative to the sample median. Panel (a) plots country-specific implied growth effects during 1960–1990 (Cold War) versus 1991–2024 (post-Cold War), with colors indicating regions; countries above the diagonal experienced more favorable geopolitical effects in the recent period. Panel (b) shows the implied percentage contribution of geopolitical relations to GDP in 2024, plotted against GDP per capita (log scale). The horizontal dashed line marks zero contribution.

Panel (b) translates these cumulative growth effects into implied income differences for 2024.⁴² Singapore emerges as the largest beneficiary, with an implied contribution of approximately 12% of GDP from favorable geopolitical positioning. Other notable gainers include small open economies with broad international engagement: Laos, Mongolia, Mozambique, and Croatia show positive implied contributions of 7–10%. The Baltic states illustrate the transformation most clearly: Estonia, Latvia, and Lithuania moved from negative contributions of 10–12% in 1990 to positive territory in 2024, reflecting successful integration into Western institutions. At the other extreme, Russia’s implied contribution collapsed to approximately –27% following the Ukraine invasion, with Belarus at –25% and China at –23%. Several major developed economies also show negative implied contributions: the United States at approximately –15%, Israel at –17%, and Canada at –11%—a reversal from 1990 when most developed economies showed positive or neutral effects.

These implied contributions show no systematic correlation with income levels—rich countries are as likely as poor countries to experience geopolitical headwinds. The magnitude of implied effects, ranging from approximately –27% to +12% of GDP, suggests that geopolitical positioning is a quantitatively important dimension of cross-country income variation.⁴³

⁴²The geopolitical contribution is calculated as the percentage difference between actual GDP and the counterfactual GDP under median geopolitical relations: $100 \times [\exp(\Delta y_{c,t}^{\text{geo}}/100) - 1]$.

⁴³For comparison, Acemoglu et al. (2019) estimate that democratization increases GDP per capita by approximately 20% in the long run. Our within-country estimates imply effects of similar magnitude: a one-standard-deviation permanent improvement generates 10 log points over 25 years.

6. Additional Robustness Results

6.1. Dynamic Panel Estimates

We complement the local projection estimates with a dynamic panel model that exploits autoregressive structure for more precise long-run inference (Olea et al. 2024):

$$y_{ct} = \alpha p_{ct} + \sum_{\ell=1}^J \beta_{\ell} y_{c,t-\ell} + \sum_{\ell=1}^J \gamma_{\ell} p_{c,t-\ell} + \delta_c + \delta_{r(c)t} + \mu_{ct}$$

where y_{ct} denotes log GDP per capita, p_{ct} represents geopolitical relations, and δ_c and $\delta_{r(c)t}$ capture country and region-year fixed effects. We set $J = 4$ to match our baseline specification.

Under Assumption 1, the impulse response function is:

$$\phi_0 = \alpha, \quad \phi_k = \sum_{j=1}^{\min(k,J)} \beta_j \phi_{k-j} + \gamma_k \cdot \mathbf{1}[k \leq J] \text{ for } k \geq 1, \quad \phi_{\infty} = \frac{\alpha + \sum_{\ell=1}^J \gamma_{\ell}}{1 - \sum_{\ell=1}^J \beta_{\ell}}$$

This formulation yields identical population impulse responses to our local projection approach (Plagborg-Møller and Wolf 2021) while offering improved small-sample precision through parametric structure.

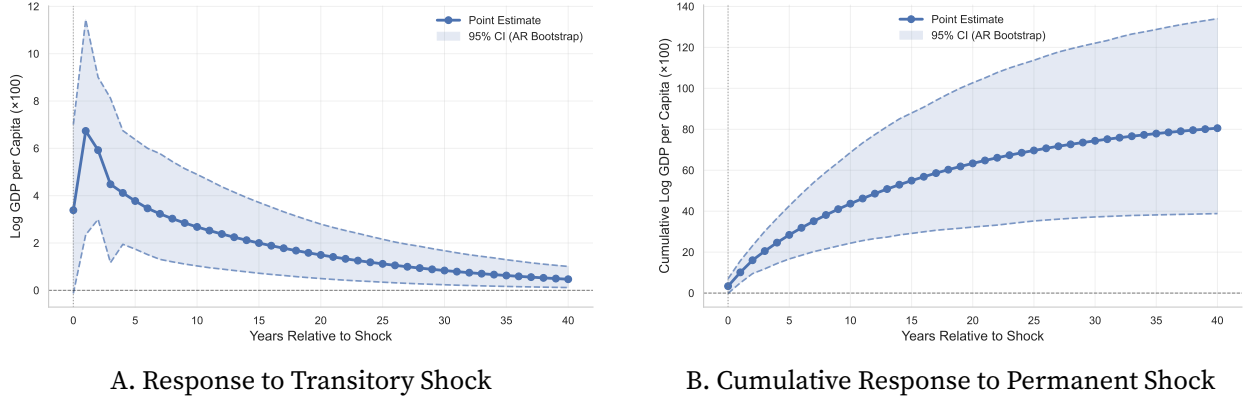


FIGURE 19. Dynamic Panel Estimates: GDP Responses to Transitory and Permanent Geopolitical Shocks

Panel (a) shows the impulse response of log GDP per capita ($\times 100$) to a purely transitory unit shock in geopolitical relations. Panel (b) displays the cumulative response to a permanent unit shock. Specifications include four lags of both variables, country fixed effects, and region-year fixed effects. Shaded areas represent 95% confidence intervals from 1,000 bootstrap iterations using country-block resampling.

Figure 19 presents the dynamic panel estimates. Panel (a) shows the response to a purely transitory shock: the initial impact of approximately 3.5 log points rises to a peak of nearly 7 log points at year 1 before declining monotonically thereafter. The response remains positive and statistically significant for approximately 25–30 years, demonstrating that even temporary

diplomatic improvements generate long-lasting economic benefits. The gradual decay—reaching approximately 2 log points by year 15 and 1 log point by year 25—reflects the persistence of growth effects as improved geopolitical relations trigger sustained investment and productivity gains.

Panel (b) shows cumulative gains from permanent improvements in geopolitical relations. GDP per capita rises steadily, reaching approximately 25 log points after 5 years, 55 log points after 15 years, and stabilizing around 80 log points by year 40.⁴⁴ The steady-state multiplier of approximately 80 log points implies that a one-standard-deviation improvement in geopolitical relations (0.143 units) generates a long-run GDP gain of approximately 11.4 log points.

6.2. Alternative Measures of Geopolitical Relations

We next test the event-based measure against alternative specifications and existing approaches: unsmoothed geopolitical events (verifying that the smoothing procedure does not drive results), UN General Assembly voting patterns (the predominant alternative), and horse-race specifications against economic sanctions and formal military alliances (the most direct bilateral proxies for coercive and cooperative relations, respectively).

6.2.1. Impulse Responses to Average Event Scores

Our main analysis employs a dynamic geopolitical score that smooths volatile bilateral events to capture persistent trends. We examine robustness by estimating impulse responses to unsmoothed average event scores $\tilde{S}_{ct}$, which correspond to setting the depreciation rate $\delta = 1$ in equation (1).⁴⁵

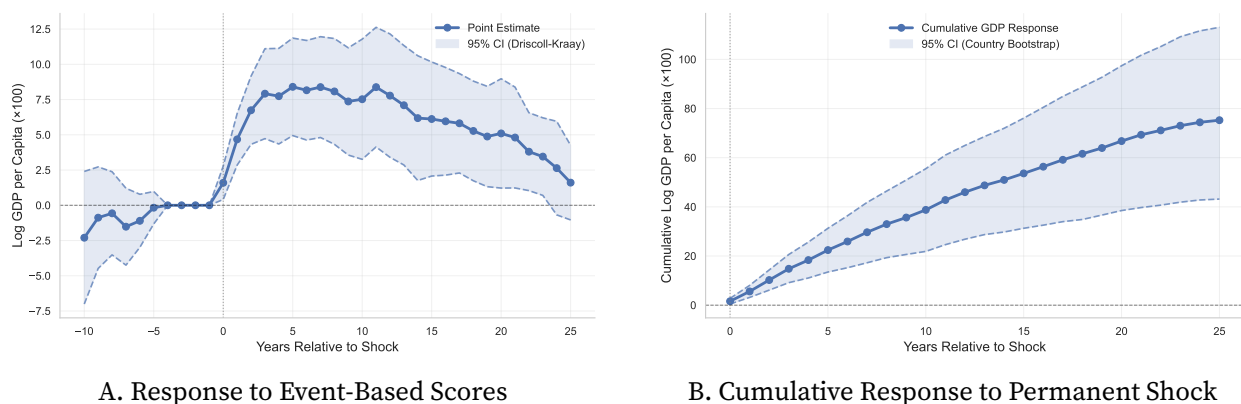


FIGURE 20. GDP Responses to Geopolitical Events

Panel (a) shows the impulse response of GDP per capita to a unit shock in event-based geopolitical scores. Panel (b) displays the cumulative response to a permanent unit shock in event scores. Shaded areas represent 95% confidence intervals.

⁴⁴The parametric structure delivers narrower confidence intervals compared to local projections. At the 25-year horizon, the 95% confidence interval spans approximately 30 to 110 log points, improving precision about long-run effects.

⁴⁵We estimate $y_{c,t+h} = \alpha_h^{\text{event}} \tilde{S}_{ct} + \gamma_h' \mathbf{x}_{ct} + \mu_{c,t+h}$, where $\tilde{S}_{ct} = \sum_{j \in \mathcal{N}} \tilde{S}_{ij,t} \times \text{GDP share}_{jt}$ represents the GDP-weighted average of bilateral event scores in year t without smoothing.

Figure 20 shows that while GDP responds positively to average event scores, the magnitude is smaller than our baseline smoothed measure—peaking at approximately 8–9 log points around years 5–10 versus 20–25 log points in our baseline. This attenuation reflects the rapid transitory nature of individual events.⁴⁶ However, panel (b) shows that a permanent change in event flows yields a similar cumulative response of approximately 75 log points after 25 years. This convergence supports the interpretation that our geopolitical score captures economically meaningful variation: permanent changes in event flows mechanically generate permanent changes in the stock of geopolitical relations, yielding equivalent long-run effects, and our smoothing procedure retains the fundamental relationship between geopolitics and growth. Appendix B.13 provides additional discussion.

6.2.2. Measuring Geopolitical Relations Using UNGA Votes

Our event-based measure covers bilateral geopolitical relations with universal scope. As an alternative approach, we examine whether voting patterns in the United Nations General Assembly—the predominant measure in existing literature (Signorino and Ritter 1999)—can generate similar results. We implement our empirical analysis using the negative Ideal Point Distance (IPD) from Bailey, Strezhnev, and Voeten (2017), which measures alignment based on UNGA voting behavior.

Figure B13 in Appendix B.14 exposes fundamental limitations of UNGA-based measures for capturing growth-relevant geopolitical dynamics. When we maintain our baseline specification with country fixed effects—important for addressing time-invariant confounders (Acemoglu et al. 2019; Kremer, Willis, and You 2022)—neither the aggregate GDP-weighted IPD nor bilateral alignment with the United States yields statistically significant effects on economic growth. The impulse responses hover near zero throughout the 25-year horizon, with confidence intervals consistently spanning zero. This null result stands in stark contrast to our event-based measure, which produces robust and persistent growth effects.

The failure of UNGA measures likely reflects their focus on multilateral positioning rather than bilateral dynamics, combined with strategic voting behavior and temporal clustering in annual sessions. Appendix B.14 develops these limitations in detail.

6.2.3. Sanctions and Formal Alliances

Sanctions and formal alliances provide informative bilateral benchmarks because they represent salient but narrow margins of interstate relations. Sanctions reflect explicit episodes of bilateral coercion; formal alliances reflect institutionalized security cooperation. Both are discrete and partial relative to a continuous measure of overall bilateral geopolitical relations. A natural horse-race test asks whether these standard bilateral proxies retain independent explanatory power once our broader event-based measure is included.

⁴⁶Appendix B.13 shows that event scores exhibit rapid mean reversion, with approximately 65% of the initial impact dissipating within one year.

Using the Global Sanctions Database (Felbermayr et al. 2020; Yalcin et al. 2025), we construct a country-level sanctions exposure measure: $p_{ct}^{\text{Sanction}} = \sum_{j \in \mathcal{N}} \mathbb{1}_{cjt} \times \text{GDP share}_{jt}$, where $\mathbb{1}_{cjt}$ indicates whether a major nation j imposed sanctions on country c in year t . Figure B14 in Appendix B.15 presents the results. When we control for sanctions, the geopolitical relations impulse response is essentially unchanged, peaking at approximately 26–27 log points around year 7–8. In contrast, controlling for geopolitical relations attenuates the sanctions effect—the trough impact shrinks from approximately –10 to –7 log points, a reduction of roughly 30% at the peak—and the sanctions coefficient in the joint specification approaches zero by year 10, indicating that the broader geopolitical measure fully absorbs the longer-run sanctions signal. Sanctions represent one manifestation of deteriorating bilateral relations: they typically follow diplomatic protests, recalled ambassadors, and suspended agreements, all of which our event-based measure already incorporates.⁴⁷

Formal alliances display a parallel pattern on the cooperative side. Using the Alliance Treaty Obligations and Provisions dataset (Leeds et al. 2002), we construct an analogous GDP-weighted alliance exposure measure: $p_{ct}^{\text{Alliance}} = \sum_{j \in \mathcal{N}} \mathbb{1}_{cjt}^A \times \text{GDP share}_{jt}$, where $\mathbb{1}_{cjt}^A$ indicates whether country c maintains an active formal alliance with major nation j in year t . Figure B15 in Appendix B.16 shows that alliance exposure predicts higher GDP per capita in univariate specifications. Once the bilateral geopolitical score is included, the alliance coefficient falls by roughly one-half, while the coefficient on geopolitical relations remains stable. The timing pattern is informative: the short-run alliance coefficient largely disappears once geopolitical relations are controlled for, consistent with the event-based measure absorbing the diplomatic cooperation surrounding alliance formation, while a modest positive residual persists at longer horizons, suggesting that durable treaty commitments may contain institutional information beyond what annual event flow fully summarizes.

Taken together, the sanctions and alliance horse races point in the same direction. Standard bilateral proxies—whether measuring hostility or cooperation—contain meaningful information when considered alone, but much of that information is absorbed by the broader geopolitical score in joint specifications. The contrasting timing patterns reinforce this interpretation: sanctions effects are fully subsumed at longer horizons, while alliances retain a modest residual, consistent with their distinct roles as episodic policy instruments versus durable institutional commitments. In both cases, the event-based measure remains stable, confirming that it tracks the broader continuous evolution of bilateral geopolitical dynamics rather than any single institutional margin.

7. Conclusion

This paper has shown that geopolitical relations constitute a first-order and persistent determinant of economic growth. The effects we document are large, persistent, and robust across identification strategies—including placebo falsification tests, instrumental variables, and extensive controls for war exposure and other confounders—as well as alternative estimation methods and measures of

⁴⁷The timing difference between measures reinforces this interpretation: our event-based approach records sanctions when announced or lifted—capturing the geopolitical signal—while the sanctions database tracks their continuous enforcement.

international relations.

Our results carry a broader message for the growth literature: the traditional separation between “domestic” and “international” determinants of growth may obscure important interactions. Geopolitical relations shape growth not through a single channel but by simultaneously relaxing political, technological, and financial constraints on development. Countries that improve their geopolitical relations experience immediate gains in domestic stability and investment, followed by gradual expansions in productivity, trade, and human capital—a reinforcing dynamic that explains why even transitory diplomatic improvements generate lasting economic benefits. The democracy–geopolitics decomposition complements the institutions literature: democracy’s short-run growth effect operates in part through improved geopolitical relations, while its long-run benefits persist independently through strengthened property rights and reduced expropriation risk.

These findings are subject to important caveats. Our identification strategy exploits within-country variation and therefore speaks to the effects of changes in geopolitical relations, not the level effects of sustained cooperation or isolation. The event-based measure, while more comprehensive than alternatives based on UN voting or binary indicators, relies on LLM-compiled data that may introduce measurement error. While our IV estimates lend support to a causal interpretation, the exclusion restriction—that non-economic verbal conflicts affect growth only through bilateral relations—is plausible but not beyond dispute.

The policy implications extend beyond the usual prescription of “more openness.” Our growth accounting reveals that the gains from international integration are neither automatic nor permanent: the post-2010 reversal toward fragmentation implies measurable costs—with median geopolitical growth contributions turning negative for the first time in six decades of data. Yet the finding that relations with ideologically different major powers yield comparable economic benefits suggests that countries need not choose sides to prosper. The “connector” strategy—maintaining productive relations across geopolitical divides—offers a viable path, though one that great-power competition increasingly pressures.

As the post-Cold War order gives way to renewed strategic competition, the economic stakes of geopolitical choices will continue to grow. Our estimates provide a framework for quantifying those stakes—from the costs of bilateral deterioration to the benefits of diversified engagement.

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Appendix A. Event-based Measure of Geopolitical Relations

This section describes additional results related to the event-based measure of geopolitical relations involving all 193 United Nations member states and 24 major nations.

A.1. Major Nations

Our event-based measure of geopolitical relations is designed to be flexible and applicable to any country over time. In this analysis, we examine geopolitical relations involving all 193 countries, with a particular focus on 24 major nations selected for their significant geopolitical, military, and, most critically, economic influence. These nations were chosen based on their substantial global economic impact, which underpins their relevance to our study of how geopolitical dynamics influence economic growth.

Figure A1 presents the time series of aggregate and individual GDP shares for these 24 major nations.⁴⁸ Panel (a) illustrates the combined GDP share of these nations relative to the global total, highlighting their collective economic dominance over time. Panel (b) displays the individual GDP share for each nation, revealing variations in economic influence across countries and over the study period.

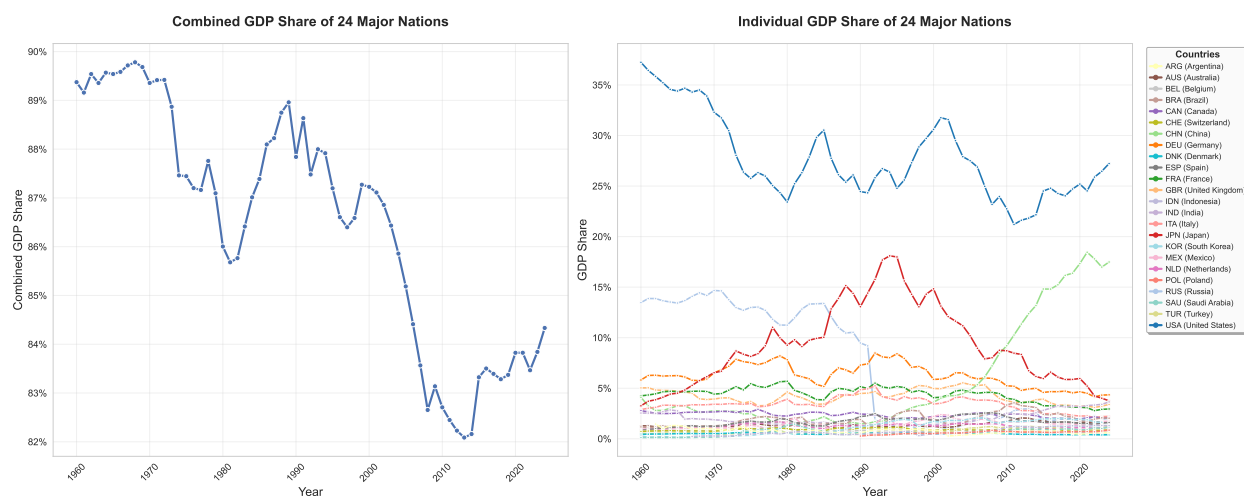


FIGURE A1. GDP Shares of Major Nations. Panel (a) shows the combined GDP share of 24 major nations (Argentina, Australia, Belgium, Brazil, Canada, Switzerland, People's Republic of China, Germany, Denmark, Spain, France, United Kingdom, Indonesia, India, Italy, Japan, Republic of Korea, Mexico, Netherlands, Poland, Russian Federation, Saudi Arabia, Turkey, and United States) relative to global GDP over time. Panel (b) presents the individual GDP share for each nation, illustrating cross-country variations and temporal trends. Data are sourced from the World Bank, with Soviet Union GDP calculated using relative GDP to the United States from the Maddison Project (Bolt and Van Zanden 2025).

⁴⁸GDP data are nominal GDP in current USD and sourced from the World Bank. For the Soviet Union, GDP is calculated using relative GDP to the United States, obtained from the Maddison Project.

A.2. Geopolitical Event Data: Methodology and Descriptive Statistics

This section provides technical details on our LLM-based methodology for compiling bilateral geopolitical events and presents comprehensive descriptive statistics of the resulting dataset spanning 1960–2024. We first describe the systematic procedure for event identification and classification, then analyze temporal patterns across 373,020 events to validate our framework’s ability to capture major shifts in international relations. Complete prompt specifications are provided in Appendix C.

A.2.1. Event Compilation and Analysis

This subsection provides additional detail on the event compilation and analysis procedure described in Section 2.1. The complete prompt structure is in Appendix C.

Entity Verification and Historical Accuracy. Our framework begins by verifying the political entities corresponding to each country pair for the target year. This step is crucial for maintaining historical accuracy, as it accounts for state succession events such as the dissolution of the Soviet Union and emergence of the Russian Federation on December 26, 1991. The LLM references authoritative sources including the Correlates of War State System Membership dataset when available. For years when specified entities did not exist, the framework identifies the primary political entities controlling the relevant territories or populations and applies them consistently throughout the analysis.

Systematic Event Identification. The LLM conducts comprehensive searches across multiple dimensions of bilateral relations to identify major geopolitical events that significantly affect or strongly indicate the state of the bilateral relationship. Table A1 summarizes the six dimensions and 22 subcategories that define the scope of geopolitical events in our analysis, with representative examples for each. Events are included only when verified through multiple credible sources to ensure authenticity. The framework prioritizes events with demonstrable significant effects on the relationship’s trajectory, with particular attention to political actions carried out through economic means and major developments concerning trade or security agreements. Detailed criteria for major geopolitical event identification are provided in Appendix C.2.

CAMEO Classification and Goldstein Scoring. Each identified event undergoes systematic classification using the Conflict and Mediation Event Observations (CAMEO) framework, which categorizes international political actions along two dimensions: cooperation versus conflict and verbal versus material. This creates four quadrant classes with hierarchical coding—root codes (two-digit) for general categories and event codes (three-digit) for specific actions. Following CAMEO classification, we assign Goldstein Scale scores ranging from -10.0 (maximum conflict) to $+10.0$ (maximum cooperation) based on the event code’s typical intensity, with contextual adjustments for the bilateral relationship’s historical context.

TABLE A1. Categories of Bilateral Geopolitical Events

Subcategory	Representative Events	Subcategory	Representative Events
<i>A. Economic Relations</i>		<i>D. Legal, Territorial & Movement</i>	
A1. Trade Policy	Tariff imposition/removal; non-tariff barriers; FTA negotiations, signing, ratification, withdrawal	D1. Legal Actions	ICJ/WTO disputes; international arbitration; extradition; law enforcement cooperation
A2. Financial Relations	Asset freezes; SWIFT exclusions; capital market denial; currency swaps; FDI restrictions; investment treaties	D2. Territorial & Maritime	EEZ/continental shelf disputes; freedom of navigation operations; border demarcation; sovereignty claims
A3. Economic Coercion	Trade embargoes; sectoral sanctions; Entity List designations; export controls; foreign aid; debt relief	D3. Movement of People	Visa regime changes; travel bans; guest worker programs; refugee/asylum policies
A4. Strategic Sectors	Energy supply agreements; pipeline projects; nuclear cooperation; 5G bans; rare earth controls	<i>E. Multilateral & Global Governance</i>	
A5. Integration	BRI/B3W infrastructure; port concessions; friend-shoring; supply chain arrangements; customs unions	E1. Int'l Organizations	UNSC confrontations; GA coalition building; regional bloc membership changes
A6. Other Economic	Regulatory harassment; boycott campaigns; government procurement restrictions; gray zone pressures	E2. Global Issues	Climate cooperation; pandemic coordination; vaccine diplomacy; human rights
<i>B. Diplomatic & Political Relations</i>		<i>F. Other Significant Events</i>	
B1. Formal Diplomacy	Embassy/consulate openings and closures; ambassador recalls; staff expulsions; démarches	F1. Historical & Symbolic	Apologies for historical wrongs; memorial visits; monument disputes
B2. High-Level Interactions	Presidential visits; bilateral summits; summit boycotts; ministerial meetings; strategic dialogues	F2. Humanitarian	Disaster aid offers/rejections; joint rescue operations; evacuation cooperation
B3. Public Diplomacy	Policy speeches; parliamentary resolutions; propaganda campaigns	F3. Sports & Events	Olympic boycotts; joint hosting; World Expo participation
B4. Cultural & Educational	Cultural agreements/boycotts; scholarship programs; university partnerships; student visa policies	F4. Technology & Space	Joint space missions; tech theft accusations; research terminations
<i>C. Security & Defense</i>		F5. Environment	Transboundary river/pollution disputes; conservation conflicts
C1. Military Cooperation	Alliances; defense pacts; base arrangements; arms sales/embargoes; joint exercises; military aid	F6. Communications	Journalist expulsions; broadcasting restrictions; information warfare
C2. Security Incidents	Border skirmishes; airspace/maritime violations; naval encounters; hybrid warfare	F7. Other	Emerging forms of bilateral interaction
C3. Intelligence & Cyber	Espionage revelations; intel officer expulsions; intel-sharing pacts; state-sponsored cyber attacks		

Additionally, we classify each event's economic content into five categories: Tariffs, Economic Sanctions, Trade Agreements and Treaties, Other Economic Policies, or Not an Economic Event. This classification enables targeted analysis of how different forms of economic diplomacy affect bilateral relations and economic outcomes. The detailed CAMEO codebook and Goldstein scoring guidelines are provided in Appendix C.3.

TABLE A2. U.S.-Soviet Union Bilateral Geopolitical Events in 1972: LLM Analysis Results

Event Name	Event Description	CAMEO Class.	Econ. Type	Goldstein Score
Declaration of Basic Principles	Joint declaration of twelve principles guiding bilateral relations, including peaceful coexistence and renouncing unilateral advantage	Verbal Coop. (01-019)	Not econ.	+5.0
US-Soviet Grain Deal	Soviet purchase of 19 million metric tons of American grain, including nearly a quarter of the US wheat harvest	Material Coop. (06-061)	Trade Agree.	+6.0
Moscow Summit	President Nixon's historic visit to Moscow, the first official US presidential visit to the USSR	Verbal Coop. (04-042)	Not econ.	+6.0
Environmental Protection Agreement	Bilateral agreement establishing cooperation on 11 areas including air and water pollution and nature preservation	Material Coop. (05-057)	Not econ.	+6.5
Biological Weapons Convention	US and USSR signed the BWC prohibiting development and stockpiling of biological weapons	Material Coop. (05-057)	Not econ.	+7.0
US-Soviet Trade Agreement	Comprehensive agreement providing reciprocal MFN tariff treatment, official trade offices, and government credits	Material Coop. (05-057)	Trade Agree.	+7.0
Apollo-Soyuz Agreement	Agreement for joint US-Soviet space mission, leading to the 1975 orbital rendezvous of American astronauts and Soviet cosmonauts	Material Coop. (05-057)	Not econ.	+7.5
Incidents at Sea Agreement	Protocols to prevent accidents between US and Soviet navies, including rules against collisions and simulated attacks	Material Coop. (05-057)	Not econ.	+8.0
SALT I Accords	ABM Treaty limiting defensive systems and Interim Agreement freezing ICBM/SLBM launchers	Material Coop. (05-057)	Not econ.	+9.0

Tables 1 and A2 illustrate the range of the methodology: 2022 U.S.-Russia events are dominated by conflict scores (−4.5 to −8.0), while the 1972 détente period features exclusively cooperative events (+5.0 to +9.0).

A.2.2. Statistics of Geopolitical Events

Table A3 and Figure A2 provide detailed statistics on the 373,020 events compiled across six decades.

TABLE A3. Summary Statistics of Geopolitical Events by Decade, 1960–2024

	1960s	1970s	1980s	1990s	2000s	2010s	2020s	Total
CAMEO Event Classification								
Verbal Cooperation	13,276	16,532	17,433	23,001	34,330	51,142	30,140	185,854
Material Cooperation	8,734	10,340	12,224	17,537	24,902	31,737	18,406	123,880
Verbal Conflict	4,700	4,639	5,980	4,805	6,468	8,571	5,226	40,389
Material Conflict	2,579	2,714	3,410	3,183	3,179	4,725	3,107	22,897
Goldstein Scale Statistics								
Mean	2.86	3.28	2.92	3.98	4.09	3.88	3.73	3.67
Std. Dev.	4.87	4.62	4.74	4.27	3.87	3.78	3.88	4.18
Median	5.00	5.00	4.50	6.00	5.00	4.50	4.50	5.00
Event Category Classification								
Economic Relations (A)	6,296	9,045	10,288	14,067	20,365	26,366	14,847	101,274
Diplomatic & Political (B)	11,559	12,516	14,082	16,058	25,591	41,086	23,529	144,421
Security & Defense (C)	3,635	3,205	3,879	4,361	5,507	8,343	5,175	34,105
Legal & Territorial (D)	1,588	1,889	1,614	2,226	3,565	4,736	2,443	18,061
Multilateral & Governance (E)	5,239	6,212	6,624	9,765	10,780	11,268	7,459	57,347
Other Significant Events (F)	974	1,362	2,561	2,052	3,073	4,377	3,426	17,825
Summary								
Total Events	29,289	34,225	39,047	48,526	68,879	96,175	56,879	373,020

CAMEO classifications follow the Conflict and Mediation Event Observations framework. Goldstein Scale ranges from –10 (most conflictual) to +10 (most cooperative). Event categories follow the six-category classification embedded in the LLM prompt: (A) Economic Relations, (B) Diplomatic & Political Relations, (C) Security & Defense, (D) Legal, Territorial & Movement, (E) Multilateral & Global Governance, and (F) Other Significant Events. See Table A1 for subcategory details. All figures represent event counts except Goldstein Scale statistics.

Table A3 and Figure A2 reveal three patterns. First, international interactions are overwhelmingly cooperative: 83% of events fall in cooperative categories (verbal and material), with conflict events comprising only 17%. Second, total event volume more than tripled from 29,289 in the 1960s to 96,175 in the 2010s, driven primarily by expanding cooperative interactions; this growth partly reflects greater availability of digitized source material for more recent decades in LLM training corpora. Third, mean Goldstein scores rose from Cold War levels (2.86–3.28) to a post-Cold War peak (3.98–4.09) before declining modestly in the 2010s–2020s (3.73–3.88), consistent with the distributional dynamics in Section 2.3. Across the six event categories, diplomatic and political interactions (category B) account for 39% of all events, followed by economic relations (A, 27%) and multilateral governance (E, 15%). Security and defense events (C) constitute 9%, while legal/territorial (D) and other events (F) account for 5% each. Appendix C provides additional visualizations of these temporal patterns.



FIGURE A2. Geopolitical Events Summary (1960–2024)

A.3. Additional Case Validation for Geopolitical Scores

This section provides additional validation of our dynamic geopolitical scores beyond the cases presented in the main text (Figures 2 and 3). We examine eight bilateral relationships spanning different regions and relationship types—great power rivalries, ideological revolutions, war-to-normalization transitions, sectarian conflicts, and diplomatic reversals. In each panel, the blue solid line shows the dynamic geopolitical score while the gray dashed line shows the raw (unsmoothed) yearly score. Shaded regions highlight key historical periods and boxed labels mark major bilateral events.

Figure A3 presents four major-power relationships. The China–U.S. trajectory (panel a) traces five distinct phases: Korean War-era hostility in the 1960s with scores near -0.4 , Nixon’s 1972 opening that lifts scores above $+0.4$, a post-Tiananmen deterioration around 1989, a period of WTO-era cooperation in the early 2000s, and the precipitous trade-war decline after 2018 that returns scores to levels comparable to the 1960s. The China–Russia case (panel b) traces one of the most dramatic

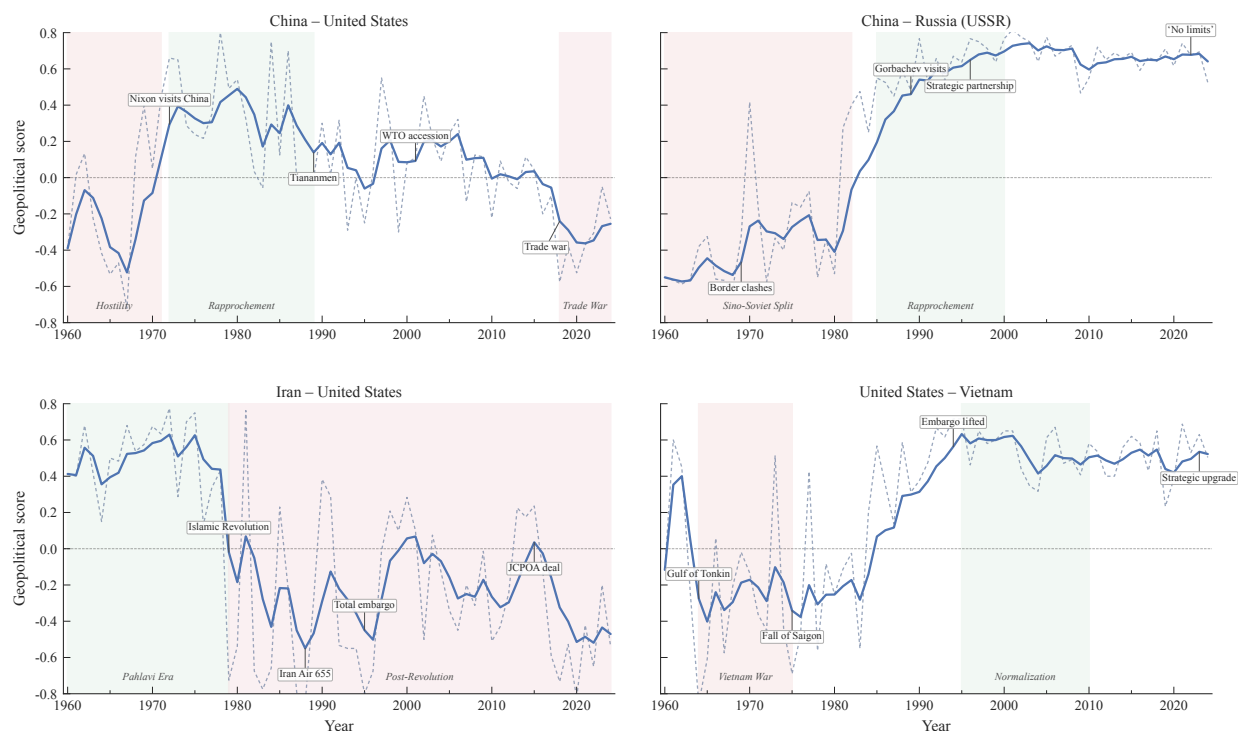


FIGURE A3. Geopolitical Scores: Major Power Dynamics

Dynamic geopolitical scores for four major-power bilateral relationships, 1960–2024. Panel (a): China–United States, from Cold War hostility through Nixon’s rapprochement and WTO-era cooperation to the recent trade war. Panel (b): China–Russia, capturing the Sino-Soviet split, border clashes, Gorbachev-era rapprochement, and the contemporary “no limits” partnership. Panel (c): Iran–United States, showing the sharp discontinuity at the 1979 Islamic Revolution and sustained hostility interrupted briefly by the JCPOA. Panel (d): United States–Vietnam, tracing the arc from war through normalization to the 2023 strategic upgrade.

arcs in Cold War geopolitics—from alliance through the Sino-Soviet split and 1969 border clashes (scores near -0.5) to Gorbachev’s 1989 visit, the strategic partnership of the late 1990s, and the contemporary “no limits” partnership with scores exceeding $+0.7$. Iran–U.S. relations (panel c) exhibit a clean regime-change discontinuity at the 1979 Islamic Revolution, with scores collapsing from approximately $+0.5$ during the Pahlavi era to sustained hostility below -0.4 , punctuated by the Iran Air 655 incident (1988), the total embargo (1995), and the brief JCPOA-related recovery around 2015. The U.S.–Vietnam case (panel d) demonstrates a complete war-to-normalization transition: scores fall from moderate levels in the early 1960s through the Gulf of Tonkin escalation and the Fall of Saigon to deep hostility, before recovering steadily after the 1994 embargo removal and reaching cooperative levels with the 2023 strategic upgrade.

Figure A4 presents four regional relationships. The Iran–Saudi Arabia case (panel a) tracks the Twin Pillar era of cooperation under the Shah (scores near $+0.7$), the sharp post-revolutionary deterioration including the 1987 Mecca clash (scores dropping to approximately -0.3), the Khatami-era thaw in the late 1990s that temporarily restored scores above $+0.5$, and the subsequent deterioration followed by the 2023 China-brokered rapprochement. China–Japan relations (panel b) show the

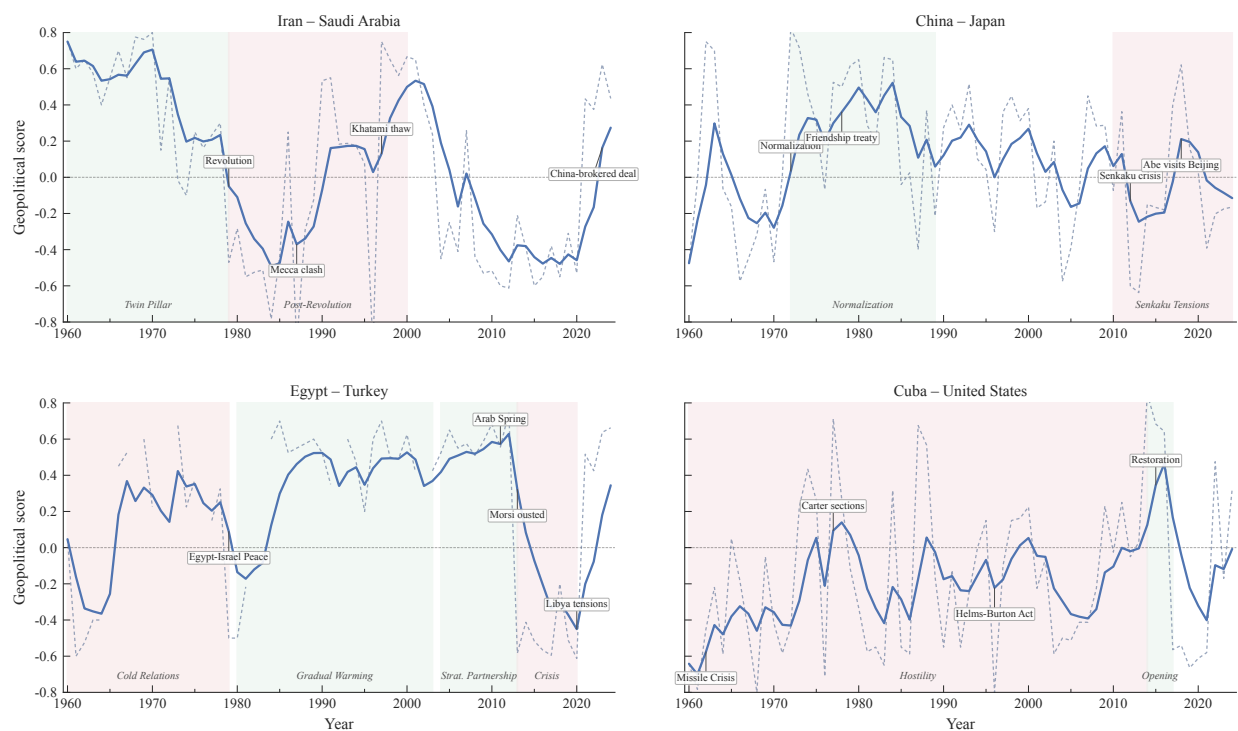


FIGURE A4. Geopolitical Scores: Regional Rivalries and Bilateral Dynamics

Dynamic geopolitical scores for four regional bilateral relationships, 1960–2024. Panel (a): Iran–Saudi Arabia, from the Twin Pillar era through post-revolutionary rivalry to the 2023 China-brokered rapprochement. Panel (b): China–Japan, capturing post-war normalization, the friendship treaty era, and the Senkaku/Diaoyu crisis. Panel (c): Egypt–Turkey, showing cold relations during the Nasser era, gradual warming, the strategic partnership peak under Morsi, and the dramatic post-2013 collapse. Panel (d): Cuba–United States, illustrating persistent hostility following the Castro revolution with the brief Obama-era opening.

post-war normalization around 1972 and the friendship treaty period of the late 1970s (scores reaching +0.8), followed by gradual erosion and the sharp deterioration after the 2012 Senkaku/Diaoyu crisis, with a modest recovery around Abe’s 2018 visit to Beijing. The Egypt–Turkey case (panel c) reflects cold relations during the Nasser era in the 1960s (scores near −0.2), gradual warming after Egypt’s 1979 peace with Israel, a strategic partnership peak during the Arab Spring and Morsi presidency in 2011–2012 (scores reaching approximately +0.6), and the collapse after Morsi’s ouster in 2013 with continuing tensions over Libya. Finally, Cuba–U.S. relations (panel d) show persistent hostility following the 1962 Missile Crisis and the Castro revolution, interrupted briefly by the Carter-era interest sections in the late 1970s, tightened further by the 1996 Helms-Burton Act, and punctuated by the sharp Obama-era restoration around 2015 before subsequent reversal.

A.4. Model Robustness and Replicability

An important concern with LLM-based measurement is whether results depend on the specific model or vary across independent runs. We evaluate both dimensions using three frontier LLMs: Gemini 2.5 Pro (our baseline), GPT 5.4, and Claude (Opus 4.6 and Sonnet 4.6).

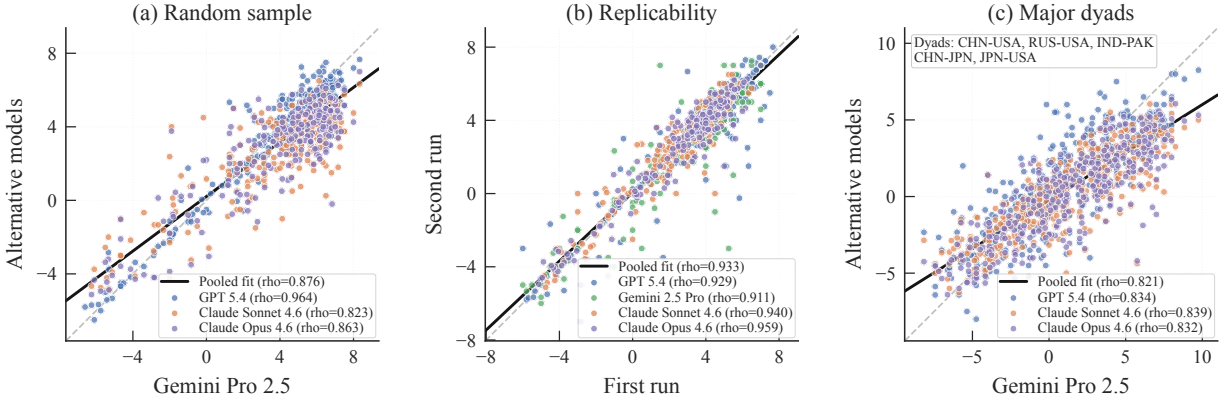


FIGURE A5. Model Robustness and Replicability of Geopolitical Scores

Notes: Each point represents the mean Goldstein score for a country-pair-year observation. Panel (a) compares scores from alternative models (vertical axis) against Gemini 2.5 Pro (horizontal axis) for 300 randomly selected dyad-years. Panel (b) tests within-model replicability by running each model twice on the same dyad-years. Panel (c) compares scores across models for five major bilateral relationships (CHN-USA, RUS-USA, IND-PAK, CHN-JPN, JPN-USA) across all years. All models are run with temperature = 0.1.

Figure A5 reports three complementary exercises. Panel (a) compares scores from alternative LLMs against our Gemini 2.5 Pro baseline for 300 randomly selected dyad-years. Scores cluster along the 45-degree line, with pairwise correlations exceeding 0.82 (pooled $\rho = 0.88$). Panel (b) tests within-model replicability by running each model twice on the same inputs.⁴⁹ Within-model correlations exceed cross-model correlations (pooled $\rho = 0.93$): the within-model residual reflects only stochastic decoding, whereas cross-model differences additionally capture variation in training data and scoring conventions. Newer frontier models achieve higher stability, with Opus 4.6 reaching $\rho = 0.96$ and Sonnet 4.6 reaching $\rho = 0.94$, compared to $\rho = 0.93$ for GPT 5.4 and $\rho = 0.91$ for Gemini 2.5 Pro. Panel (c) compares scores for five major bilateral relationships across all available years—a more demanding test, since volatile major-power dyads require greater interpretive judgment. The pooled correlation ($\rho = 0.82$) is correspondingly lower but remains high. Across all exercises, correlations exceed 0.82, indicating that the measurement is robust to the choice of LLM.

Full-Pipeline Robustness. The preceding exercises hold the prompt structure fixed and vary only the model on subsamples. We next vary the entire measurement pipeline—both the LLM and the prompt structure—to check whether the economic conclusions change. We compare our baseline index to an earlier version (V1) constructed using Gemini 2.5 Flash with a different prompt structure developed prior to the current specification.⁵⁰ To isolate measurement from aggregation differences, we reconstruct the V1 country-year index using the same 24-partner universe and the

⁴⁹All models are run with temperature = 0.1 to reduce stochastic variation while preserving some flexibility in event retrieval and summary.

⁵⁰The V1 prompt pursues the same objective of identifying and scoring bilateral geopolitical events using the CAMEO/Goldstein framework but differs in scope, organization, and event taxonomy. Complete V1 prompt specifications are available upon request.

same time-varying GDP-share weights as the baseline.

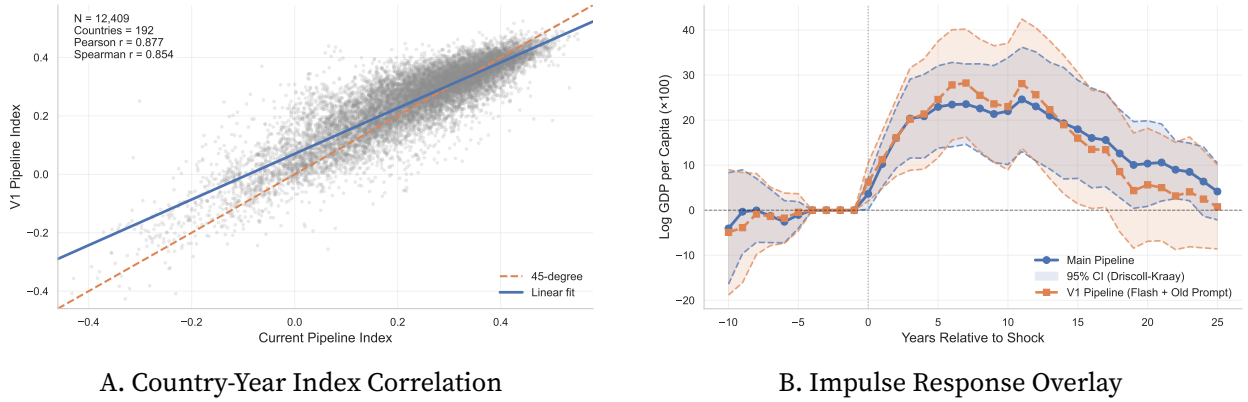


FIGURE A6. Full-Pipeline Robustness: Alternative LLM and Prompt Structure

Panel (a) plots the country-year geopolitical index from the V1 pipeline (Gemini 2.5 Flash with an earlier prompt structure) against the current baseline (Gemini 2.5 Pro) for all 12,409 overlapping country-year observations across 192 countries. The blue solid line shows the linear fit; the orange dashed line shows the 45-degree line. Panel (b) overlays the baseline impulse response of log GDP per capita ($\times 100$) from the main pipeline (blue) with the corresponding estimate using the V1 pipeline index (orange), both estimated on a common sample of 146 countries. Shaded areas represent 95% confidence intervals based on Driscoll-Kraay standard errors.

Figure A6 reports the results. Panel (a) shows that the two country-year indices are strongly correlated (Pearson $\rho = 0.877$, Spearman $\rho = 0.854$), with the linear fit closely tracking the 45-degree line. Panel (b) overlays the baseline impulse response with the V1 pipeline estimate on a common sample of 146 countries. Both pipelines produce the same qualitative dynamics: no pre-trends, a hump-shaped response peaking around years 5–8, and gradual decay thereafter. The V1 point estimates fall within the 95% confidence interval of the main pipeline at all 26 post-shock horizons, and the horizon-by-horizon correlation of point estimates is 0.97. Because this exercise varies both the LLM and the prompt structure simultaneously, it provides a joint test of sensitivity to the complete measurement pipeline rather than any single component.

A.5. Dynamics of Geopolitical Relations

This section examines the dynamic properties of our geopolitical measures using local projection methods. We analyze both the persistence of bilateral geopolitical scores and country-level geopolitical relations to understand how these measures evolve over time.

Methodology. We estimate impulse response functions using local projections for two key measures. For country-level geopolitical relations, we estimate:

$$p_{c,t+h} = \alpha_h^{\text{Geo. Relation}} p_{ct} + \gamma_h' \mathbf{x}_{ct} + \mu_{c,t+h}, \quad h = 0, 1, \dots, 30$$

where p_{ct} denotes either our dynamic geopolitical relation measure or the contemporaneous average, $\mathbf{x}_{ct}$ includes four lags of GDP and the respective geopolitical measure, and we control for

country and year fixed effects.

For bilateral geopolitical scores, we estimate:

$$S_{ij,t+h} = \alpha_h^{\text{Score}} S_{ij,t} + \gamma_h' \mathbf{x}_{ij,t} + \mu_{ij,t+h}, \quad h = 0, 1, \dots, 30$$

where $S_{ij,t}$ represents either the yearly average score or our dynamic score, $\mathbf{x}_{ij,t}$ contains four lags of the score variable, and we include country-pair and year fixed effects with Driscoll-Kraay standard errors.

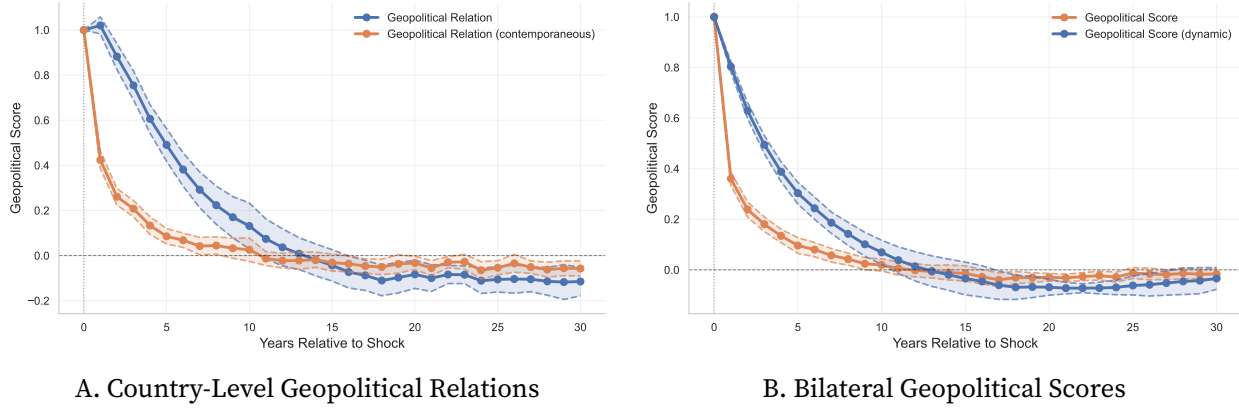


FIGURE A7. Impulse Response Functions of Geopolitical Measures

Local projection impulse responses to own shocks. Panel (a): Country-level geopolitical relations with dynamic measure (blue) and contemporaneous average (orange). Panel (b): Bilateral geopolitical scores with dynamic score (blue) and yearly average (orange). Shaded areas represent 95% confidence intervals. Panel (a) includes four lags of GDP growth and the respective geopolitical measure, plus country and year fixed effects with Driscoll-Kraay standard errors. Panel (b) includes four lags of the respective score variable, plus country-pair and year fixed effects with Driscoll-Kraay standard errors.

Results. Figure A7 presents the impulse responses for both country-level relations and bilateral scores. Panel (a) shows that the dynamic geopolitical relation measure exhibits substantial persistence with a half-life of approximately 5 years, while the contemporaneous average displays rapid mean reversion, returning to baseline within 2–3 years. Notably, the dynamic measure shows slight overshooting between years 15–20 before converging to zero, consistent with cyclical patterns in international relations.

Panel (b) reveals similar patterns at the bilateral level. The yearly average score exhibits strong mean reversion with a half-life under one year, reflecting the transitory nature of individual geopolitical events. In contrast, our dynamic score demonstrates markedly higher persistence, with effects remaining significant for over a decade. This enhanced persistence stems from our decay parameter $\delta = 0.3$, calibrated to capture the typical four-year political cycle while allowing past events to influence current relations. These findings support our dynamic specification’s ability to capture the inherent persistence in geopolitical relationships while filtering out short-term noise from individual events.

A.6. Maps of Bilateral Geopolitical Relations with the US and Its Rivals

This section provides geographic visualization of geopolitical relations, comparing Cold War patterns (1980) with the contemporary landscape (2024) for both the United States and its principal rivals.

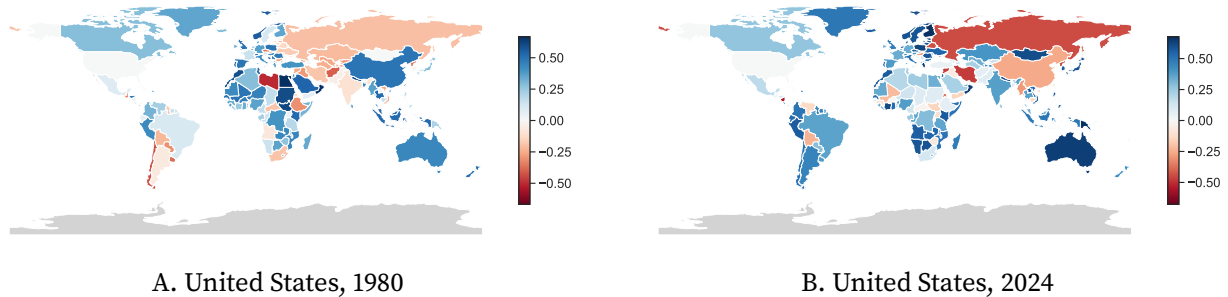


FIGURE A8. Map of Geopolitical Relations with the United States

World maps showing bilateral geopolitical scores with the United States in 1980 and 2024. Blue colors indicate positive relations; red colors indicate negative relations. The transformation from stark Cold War divisions to a more nuanced contemporary pattern is evident, though the re-emergence of great power rivalry with Russia and China is clearly visible in 2024.

Figure A8 shows the transformation from Cold War bipolarity (1980: strong NATO/Pacific alliance bloc vs. hostile Soviet/Eastern European bloc) to a more complex 2024 landscape where core Western alliances remain intact but Russia and China appear in deep red, while much of Latin America, Africa, and Southeast Asia maintain moderate or neutral relations.

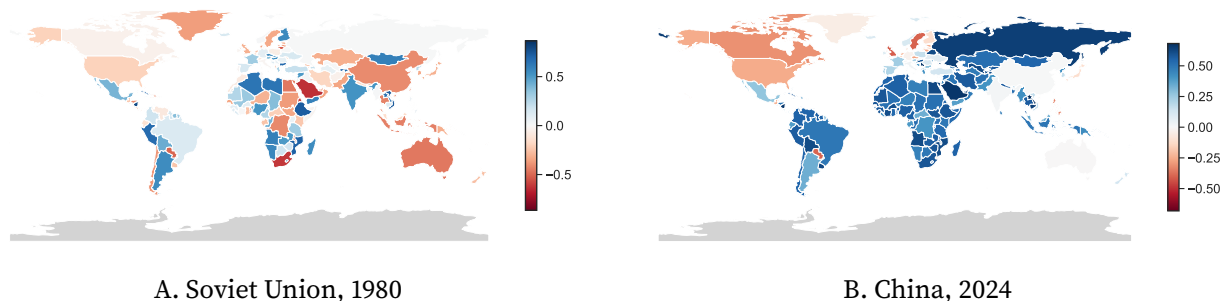


FIGURE A9. Map of Geopolitical Relations: Soviet Union (1980) vs. China (2024)

Comparison of America's principal rival's geopolitical reach in 1980 (Soviet Union) and 2024 (China). While the Soviet Union's positive relations were concentrated in Eastern Europe and select developing countries, China's positive relations extend broadly across Asia, Africa, and Latin America through economic engagement.

Figure A9 contrasts the geographic reach of America's principal rivals. The Soviet Union's 1980 positive relations concentrated in Eastern Europe, Cuba, and select African and Middle Eastern allies, with China notably hostile (reflecting the Sino-Soviet split). China's 2024 map shows a fundamentally different pattern: positive relations extend broadly across Sub-Saharan Africa, Central Asia, and Latin America through economic engagement, though China faces hostile relations with

the United States, Australia, and India. The contrast between the near-mirror-image U.S./Soviet maps of 1980 and the overlapping U.S./China maps of 2024—where many developing countries maintain positive relations with both—corroborates the shift from Cold War bipolarity to the more complex distributional patterns documented in Figure 5.

A.7. Statistics of Country-level Geopolitical Relations

Table A4 presents summary statistics for country-level geopolitical relations across seven decades, revealing systematic patterns in the evolution of global geopolitical alignment.

TABLE A4. Summary Statistics of Geopolitical Relations by Decade, 1960–2024

	1960s	1970s	1980s	1990s	2000s	2010s	2020s
<i>Summary Statistics</i>							
Mean	0.177	0.193	0.182	0.260	0.287	0.296	0.281
Median	0.194	0.207	0.209	0.284	0.315	0.319	0.302
Std. Dev.	0.141	0.119	0.138	0.152	0.132	0.108	0.114
Min	−0.323	−0.315	−0.275	−0.430	−0.256	−0.335	−0.160
Max	0.499	0.531	0.466	0.550	0.548	0.479	0.482
5th Pct.	−0.055	−0.013	−0.103	−0.027	0.023	0.082	0.059
25th Pct.	0.069	0.111	0.111	0.188	0.226	0.251	0.223
75th Pct.	0.286	0.281	0.284	0.371	0.378	0.368	0.367
95th Pct.	0.379	0.368	0.359	0.451	0.446	0.425	0.422
N Country-Years	1,883	1,911	1,925	1,930	1,930	1,930	965
<i>Countries with Lowest Geopolitical Relations (Bottom 5)</i>							
1.	S. Africa	S. Africa	Afghanistan	Libya	Myanmar	N. Korea	N. Korea
2.	China	Albania	Libya	Iraq	Belarus	Eritrea	Nicaragua
3.	Zimbabwe	Zimbabwe	S. Africa	Myanmar	Zimbabwe	Syria	China
4.	Albania	Timor-L.	Latvia	Serbia	Iraq	Venezuela	Syria
5.	Timor-L.	Chile	Lithuania	Montenegro	Gambia	Iran	Russia
<i>Countries with Highest Geopolitical Relations (Top 5)</i>							
1.	Colombia	Colombia	Mali	Mongolia	Albania	Senegal	Laos
2.	Afghanistan	Belgium	Senegal	Kuwait	Vietnam	Liberia	Senegal
3.	Nepal	Bangladesh	Bangladesh	Mozambique	Ghana	Sierra Leone	Singapore
4.	Italy	Finland	Egypt	Poland	Romania	Singapore	Qatar
5.	Ethiopia	Romania	Mozambique	Hungary	Mozambique	Peru	Mozambique

The geopolitical relations index is the GDP-weighted average of bilateral geopolitical scores for each country. Statistics are calculated using all available country-year observations within each decade. Countries are ranked by their decade-average geopolitical relations scores.

The summary statistics reveal three key patterns. First, mean geopolitical relations improved from 0.177 in the 1960s to 0.296 in the 2010s—an increase of approximately 0.85 standard deviations—before declining to 0.281 in the 2020s. This trajectory aligns with our distributional analysis showing post-Cold War convergence followed by recent fragmentation. Second, the standard deviation declined from 0.141 in the 1960s to 0.108 in the 2010s, indicating reduced heterogeneity as countries converged toward more cooperative average relations, though it has since risen slightly to 0.114 in the 2020s as geopolitical tensions resurface. Third, the 5th percentile improved dramatically from −0.055 in the 1960s to 0.082 in the 2010s, demonstrating that even the most isolated countries expe-

rienced substantial improvement in their average geopolitical relations during the globalization era, though this progress partially reversed to 0.059 in the 2020s.

The countries occupying extreme positions provide additional validation of our measure. During the Cold War decades, apartheid-era South Africa, China, and Albania consistently ranked among the most geopolitically isolated states, reflecting ideological divisions and international ostracism. The contemporary period's most isolated countries—North Korea, Syria, and Iran in the 2010s, joined by China, Russia, and Nicaragua in the 2020s—are all subjects of extensive international sanctions or geopolitical confrontation with Western powers. Conversely, the highest-scoring countries often represent small states successfully maintaining positive relations across geopolitical divides (Singapore, Senegal, Qatar) or beneficiaries of particular historical moments (Poland and Hungary during post-Cold War democratization, Vietnam and Albania following market reforms, African states like Mozambique and Ghana engaging multiple development partners). These patterns confirm that our measure captures both systematic features of the international system and country-specific geopolitical strategies.

A.8. Instruments: Non-Economic Verbal Conflict Events

Our instrumental variable strategy exploits variation from non-economic verbal conflicts—diplomatic disputes and political tensions that affect bilateral relations without directly impacting economic activity. This section provides a detailed analysis of these events, which underpin our instrument and motivate the exclusion restriction.

Figure A10 synthesizes patterns across 37,519 non-economic verbal conflict events from our dataset. These events exhibit three key characteristics that validate their use as instruments. First, they show substantial temporal variation, averaging 577 events annually with considerable year-to-year fluctuation and peaking at 1,044 events in 2022, providing rich identifying variation across time. Second, the distribution across CAMEO categories reveals that verbal disapproval (code 11) comprises 74.3% of events, followed by rejections (code 12) at 13.6%, demonstrating a spectrum of diplomatic intensity that generates heterogeneous effects on bilateral relations. Third, the negative mean Goldstein score of -3.91 (with a standard deviation of 1.16) confirms that these events consistently deteriorate geopolitical relations without involving economic content.

Table A5 details the taxonomy of non-economic verbal conflicts that constitute our instrumental variable. The predominance of “Disapprove” events (27,867 occurrences, 74.3%) and “Reject” events (5,093 occurrences, 13.6%) reflects the rhetorical nature of most bilateral tensions. These events—ranging from human rights criticisms to diplomatic protests and refusals of cooperation—generate substantial variation in our geopolitical relations measure while remaining orthogonal to economic fundamentals. We exclude CAMEO codes 15–16 (Exhibit Force and Reduce Relations) because these material conflict events may have direct economic consequences through disrupted diplomatic channels, expelled personnel, or signaling effects that influence investment decisions. We also exclude codes 17–20 (Coerce and Assault), as these severe conflict events may directly affect economic activity through disruption of commerce, destruction of property, or humanitarian

TABLE A5. Non-Economic Verbal Conflict Events: CAMEO Root Codes 09–14

Root Code & Category	Event Types	Primary Causes
09 INVESTIGATE	Investigations into crime, corruption, human rights abuses, military actions, and war crimes	Accountability demands, transparency requirements, moral obligations, violation of international norms
10 DEMAND	Diplomatic cooperation, political reform, compliance, meetings, negotiations, dispute settlement, mediation initiatives	Political disagreements, sovereignty disputes, governance failures, diplomatic tensions
11 DISAPPROVE	Criticism, accusations, opposition mobilization, official complaints, legal proceedings, guilt determinations	Ideological differences, policy disagreements, human rights concerns, norm violations
12 REJECT	Refusal of cooperation, political reform, negotiations, mediation; defiance of norms and laws	Political incompatibility, sovereignty protection, ideological resistance, strategic positioning
13 THREATEN	Diplomatic threats, administrative sanctions, political dissent support, negotiation suspension, military ultimatums	Deterrence strategies, power projection, diplomatic leverage, security concerns
14 PROTEST	Political dissent, demonstrations, hunger strikes, boycotts, obstructions	Political grievances, social justice issues, governance problems, rights violations

This table presents the classification scheme for geopolitical conflict events using the Conflict and Mediation Event Observations (CAMEO) framework, focusing on root codes 09–14 representing verbal conflict. These events involve rhetorical confrontation without material force or tangible demonstrations of power. Events are classified based on primary action type rather than underlying motivations, which may include multiple concurrent factors. Economic conflict events (e.g., trade disputes, sanctions), material conflict events (CAMEO codes 15–16 involving force exhibition and relation reduction), and severe conflict events (CAMEO codes 17–20 involving coercion and assault) are excluded from this classification to focus on purely verbal diplomatic tensions that satisfy our exclusion restriction.

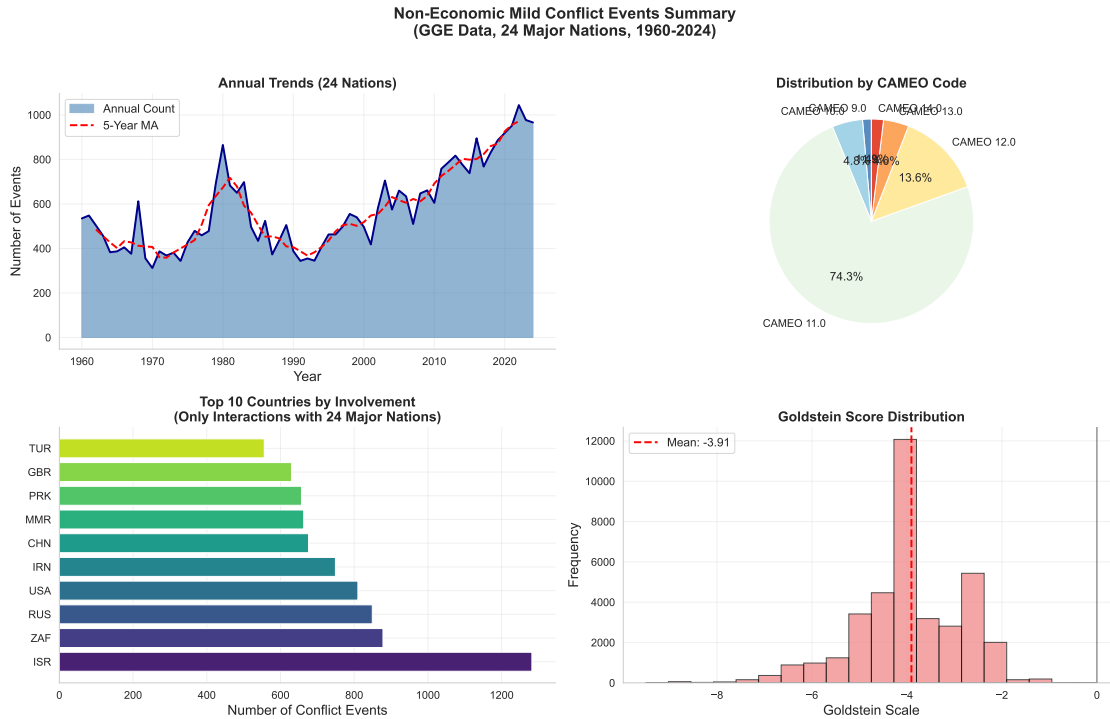


FIGURE A10. Non-Economic Verbal Conflict Events: Summary Statistics (1960–2024)

This figure presents four panels analyzing 37,519 non-economic verbal conflict events. Panel A shows annual trends with a 5-year moving average, revealing increasing frequency over time with a peak of 1,044 events in 2022. Panel B displays the distribution across CAMEO root codes, with “Disapprove” (74.3%) and “Reject” (13.6%) dominating. Panel C identifies the top 10 initiating countries, led by Israel, South Africa, and Russia. Panel D presents the Goldstein score distribution, with a mean of -3.91 , confirming the conflictual nature of these events. Data include only events with Goldstein scores ≤ 0 and exclude economic events.

crises.

The exclusion restriction requires that these non-economic verbal conflicts affect GDP only through their impact on overall geopolitical relations. This assumption is plausible for several reasons. First, by construction, we exclude all events with direct economic content (tariffs, sanctions, trade agreements). Second, the events consist exclusively of verbal actions—diplomatic protests, investigations into human rights violations, or ideological disagreements—that lack immediate economic consequences. Third, by restricting attention to verbal conflicts (codes 09–14), we exclude both material actions (force exhibitions, relation severance) and violent events that could directly affect economic activity through physical destruction or displacement. Fourth, the distribution of initiating countries spans diverse political systems and development levels, suggesting these conflicts arise from idiosyncratic political factors rather than systematic economic conditions.

Importantly, while these verbal conflicts generate negative Goldstein scores (mean = -3.91 , median = -4.0), they create sufficient variation in bilateral relations to identify causal effects. A typical diplomatic criticism scoring -4.0 on the Goldstein scale meaningfully deteriorates the bilateral relationship, affecting the country-level geopolitical measure through our GDP-weighted aggregation. The instrument’s strength derives from both the frequency of these events (averaging

577 per year) and their cumulative impact on diplomatic relations.

The temporal trend reveals substantial variation in verbal diplomatic conflict over our sample period. Events exhibit considerable year-to-year fluctuation around a median of 524 per year, with notable peaks during periods of heightened geopolitical tension such as the early 1980s, mid-2010s, and early 2020s. The spike to 1,044 events in 2022 reflects intensified diplomatic disputes associated with contemporary great power competition. For identification, these temporal patterns exhibit substantial year-to-year variation beyond smooth trends, providing the within-country variation that our instrumental variables strategy exploits.

Appendix B. Additional Empirical Results

B.1. Economic Data

This appendix documents the economic variables used in our analysis. Our dataset extends Acemoglu et al. (2019) to cover 193 countries over 1960–2024, incorporating updated democracy measures from Acemoglu et al. (2025) and economic indicators from the Penn World Tables (Feenstra, Inklaar, and Timmer 2015).

Table B1 organizes variables into four categories following the growth literature: enhanced Solow fundamentals (GDP, investment, capital, demographics), political institutions and governance (democracy, unrest, regime indicators), market institutions and reforms (liberalization indices, fiscal measures), and human capital and labor markets (education, employment, productivity measures). Coverage varies from 108 countries (TFP) to 192 countries (demographic variables), reflecting differences in data availability and statistical capacity across countries and time periods. All monetary variables are in constant prices for temporal comparability.

B.2. Cross-Sectional Correlation with Economic Fundamentals

Table B2 reports within-year cross-sectional correlations between economic fundamentals and (i) GDP per capita (Panel A) and (ii) the geopolitical index (Panel B). GDP per capita correlates strongly with human capital (0.7–0.8), capital stock (0.5), and investment (0.4). In contrast, the geopolitical index shows near-zero and sign-unstable correlations with the same variables across all decades, indicating that it captures a dimension of international relations orthogonal to standard measures of economic development.

B.3. Additional Results for Baseline Estimates

This section provides supplementary evidence for our baseline results, including detailed coefficient estimates and robustness checks addressing potential concerns about sample composition, inference methods, and lag specifications.

Table B3 presents the complete local projection coefficients underlying our baseline impulse response function. The results reveal a clear dynamic pattern: insignificant pre-trends for negative horizons are consistent with the absence of reverse causality, while the contemporaneous effect of 3.3 log points grows to peak at 21.9 log points after 5 years before gradually declining. The coefficient remains statistically significant through horizon 20, demonstrating persistence. The within- R^2 values follow an inverse U-shape, reaching near-unity around horizons 0 and –5 where lagged dependent variables have maximum predictive power, then declining at longer horizons as prediction becomes more challenging.

Balanced versus Unbalanced Panel Estimates. Our baseline analysis employs an unbalanced panel where country coverage varies across horizons. To ensure compositional changes do not drive our

TABLE B1. Data Description and Coverage

Variable	Data Source	Country Coverage	Data Period
<i>Enhanced Solow Fundamentals</i>			
GDP per capita (Constant US Dollar)	WDI	184 countries	1960–2024
Real GDP per capita	PWT	172 countries	1960–2023
Trade as Share of GDP	WDI	176 countries	1960–2024
Investment as Share of GDP	PWT	172 countries	1960–2023
Capital Stock	PWT	169 countries	1960–2023
Population	PWT	172 countries	1960–2023
0–14 Population Share	WDI	192 countries	1960–2023
15–65 Population Share	WDI	192 countries	1960–2023
<i>Political Institutions and Governance</i>			
Democracy (Acemoglu et al. 2019)	Acemoglu et al. (2019)	191 countries	1960–2019
Democracy (Acemoglu et al. 2025)	Acemoglu et al. (2025)	172 countries	1960–2019
Unrest	Acemoglu et al. (2019)	179 countries	1960–2010
Soviet Union	Acemoglu et al. (2019)	183 countries	1960–2024
Region	Acemoglu et al. (2025)	192 countries	1960–2024
<i>Market Institutions and Reforms</i>			
Market Reform Index	Acemoglu et al. (2019)	152 countries	1960–2005
Tax-to-GDP	Acemoglu et al. (2019)	136 countries	1960–2005
<i>Human Capital and Labor Markets</i>			
Human Capital Index	PWT	142 countries	1960–2023
Employment	PWT	172 countries	1960–2023
Labor Share	PWT	132 countries	1960–2023
Primary Enrollment	Acemoglu et al. (2019)	176 countries	1970–2010
Secondary Enrollment	Acemoglu et al. (2019)	176 countries	1970–2010
<i>Productivity and Returns</i>			
TFP	PWT	108 countries	1960–2010
Internal Rate of Return	PWT	132 countries	1960–2023
Real Consumption	PWT	172 countries	1960–2023
Real Domestic Absorption	PWT	172 countries	1960–2023

This table summarizes all variables used in the analysis, organized into four categories: enhanced Solow fundamentals, political institutions and governance, market institutions and reforms, and human capital measures. Country coverage represents the number of countries with at least one non-missing observation for each variable. Data sources: WDI = World Development Indicators; PWT = Penn World Tables.

TABLE B2. Cross-Sectional Correlation with Economic Fundamentals

	Overall	1960s	1970s	1980s	1990s	2000s	2010s	2020–2024
<i>Panel A: Correlation with GDP per Capita</i>								
Investment Share	0.388	0.404	0.311	0.374	0.430	0.444	0.406	0.438
Capital Stock	0.486	0.602	0.469	0.477	0.473	0.475	0.474	0.465
Internal Rate of Return	−0.196	−0.429	−0.302	−0.250	−0.101	−0.083	−0.128	−0.070
TFP	0.027	0.049	0.166	−0.105	0.030	−0.030	−0.437	
Trade Openness	0.313	0.092	0.235	0.294	0.282	0.346	0.387	0.409
Human Capital	0.739	0.764	0.628	0.732	0.755	0.767	0.792	0.775
<i>Panel B: Correlation with Geopolitical Relations</i>								
GDP per Capita (PWT)	0.095	0.094	0.112	−0.168	0.239	0.128	0.134	0.007
Investment Share	0.102	−0.036	0.147	0.018	0.130	0.186	0.195	0.100
Capital Stock	0.127	0.080	0.257	−0.006	0.256	0.163	0.060	−0.153
Internal Rate of Return	−0.013	0.001	−0.082	−0.010	0.001	−0.041	0.005	0.137
TFP	−0.069	0.018	−0.146	−0.085	−0.077	−0.173	−0.160	
Trade Openness	0.114	−0.045	0.089	0.024	0.181	0.160	0.146	0.120
Human Capital	0.079	−0.009	0.067	−0.188	0.333	0.163	0.056	−0.094

Entries are pooled within-year cross-sectional correlations. Panel A reports correlations between GDP per capita and economic fundamentals. Panel B reports correlations between the geopolitical index and the same variables. For each period, variables are first demeaned within year, and correlations are computed across country-year observations. Only years with at least 20 countries and nonzero cross-country variation are included.

TABLE B3. Local Projection Estimates: Effect of Geopolitical Relations on GDP per Capita

Horizon (years)	−15	−10	−5	+0	+5	+10	+15	+20	+25
Geopolitical Relations	−10.031 (8.040)	−6.688 (6.745)	−1.226 (1.527)	3.341* (1.779)	21.914*** (4.536)	19.105*** (5.607)	14.412*** (5.265)	10.268** (4.773)	3.134 (2.927)
Within- R^2	0.525	0.767	0.980	0.981	0.775	0.538	0.353	0.221	0.127
Observations	6,997	7,899	8,813	8,971	8,057	7,155	6,260	5,373	4,493
Countries	179	182	183	183	182	179	179	177	172

This table presents local projection estimates from equation (4) for the effect of geopolitical relations on log GDP per capita ($\times 100$). Each column represents a separate regression for horizon h , where negative horizons test for pre-trends. All specifications include country fixed effects, region-year fixed effects, and four lags of both log GDP per capita and geopolitical relations. Driscoll-Kraay standard errors in parentheses account for cross-sectional and temporal dependence. The sample spans 1975–2024 with varying coverage across horizons due to lag requirements and data availability. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

results, we construct a balanced panel of 146 countries with complete data from $h = -10$ to $h = 25$. Panel (a) of Figure B1 shows stability: impulse responses from both panels closely track each other, with overlapping confidence intervals throughout most horizons. Both specifications exhibit the characteristic hump-shaped response, peaking around years 6–8 at approximately 20 log points. The absence of pre-trends in both samples reinforces our identification strategy.

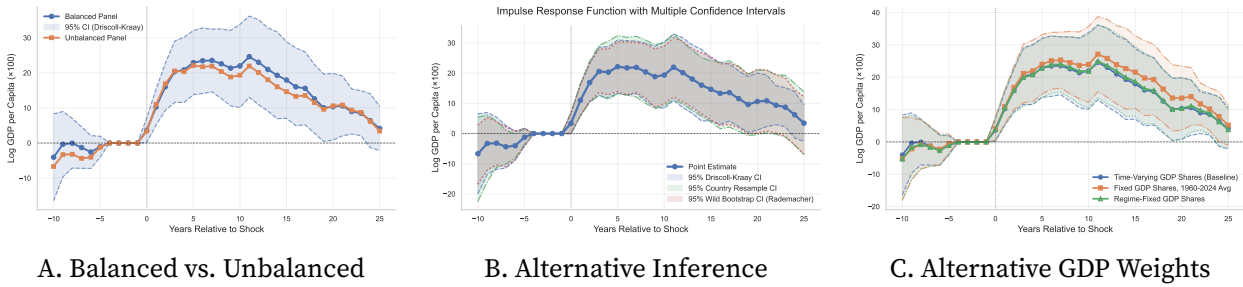


FIGURE B1. Robustness Checks for Baseline IRF Estimates

Panel (a) compares impulse responses using balanced (146 countries with complete data, blue) and unbalanced panels (all available observations, orange). Panel (b) displays the baseline IRF with three types of confidence intervals: Driscoll-Kraay standard errors (blue shaded), 1,000 country-block bootstrap iterations (green shaded), and 1,000 wild bootstrap iterations using Rademacher weights (red shaded). Panel (c) compares the baseline using time-varying GDP shares (blue) with two alternatives: fixed GDP shares averaged over 1960–2024 (orange) and regime-fixed GDP shares using Cold War versus post-Cold War averages (green). All panels show the response of log GDP per capita ($\times 100$) to a unit shock in geopolitical relations with country and region-year fixed effects.

Bootstrap-Based Inference. Our baseline Driscoll-Kraay standard errors treat the geopolitical relations measure as fixed, potentially understating uncertainty. Panel (b) examines robustness using two alternative bootstrap procedures. First, we implement a block bootstrap that resamples entire countries with replacement, capturing both within-country serial correlation and measurement uncertainty in the geopolitical index. Second, we employ a wild bootstrap with Rademacher weights that preserves the panel structure while accounting for potential heteroskedasticity and

within-cluster correlation.⁵¹

The confidence intervals from 1,000 iterations of each bootstrap method are marginally wider than the Driscoll-Kraay bands, particularly at medium horizons (5–15 years), but the differences are economically modest. The wild bootstrap intervals (red shaded area) are slightly narrower than the country-resampling bootstrap (green shaded area) at most horizons, suggesting that accounting for heteroskedasticity provides efficiency gains while maintaining valid inference. All three inference methods yield statistically significant positive effects throughout the key horizons, with the impulse response remaining well above zero even at the widest confidence bounds. The convergence of results across these diverse inference approaches—parametric clustered standard errors, block resampling, and wild bootstrap—confirms the robustness of our inference.

Alternative GDP Weights. Panel (c) of Figure B1 addresses the concern that time-varying GDP weights in equation (2) could generate mechanical variation correlated with economic outcomes. We compare the baseline (time-varying GDP shares) with two alternatives: fixed GDP shares averaged over 1960–2024 and regime-fixed shares using Cold War (1960–1989) versus post-Cold War (1990–2024) averages. All three weighting schemes produce nearly identical impulse responses, confirming that the identifying variation comes from changes in bilateral geopolitical scores rather than from shifts in GDP weights.

Alternative Lag Specifications. Figure B2 examines sensitivity to lag length selection. Parsimonious specifications with one lag exhibit problematic pre-trends—GDP declines before geopolitical improvements—indicating inadequate control for growth dynamics. Our baseline four-lag specification eliminates these pre-trends while maintaining precision. The eight-lag specification yields nearly identical point estimates with moderately wider confidence intervals due to additional parameters. The convergence between four- and eight-lag results validates our baseline choice, confirming adequate capture of relevant dynamics without overfitting.

These robustness checks collectively reinforce our main findings. The stability of impulse responses across different samples, inference methods, and lag structures supports our conclusion that geopolitical alignment generates substantial and persistent economic returns. The consistency of results is particularly noteworthy given the stringent demands of within-country identification in growth empirics.

B.4. Impulse Responses to Transitory and Persistent Shocks

The impulse responses presented in Section 3.3 reflect both the direct impact of initial geopolitical shocks and the subsequent effects of geopolitical persistence. To isolate the direct effects, we construct responses to a counterfactual scenario where geopolitical improvements are purely

⁵¹The wild bootstrap generates weights $w_i \in \{-1, +1\}$ with equal probability for each observation, then constructs bootstrap samples as $y_{it}^* = \hat{y}_{it} + w_i \hat{\epsilon}_{it}$, where $\hat{y}_{it}$ and $\hat{\epsilon}_{it}$ are fitted values and residuals from the baseline specification. This approach is particularly robust to heteroskedasticity of unknown form in panel settings (Cameron, Gelbach, and Miller 2008).

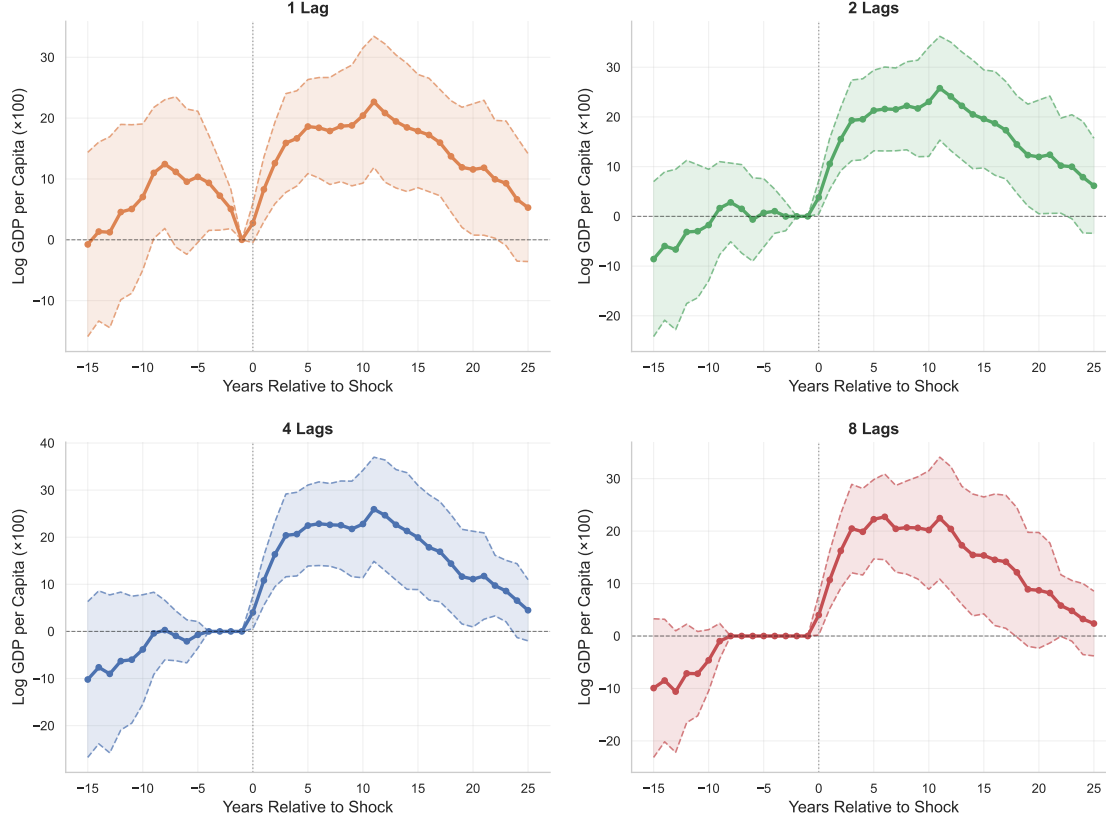


FIGURE B2. Impulse Response Functions Under Alternative Lag Specifications

IRF of log GDP per capita ($\times 100$) to a unit improvement in geopolitical relations. Each panel shows results with the indicated number of lags for both GDP and geopolitical relations. All specifications include country and region-year fixed effects. Shaded areas represent 95% confidence intervals with Driscoll-Kraay standard errors. The spurious pre-trends in one- and two-lag specifications highlight the importance of adequate lag structure for identification.

transitory—increasing by one unit on impact and returning to zero immediately thereafter. Following Sims (1986) and Bilal and Känzig (2024), we combine the impulse responses of geopolitical relations and GDP to construct this counterfactual transitory response.

We begin by estimating the dynamics of geopolitical relations using local projections:

$$(B1) \quad p_{c,t+h} = \phi_h^p p_{ct} + \gamma_h' \mathbf{x}_{ct} + \mu_{c,t+h}, \quad h = 0, 1, \dots, H$$

where $\{\phi_h^p\}_{h=0,\dots,H}$ represents the impulse response of geopolitical relations to its own shock. To construct the purely transitory shock, we introduce a series of auxiliary shocks $\{p_h^{\text{shock}}\}_{h=0}^H$ at each horizon that impose the desired transitory response pattern $\tilde{\Phi}^p = (1, 0, \dots, 0)'$. The required shock

series $\mathbf{p}^{\text{shock}}$ is obtained by solving:

$$(B2) \quad \underbrace{\begin{pmatrix} p_0^{\text{shock}} \\ p_1^{\text{shock}} \\ \vdots \\ p_H^{\text{shock}} \end{pmatrix}}_{\mathbf{p}^{\text{shock}}} = \underbrace{\begin{pmatrix} 1 & 0 & \dots & 0 \\ \phi_1^p & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \phi_H^p & \phi_{H-1}^p & \dots & 1 \end{pmatrix}}_{(\Phi^p)^{-1}}^{-1} \underbrace{\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}}_{\tilde{\Phi}^p}$$

Given this shock series, the corresponding GDP impulse responses $\tilde{\alpha}$ to the purely transitory geopolitical shock are:

$$(B3) \quad \underbrace{\begin{pmatrix} \tilde{\alpha}_0 \\ \tilde{\alpha}_1 \\ \vdots \\ \tilde{\alpha}_H \end{pmatrix}}_{\tilde{\alpha}} = \underbrace{\begin{pmatrix} p_0^{\text{shock}} & 0 & \dots & 0 \\ p_1^{\text{shock}} & p_0^{\text{shock}} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ p_H^{\text{shock}} & p_{H-1}^{\text{shock}} & \dots & p_0^{\text{shock}} \end{pmatrix}}_{\mathbf{p}^{\text{shock}}} \underbrace{\begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_H \end{pmatrix}}_{\alpha}$$

The resulting impulse responses $\tilde{\alpha}$ represent the GDP effects following a one-time, purely transitory improvement in geopolitical relations. This decomposition enables computation of responses to geopolitical shocks with arbitrary persistence patterns. For the permanent shock response shown in Figure 7, we compute the cumulative effect as $\sum_{s=0}^h \tilde{\alpha}_s$, which represents the total GDP impact when geopolitical relations permanently increase by one unit. Statistical inference employs block bootstrap resampling across countries to account for estimation uncertainty in both stages.

Methodological Caveat. This approach assumes that the economic effects of a sequence of unanticipated geopolitical shocks equal those of an anticipated path announced at time zero. While this assumption facilitates decomposition analysis and is standard in the impulse response literature, it abstracts from forward-looking behavior that might differ under anticipation of future geopolitical changes. The assumption is most plausible for transitory shocks where agents have limited ability to anticipate persistence. We employ this decomposition primarily to illustrate the economic importance of geopolitical persistence rather than for structural policy analysis.

B.5. Controlling for Economic Fundamentals

Figures B3A and B3B show that our baseline results survive controlling for a broad set of observable economic fundamentals from the Penn World Tables.⁵² The Solow/PWT specification adds four

⁵²The Solow/PWT specification covers 99 countries and the expanded specification covers 85 countries. Both baselines are re-estimated on the same country sample to ensure comparability.

lags of trade openness, population, age-structure variables, the investment share, capital stock, and human capital index. The expanded specification further adds lagged internal rates of return, labor shares, employment-to-population ratios, real consumption, and real domestic absorption per capita. In both cases, the impulse response closely tracks the baseline, with peak effects of approximately 20 log points and similar hump-shaped dynamics. The stability of the estimates after absorbing these growth determinants indicates that geopolitical relations operate through channels beyond those captured by standard observable fundamentals.

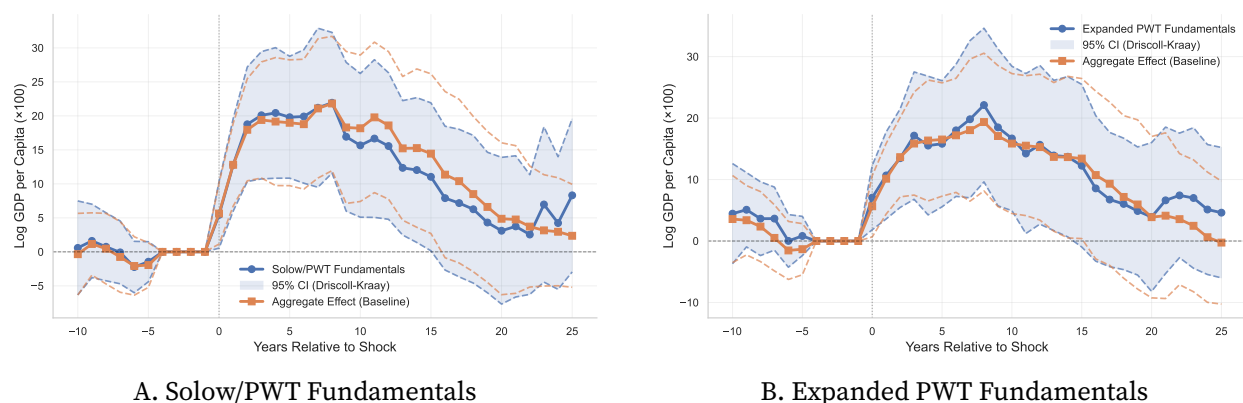


FIGURE B3. Robustness to Controlling for Economic Fundamentals

Impulse responses of log GDP per capita ($\times 100$) to a unit improvement in geopolitical relations. Blue lines with shaded 95% confidence intervals show specifications controlling for lagged economic fundamentals. Orange lines show the baseline re-estimated on the same country sample. Panel (a) adds four lags of trade openness, log population, age-structure shares, investment share, log capital stock, and log human capital index (99 countries). Panel (b) further adds four lags of internal rate of return, labor share, employment-to-population ratio, real consumption per capita, and real domestic absorption per capita (85 countries). All specifications include country and region-year fixed effects with Driscoll-Kraay standard errors.

B.6. Additional Robustness Results

This section provides additional evidence supporting the robustness of our main findings through extended visualizations and decomposition analyses.

Binscatter at Different Horizons. We present binscatter plots at various horizons to demonstrate that the positive relationship between geopolitical relations and economic growth is not driven by outliers or specific subsamples.

Figure B4 examines the relationship at the 10-year horizon from two perspectives. Panel (a) shows that geopolitical relations continue to predict GDP growth 10 years later, with a slope coefficient of 19.4 log points. This modest decline from the 5-year coefficient aligns with the hump-shaped impulse response in our main results. Panel (b) examines a shorter prediction window by relating GDP at $t + 10$ to geopolitical relations at $t + 5$, yielding a slope of 17.3. The similarity of these coefficients—whether using a 10-year or 5-year prediction horizon—reinforces our interpretation that geopolitical alignment generates persistent economic benefits.

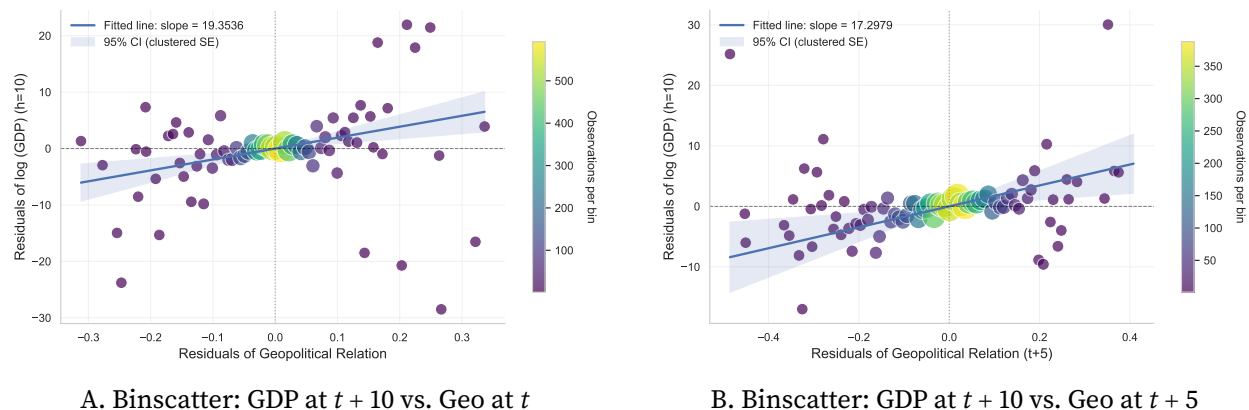


FIGURE B4. Binscatter Plots at Extended Horizons

This figure extends the binscatter analysis to the 10-year horizon. Panel (a) relates log GDP per capita at $t + 10$ to geopolitical relations at t , while panel (b) examines GDP at $t + 10$ against forward geopolitical relations at $t + 5$. Each dot represents the mean of approximately 100 observations within each bin, with size proportional to the number of observations. The specification includes four lags of both variables, country fixed effects, and region-year fixed effects. Fitted lines use Driscoll-Kraay standard errors.

The binscatter visualization reveals a smooth, approximately linear relationship across the entire distribution. Using forward geopolitical relations—which capture cumulative changes over 5-year windows—yields consistent results across all horizons. The slopes remain stable whether examining GDP at $t + 5$, $t + 10$, or $t + 15$, ranging from 17 to 22 log points. This consistency across different shock sizes and time windows confirms that our results reflect a genuine economic relationship rather than the influence of extreme events or specific subsamples.

B.7. Geopolitical Relations with Western and Non-Western Countries

Building on our decomposition of geopolitical effects, we examine whether growth impacts differ between alignment with Western democracies versus non-Western powers. We partition our geopolitical relations measure into two components: relations with Western countries (United States, United Kingdom, Germany, France, Canada, Australia, Belgium, Denmark, Italy, Netherlands, Spain, Switzerland, and Poland) and relations with non-Western countries (China, Russia, India, Japan, South Korea, Brazil, Mexico, Argentina, Indonesia, Turkey, and Saudi Arabia). This classification captures the traditional West/non-West divide in international relations while accounting for the economic weight of each group.

Figure B5 presents impulse responses from joint estimation of both components. Panel (a) reveals that improved relations with Western countries generate substantial growth effects, with GDP per capita increasing by approximately 21–23 log points at the peak around years 5–10. The response exhibits our characteristic hump shape, with effects persisting significantly through year 20 before gradually declining. Panel (b) demonstrates similar dynamics for relations with non-Western countries, though with higher peak effects: the impulse response reaches approximately 28–35 log points around years 3–5 before declining more rapidly, with wider confidence intervals reflecting

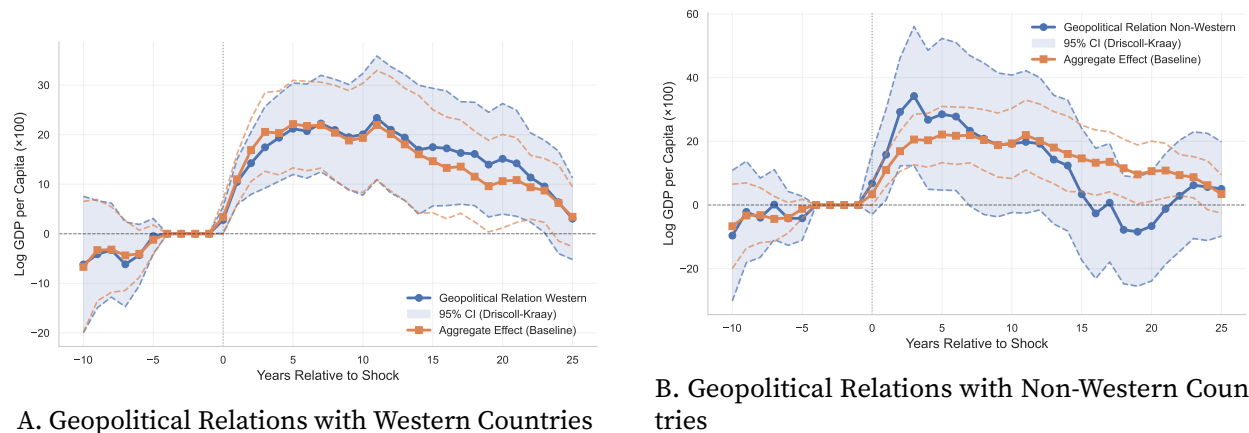


FIGURE B5. Decomposing Geopolitical Relations: Western versus Non-Western Countries

This figure shows impulse responses of log GDP per capita ($\times 100$) to unit improvements in geopolitical relations with Western countries (panel a) and non-Western countries (panel b). Blue lines with shaded areas show the decomposed effects from joint estimation, while orange lines display the baseline aggregate effect for comparison. Both specifications include four lags of all variables, country fixed effects, and region-year fixed effects. Shaded areas represent 95% confidence intervals based on Driscoll-Kraay standard errors.

greater volatility in non-Western diplomatic relationships. Both components track closely with the baseline aggregate measure (shown in orange), confirming that each contributes meaningfully to overall geopolitical effects.

This symmetry extends our main finding that geopolitical benefits transcend ideological boundaries.⁵³ Whether countries improve relations with traditional Western democracies or emerging non-Western powers, the growth dividends are economically substantial. The Western component shows more persistent effects with tighter confidence intervals, while the non-Western component exhibits higher but more volatile peak effects. These patterns likely reflect differences in the nature of diplomatic engagement: Western relationships tend to be anchored in institutional frameworks (trade agreements, alliance structures) that generate sustained benefits, while non-Western relationships may depend more on bilateral political dynamics that fluctuate with leadership changes. For developing countries navigating an increasingly multipolar world, this finding suggests that diversified diplomatic engagement across both Western and non-Western powers can yield substantial economic returns without requiring exclusive alignment with either bloc.

B.8. Sub-Period Stability

We assess whether the growth effects of geopolitical relations are stable across historical periods by estimating our baseline specification separately for 1960–1989 and 1990–2019. Each sub-period spans 30 years, and we restrict the horizon to $h = 0, \dots, 10$ to preserve adequate sample size within each window.

⁵³The conditional correlation between Western and non-Western relations—after partialling out country fixed effects, region-year fixed effects, and lagged controls—is only 0.27, confirming that the identifying variation across these groups is largely independent.

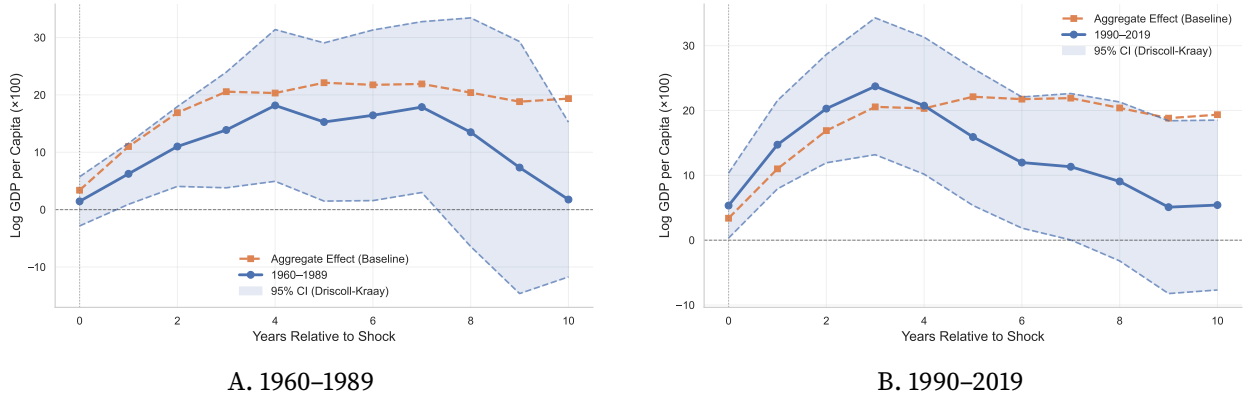


FIGURE B6. Sub-Period Stability of Geopolitical Growth Effects

This figure shows impulse responses of log GDP per capita ($\times 100$) to a unit improvement in geopolitical relations, estimated separately for 1960–1989 (panel a) and 1990–2019 (panel b). Blue lines with shaded areas show sub-period estimates; orange dashed lines display the full-sample baseline for comparison. Both specifications include four lags of all variables, country fixed effects, and region-year fixed effects. Shaded areas represent 95% confidence intervals based on Driscoll-Kraay standard errors. Horizons are restricted to $h = 0, \dots, 10$ due to reduced sample size in each sub-period.

Figure B6 shows that the effect is present in both periods. Point estimates track the full-sample baseline through year 5, with both sub-periods generating peak effects of 15–24 log points. The full-sample estimate lies within the 95% confidence interval throughout. Estimates attenuate at longer horizons, reflecting reduced statistical power as the forward projection consumes a larger share of the 30-year window rather than a genuine decline in persistence.

B.9. Placebo Design and Randomization Inference

This appendix documents the placebo exercises summarized in Section 3.5.3 and reports scalar randomization-inference results.

Implementation. Both exercises use 500 independent placebo draws. In each draw, the full baseline local projection is re-estimated from scratch using the same specification—four lags of GDP and geopolitical relations, country fixed effects, and region-year fixed effects—with Driscoll-Kraay standard errors. The summary statistic is the average estimated impulse response over horizons $h = 0, \dots, 10$, computed from the re-estimated LP in each draw.

Placebo A: within-region-year reassignment. For each draw, we randomly permute the realized values of the geopolitical shock variable across countries within each region-year cell. This preserves the region-year distribution of shocks and the panel structure of outcomes and controls while breaking the country-specific assignment of the geopolitical shock.

Placebo B: future-year timing. For each draw and each country, we replace the contemporaneous shock with the shock observed 8 to 15 years in the future, where the lead is drawn uniformly

at random. To ensure that all leads are well defined, the base-year support is restricted to 1960–2009. This preserves each country’s own shock distribution (up to a time shift) while breaking the contemporaneous timing between the shock and subsequent GDP dynamics.

Randomization inference. Figure B7 presents the distribution of the average response over horizons 0 to 10 across 500 placebo draws under each design. In the within-region-year reassignment, the placebo distribution is centered near zero (median = 0.09) with a 95th percentile of 1.92, while the realized statistic is 17.87. In the future-year timing placebo, the placebo distribution is again centered near zero (median = –0.56) with a 95th percentile of 3.55, while the realized statistic is 18.16. In both cases, the randomization p -value is below 0.002 (no placebo draw exceeds the realized value in 500 draws).

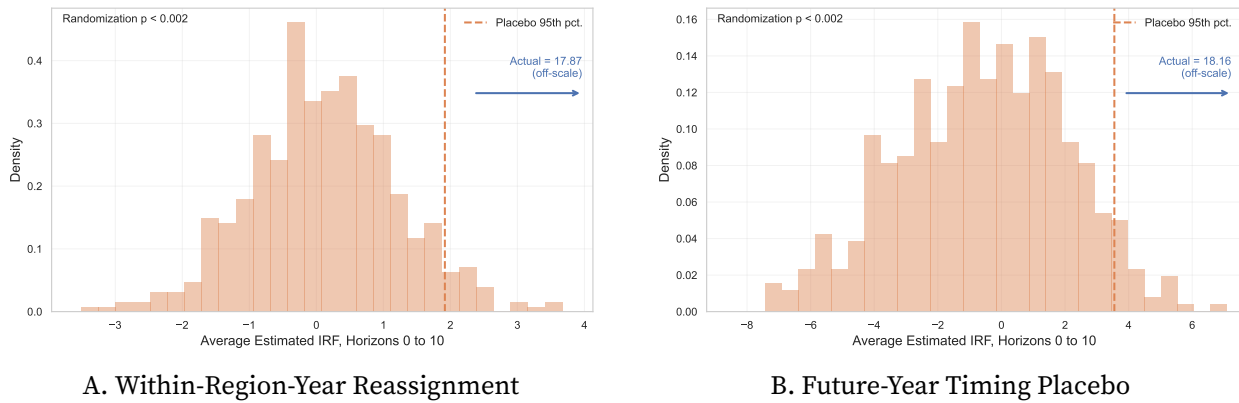


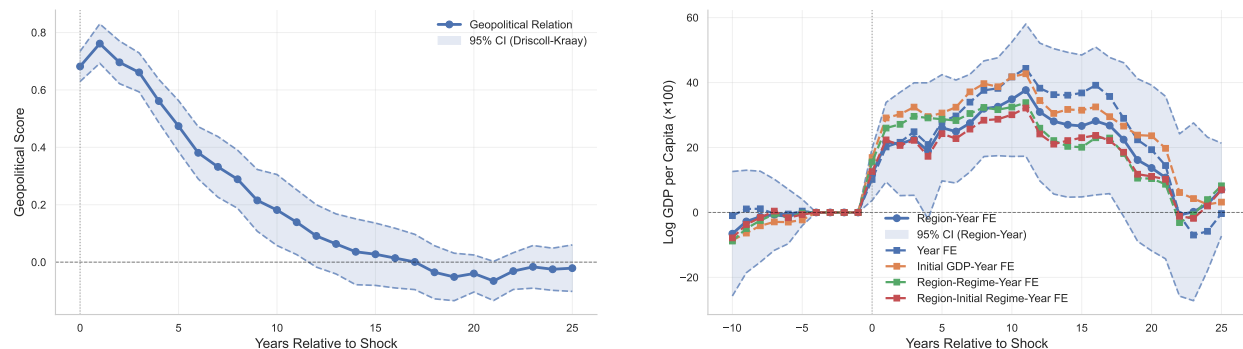
FIGURE B7. Randomization Inference: Distribution of Placebo Average Responses

Each panel shows the distribution of the average estimated impulse response over horizons 0 to 10 across 500 placebo draws. Panel (a) randomly reassigns geopolitical shocks within region-year cells; panel (b) replaces contemporaneous shocks with future shocks drawn 8 to 15 years ahead. The dashed vertical line marks the 95th percentile of the placebo distribution. The realized average response (17.87 in panel a, 18.16 in panel b) lies far off-scale to the right. All specifications re-estimate the full baseline local projection in each draw with four lags of GDP and geopolitical relations, country fixed effects, region-year fixed effects, and Driscoll-Kraay standard errors.

B.10. Additional Results for IV Estimates

This appendix provides supplementary results for our instrumental variables analysis, including the first-stage relationship between non-economic verbal conflicts and overall geopolitical relations, as well as LP-IV estimates under alternative fixed effects specifications.

First-Stage Dynamics. Panel (a) of Figure B8 demonstrates the strength and persistence of our first-stage relationship. The instrument generates a strong positive response in geopolitical relations: a one-unit increase in the instrument raises overall geopolitical relations by approximately 0.69 units on impact. This effect peaks at horizon 1 (reaching approximately 0.76) before gradually decaying—falling to approximately 0.48 by year 4, 0.22 by year 8, and crossing zero around year 16. The



A. First-Stage: Instrument → Geopolitical Relations

B. LP-IV Estimates with Alternative Fixed Effects

FIGURE B8. First-Stage Relationship and LP-IV Robustness

Panel (a) displays the first-stage impulse response of overall geopolitical relations to a unit shock in the instrument (non-economic verbal conflicts). The specification includes four lags of geopolitical relations, GDP, and the instrument, plus country and region-year fixed effects. Panel (b) presents LP-IV estimates of GDP responses under alternative fixed effects specifications. All specifications include four lags of core variables and the instrument with Driscoll-Kraay standard errors. Shaded areas represent 95% confidence intervals.

persistence of the first-stage relationship—remaining statistically significant for approximately 15–16 years—reflects how initial verbal diplomatic conflicts propagate through bilateral relationships. The effect eventually dissipates, with modest negative values at longer horizons (approximately -0.05 by year 21), consistent with the transitory nature of many diplomatic developments.

LP-IV Robustness Across Fixed Effects. Panel (b) of Figure B8 examines the sensitivity of our IV estimates to different assumptions about unobserved heterogeneity. The stability across specifications supports our causal interpretation. All specifications show insignificant pre-trends and generate similar impulse response patterns.

The baseline region-year specification shows GDP increasing by approximately 14 log points on impact, rising to approximately 25 log points by year 5, and reaching a peak of approximately 38 log points around year 10, before declining to approximately 10 log points by year 20. When we replace region-year fixed effects with only year effects—allowing for global shocks but not regional ones—the IV estimates follow a similar trajectory with slightly higher point estimates, peaking around 42 log points at year 10, suggesting our results are not driven by regional confounders. The initial GDP quintile-year specification, which compares countries at similar development stages, yields point estimates that track closely with the baseline, with effects reaching approximately 32–35 log points around years 8–10 before declining. The region-regime-year and region-initial regime-year specifications—our most demanding tests that account for political institutions—produce similar patterns, with peak effects of approximately 32–35 log points around years 8–10, though both show somewhat earlier attenuation compared to the baseline.

The consistency of LP-IV estimates across these diverse fixed effects structures is noteworthy. Despite varying assumptions about relevant comparison groups and sources of unobserved heterogeneity, all specifications yield effects between 20 and 42 log points at the 10-year horizon, with

substantial effects persisting (5–15 log points) even at the 20-year horizon. This stability, combined with the strong first-stage relationship and absence of significant pre-trends, suggests that the instrument provides an informative source of identifying variation. The similarity between IV and OLS estimates across these specifications is consistent with a causal interpretation of the baseline relationship and indicates that the results are not driven by regional patterns or dependent on specific institutional contexts.

B.11. Other Correlates of Growth

This section extends our analysis by examining additional growth correlates beyond those presented in the main text. We investigate two sets of outcomes: (i) institutional and human capital variables emphasized by Acemoglu et al. (2019) in their study of democracy’s economic effects, and (ii) labor market and absorption measures from the Penn World Tables that capture alternative dimensions of economic development.

Market Reforms and Human Capital Formation. Panel (a) of Figure B9 reveals heterogeneous institutional and educational responses to geopolitical alignment. The market reform index shows no statistically significant response, with point estimates fluctuating around zero throughout the horizon and confidence intervals consistently spanning zero. Government expenditure exhibits a positive response, rising by approximately 20–30 log points over the first decade, though wide confidence intervals reflect substantial cross-country heterogeneity in fiscal responses to improved international relations. Education outcomes display contrasting patterns: primary school enrollment shows a gradual and persistent improvement, rising by approximately 10–15 log points over 25 years as sustained international stability enables educational investments. In contrast, secondary school enrollment exhibits no significant effect, with point estimates remaining statistically indistinguishable from zero throughout the horizon.

Labor Markets and Domestic Absorption. Panel (b) demonstrates that growth from geopolitical alignment translates into broad-based welfare improvements, though labor market structure remains largely unchanged. The employment-to-population ratio exhibits a modest upward trend but remains statistically insignificant throughout the horizon, suggesting that geopolitical alignment does not generate substantial extensive-margin employment effects. Similarly, the labor share shows no significant response, fluctuating around zero with confidence intervals consistently spanning zero—indicating that the functional distribution of income remains stable following geopolitical improvements. In contrast, both consumption measures display robust positive responses that closely track GDP dynamics documented in our main analysis: real consumption per capita rises sharply, peaking at approximately 30 log points around year 5 before gradually declining, while domestic absorption per capita follows a similar pattern with comparable magnitude. These results confirm that growth translates into household welfare improvements and validate the investment boom documented in Section 4.

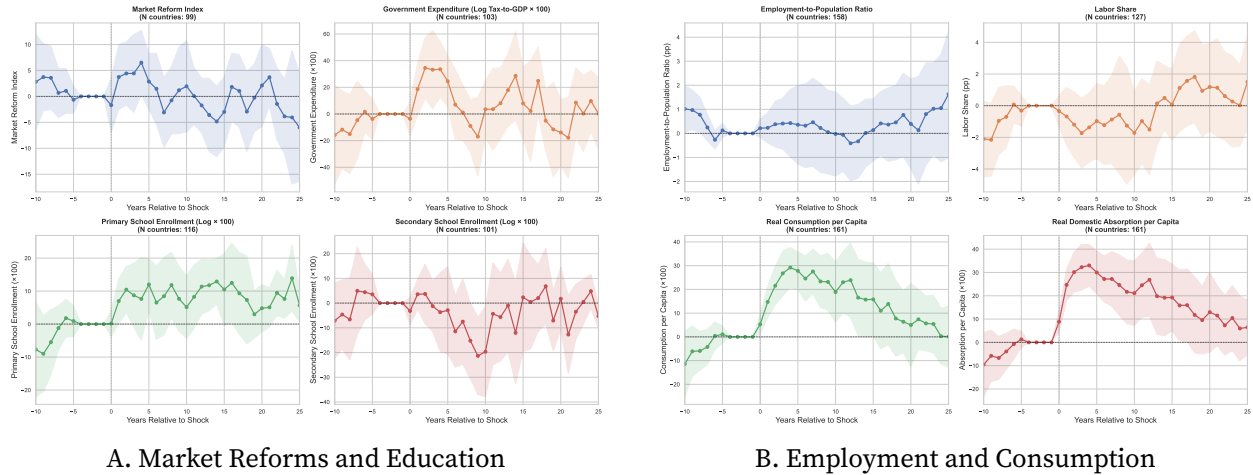


FIGURE B9. Dynamic Effects of Geopolitical Relations on Additional Growth Correlates

Panel (a) displays impulse responses of market reform index, government expenditure (log tax-to-GDP $\times 100$), primary and secondary school enrollment (log $\times 100$) to a unit improvement in geopolitical relations. Panel (b) shows responses for employment-to-population ratio, labor share, real consumption and domestic absorption per capita (log $\times 100$). All specifications follow equation (5) with four lags of the dependent variable, GDP, and geopolitical relations, plus country and region-year fixed effects. Sample restricted to countries with complete data (N in parentheses). Shaded areas represent 95% confidence intervals based on Driscoll-Kraay standard errors.

Synthesis. These results reinforce our main findings while revealing important heterogeneity across outcomes. The positive responses in government expenditure, primary education, consumption, and domestic absorption confirm that geopolitical alignment generates tangible benefits across multiple dimensions of economic development. Notably, the absence of significant effects on market reforms, secondary education, employment rates, and labor share suggests that geopolitical improvements operate primarily through existing economic structures rather than through fundamental institutional transformation or labor market restructuring. The strong consumption and absorption responses, combined with stable labor market indicators, indicate that geopolitical alignment enhances economic efficiency and resource availability without necessarily altering the underlying distribution of economic activity or income.

B.12. Additional Results for Democracy and Geopolitics

This appendix provides supplementary results for our analysis of democracy and geopolitical relations, including decompositions of transitory versus permanent democratization effects and the conditional correlation between these variables.

B.12.1. Transitory versus Permanent Democratization Shocks

To disentangle the dynamic effects of democratization, we decompose democracy's growth impact into responses to transitory versus permanent institutional changes. Following the methodology in Appendix B.4, we construct counterfactual impulse responses that isolate the effects of purely

transitory democratization (a one-time shock that immediately reverts) from permanent democratic transitions.

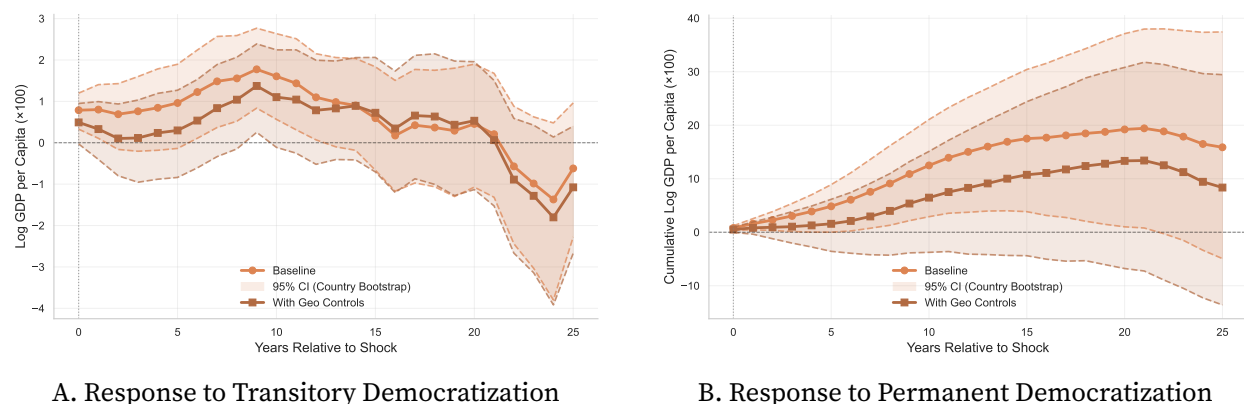


FIGURE B10. GDP Responses to Transitory and Permanent Democracy Shocks

Panel (a) shows the impulse response of log GDP per capita ($\times 100$) to a purely transitory democratization shock. Panel (b) displays the cumulative response to a permanent democratic transition. Both panels compare the baseline specification (controlling for 4 lags of GDP) with the specification controlling for geopolitical relations. Shaded areas represent 95% confidence intervals from 1,000 bootstrap iterations using country-block resampling.

Figure B10 reveals clear patterns in how geopolitical relations mediate democracy's growth effects. Panel (a) demonstrates that even transitory democratization generates persistent economic gains in the baseline specification, with GDP remaining 1–2 log points higher for several years. However, when we control for geopolitical relations, the short-run effect virtually disappears—the point estimates hover near zero. This stark attenuation suggests that temporary democratic episodes generate immediate growth primarily through improved international relations rather than domestic institutional changes.

Panel (b) presents the cumulative effects of permanent democratization. The baseline specification shows GDP rising steadily to approximately 20 log points after 25 years. Controlling for geopolitical relations reduces but does not eliminate these gains: the long-run effect remains economically significant at 10–15 log points. This persistence indicates that while geopolitical improvements explain roughly 30% of democracy's growth impact, sustained democratic institutions generate additional benefits through domestic channels—improved property rights, reduced expropriation risk, and enhanced contract enforcement—that operate independently of international alignment.

B.12.2. Democracy and Geopolitical Relations: Conditional Correlation

While our main analysis examines how democracy and geopolitics jointly affect growth, understanding their mutual relationship provides additional insights. Figure B11 presents two complementary perspectives.

Panel (a) reveals that democratization generates sustained improvements in geopolitical relations, with alignment increasing gradually to peak at 0.035 units (approximately 0.3 standard

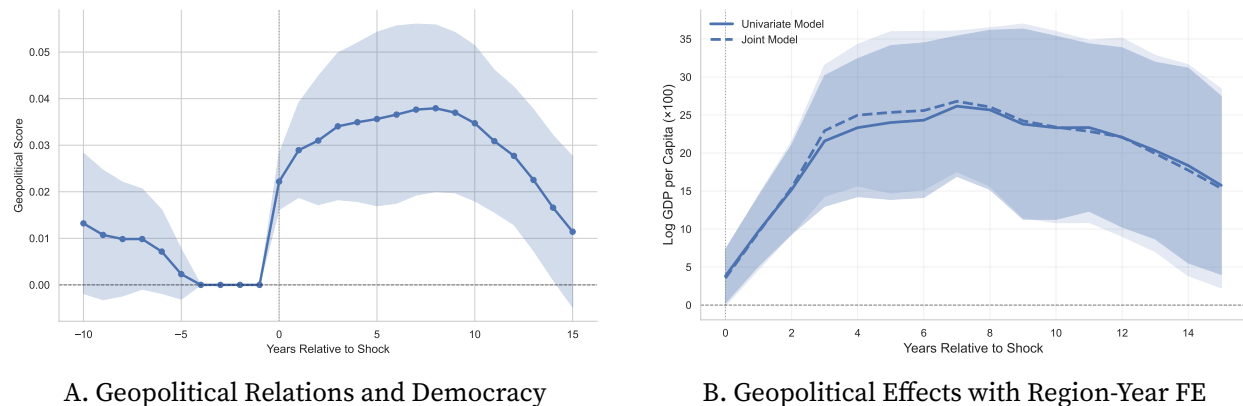


FIGURE B11. Democracy-Geopolitics Nexus: Additional Evidence

Panel (a) shows the impulse response of overall geopolitical relations to a democratization shock, estimated using local projections with four lags of geopolitical relations, country fixed effects, and year fixed effects. Panel (b) compares the growth effects of geopolitical relations with and without controlling for democracy, using our baseline region-year fixed effects specification. Both panels use Driscoll-Kraay standard errors with 95% confidence intervals.

deviations) after 8 years—roughly half the difference between neutral relations and moderate cooperation. The absence of pre-trends confirms that international improvements follow rather than precede democratization, with effects persisting through year 15. This complements our bilateral analysis: while democratization primarily improves Western relations, these gains are sufficient to raise the GDP-weighted aggregate measure.

Our main democracy analysis follows ANRR using year fixed effects to preserve variation from regional democratization waves, while our baseline employs region-year fixed effects. Panel (b) demonstrates that geopolitical relations drive growth regardless of specification choice. The growth effects remain virtually identical whether controlling for democracy or not, with both specifications peaking around 25 log points. This stability confirms that geopolitical relations capture distinct variation from democratic institutions within region-year cells. Geopolitical alignment generates substantial growth differences unexplained by shared democratization waves.

These results clarify the complex interplay between democratic institutions and international relations. The transitory shock analysis reveals that temporary democratic episodes operate almost exclusively through geopolitical channels, while permanent democratization generates additional domestic benefits. Combined with our evidence on bilateral heterogeneity and robustness across specifications, these findings support both the complementarity and independence of political institutions and geopolitical alignment as drivers of economic development.

B.13. Dynamics of Average Event Scores

This appendix provides a detailed analysis of the average event scores $\tilde{S}_{ct}$ examined in Section 6.2.1. We document the persistence properties of these unsmoothed scores and demonstrate how transitory event shocks aggregate into permanent effects on GDP, providing insight into the dynamic relationship between diplomatic events and economic outcomes.

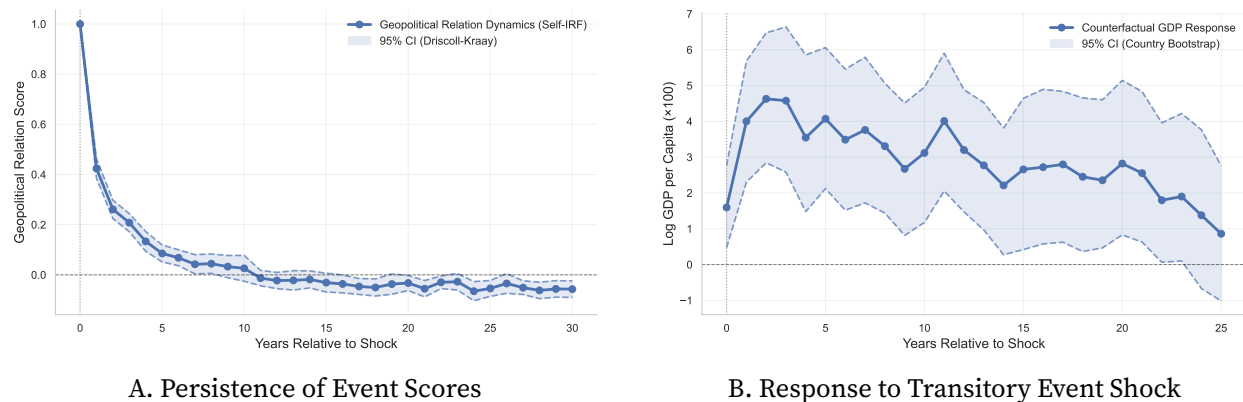


FIGURE B12. Event Score Dynamics and Transitory Shock Responses

Panel (a) displays the impulse response of event-based geopolitical scores to their own shock, revealing rapid mean reversion. Panel (b) shows the GDP response to a purely transitory event shock (1 at $h = 0$, 0 thereafter), constructed using auxiliary shock methodology. Shaded areas represent 95% confidence intervals from 1,000 bootstrap iterations.

Panel (a) of Figure B12 reveals the fundamental difference between event scores and our smoothed geopolitical relations measure. Following a unit shock, event scores exhibit strong mean reversion: approximately 63% of the initial impact dissipates within one year, falling to roughly 0.37, and the effect continues declining to approximately 0.10 by year 5. The effect becomes statistically indistinguishable from zero after approximately 7–8 years and turns slightly negative thereafter, suggesting mild overshooting before full dissipation. This rapid decay reflects the inherently episodic nature of diplomatic events. The mean reversion pattern suggests that individual geopolitical events, while impactful in the moment, lack the institutional persistence that characterizes broader bilateral relationships. In contrast, our baseline geopolitical relations measure (Figure 6A) shows substantially greater persistence, capturing the institutional memory and path dependence in international relations.

Panel (b) isolates the GDP response to a purely transitory event shock using the decomposition in Appendix B.4. GDP rises to a peak of 4–5 log points around years 2–3, then gradually declines while remaining positive throughout the 25-year horizon. The rapid event-score decay in panel (a) explains why the direct response to $\tilde{S}_{ct}$ in Section 6.2.1 appears muted relative to our baseline: the observed IRF combines the direct impact with the indirect effects through event persistence. When we construct the response to a permanent change in event flows, we recover comparable long-run effects to our smoothed measure, confirming that the baseline approach filters noise from isolated events while preserving the economically relevant variation.

B.14. UNGA Voting and Economic Growth: Detailed Results

This appendix presents comprehensive results using UNGA voting alignment as an alternative measure of geopolitical relations. We employ the negative Ideal Point Distance (IPD) from Bailey, Strezhnev, and Voeten (2017), which ranges from –5 (complete disagreement) to 0 (perfect align-

ment). Higher values thus indicate closer alignment in voting behavior. The IPD measure has been widely used in the international relations literature as a proxy for foreign policy similarity, making it a natural benchmark for our event-based approach.

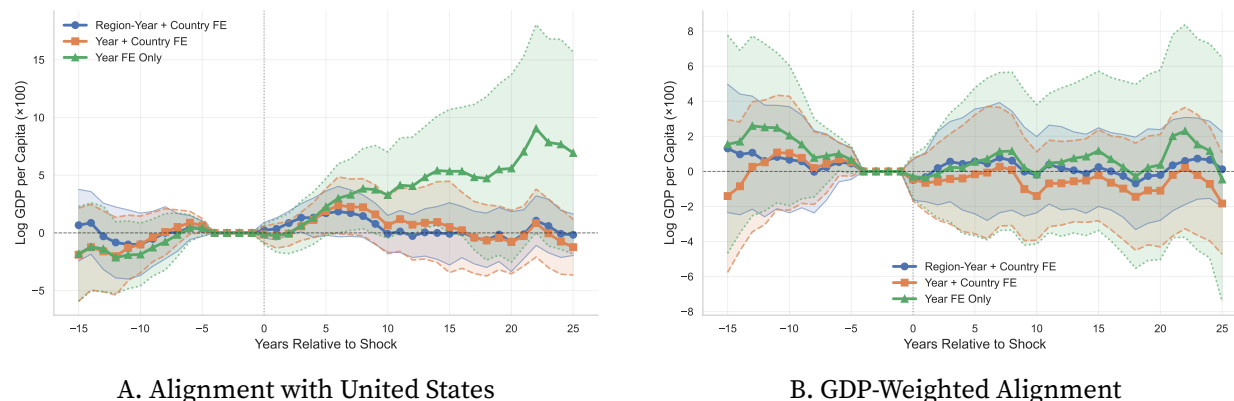


FIGURE B13. Impulse Responses Using UNGA Voting Alignment

This figure displays impulse responses of log GDP per capita to improvements in UNGA voting alignment. Panel (a) shows responses to closer alignment with US positions (negative IPD with USA). Panel (b) presents responses to improved GDP-weighted alignment with all major powers. Three specifications are shown: region-year plus country fixed effects (blue, our baseline), year plus country fixed effects (orange), and year fixed effects only (green). All specifications include four lags of GDP and the IPD measure. Shaded areas represent 95% confidence intervals with Driscoll-Kraay standard errors.

Figure B13 reveals why UNGA voting patterns fail to capture the economic effects of geopolitical relations. Panel (a) examines alignment with US voting positions. With our baseline specification including country fixed effects (blue line), the impulse response hovers near zero throughout the horizon, with confidence intervals consistently spanning zero. This null result persists when we relax to year plus country fixed effects (orange line). Only when we remove country fixed effects entirely (green line) does a positive relationship emerge, reaching approximately 6 log points after 20 years. However, this cross-sectional correlation likely reflects omitted variables rather than a causal effect of voting alignment on growth.

Panel (b) presents results for GDP-weighted alignment with all major powers, constructed analogously to our main geopolitical measure but using IPD rather than bilateral events. The pattern is even clearer: all three specifications yield economically small and statistically insignificant effects. Even without country fixed effects, which should maximize the chance of finding a relationship, the impulse response remains indistinguishable from zero. This complete absence of growth effects for the aggregate measure suggests that UNGA voting patterns fail to capture the economically relevant aspects of international relations.

These null results reflect three limitations of UNGA voting as a bilateral measure: votes primarily address multilateral issues rather than bilateral economic or security concerns; strategic bloc voting and vote trading add noise unrelated to bilateral dynamics; and the annual session calendar creates temporal misalignment with the continuous flow of diplomatic interactions. The positive cross-sectional relationship between US alignment and GDP (green line, panel a) likely reflects

reverse causality—wealthier countries tend to share US positions on international law and market economics—rather than a causal effect of voting similarity on growth.

B.15. Sanctions and Geopolitical Relations: A Horse-Race Analysis

This appendix examines the relationship between our comprehensive geopolitical relations measure and economic sanctions—a categorical measure that captures the most explicit form of economic statecraft. We implement horse-race specifications to disentangle their respective contributions to economic growth and assess whether sanctions provide additional explanatory power beyond our event-based measure.

Empirical Specification. We estimate both univariate and joint specifications to assess how sanctions and geopolitical relations interact:

$$\begin{aligned}
 \text{(B4) Univariate: } y_{c,t+h} &= \alpha_h^k p_{ct}^k + \sum_{\ell=1}^4 \beta_{\ell} y_{c,t-\ell} + \sum_{\ell=1}^4 \gamma_{\ell}^k p_{c,t-\ell}^k + \delta_c + \delta_{rt} + \varepsilon_{c,t+h} \\
 \text{Joint: } y_{c,t+h} &= \alpha_h^{\text{Geo}} p_{ct} + \alpha_h^{\text{Sanc}} p_{ct}^{\text{Sanction}} + \sum_{\ell=1}^4 \beta_{\ell} y_{c,t-\ell} + \sum_{\ell=1}^4 \gamma_{\ell}^{\text{Geo}} p_{c,t-\ell} \\
 \text{(B5) } &+ \sum_{\ell=1}^4 \gamma_{\ell}^{\text{Sanc}} p_{c,t-\ell}^{\text{Sanction}} + \delta_c + \delta_{rt} + \varepsilon_{c,t+h}
 \end{aligned}$$

where $k \in \{\text{Geo}, \text{Sanction}\}$ indexes the measure type. The univariate specifications include four lags of GDP and the respective geopolitical measure, while the joint specification controls for both measures and their lags simultaneously. This approach follows our baseline methodology while accounting for the potential interdependence between sanctions and broader geopolitical relations.

Results and Interpretation. Figure B14 presents the horse-race results. Panel (a) demonstrates stability in the geopolitical relations effect: the impulse response peaks at approximately 26–27 log points around year 7–8 in both specifications, with the univariate and joint trajectories closely tracking each other throughout the 15-year horizon. Controlling for sanctions has minimal impact on the estimated effect of geopolitical relations, suggesting that our event-based approach already incorporates the economically relevant variation associated with sanctions through the diplomatic deterioration that precedes, accompanies, and follows their imposition.

Panel (b) reveals important patterns for sanctions. The univariate specification (solid line) shows sanctions reducing GDP by approximately 10 log points at the trough around year 3, with effects gradually attenuating toward –5 log points by year 10 and –3 log points by year 15. The joint specification (dashed line) shows attenuated effects: the trough is shallower at approximately –7 log points, and the effect recovers more quickly, approaching zero by year 10. Both specifications

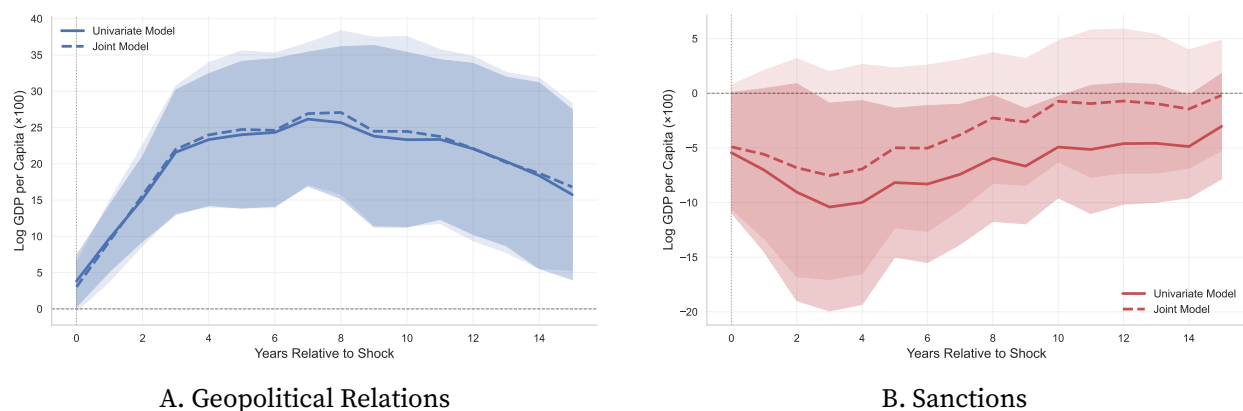


FIGURE B14. Horse-Race: Geopolitical Relations versus Sanctions

This figure compares univariate and joint specifications for geopolitical relations and sanctions. Panel (a) shows impulse responses for our geopolitical relations measure, comparing the univariate model (solid line) with the joint specification controlling for sanctions (dashed line). Panel (b) presents analogous results for sanctions exposure. Both panels include 95% confidence intervals based on Driscoll-Kraay standard errors.

show sanctions effects that remain negative but diminish over time, with the joint specification consistently closer to zero. The attenuation when controlling for geopolitical relations—reducing the peak negative effect by approximately 30%—indicates that part of the sanctions effect operates through the broader deterioration of bilateral relationships that our comprehensive measure captures.

Comprehensiveness of Event-Based Measures. Our event-based approach captures not only formal sanctions announcements but also the entire diplomatic ecosystem surrounding them. Sanctions rarely emerge suddenly; they typically follow an escalating pattern of diplomatic tensions, failed negotiations, and deteriorating bilateral trust. Our measure incorporates diplomatic protests, recalled ambassadors, suspended cooperation agreements, cancelled summits, hostile rhetoric, and other negative events that precede formal economic restrictions. Similarly, it captures the diplomatic efforts at sanctions relief, negotiated settlements, and gradual normalization that may follow. By incorporating this full spectrum of interactions, our measure subsumes much of the information content in binary sanctions indicators while providing additional variation from the broader relationship context.

The robustness of our geopolitical relations measure to controlling for sanctions—the most explicit and measurable form of economic coercion—reinforces the breadth of our event-based approach relative to narrow categorical indicators.

B.16. Formal Alliances and Geopolitical Relations: A Horse-Race Analysis

This appendix extends the sanctions horse-race analysis to formal military alliances—the cooperative counterpart to sanctions on the spectrum of bilateral geopolitical relations.

Data and Variable Construction. We use the Alliance Treaty Obligations and Provisions (ATOP) dataset, version 5.1 (Leeds et al. 2002), which codes all formal military alliances among independent states from 1815 through 2018. For each country c and year t , we construct a GDP-weighted alliance exposure measure:

$$p_{ct}^{\text{Alliance}} = \sum_{j \in \mathcal{N}} \mathbb{1}_{cjt}^A \times \text{GDP share}_{jt},$$

where $\mathbb{1}_{cjt}^A$ equals one if country c maintains any active formal alliance—defense pact, offensive pact, neutrality agreement, non-aggression pact, or consultation agreement—with major nation j in year t . The measure ranges from zero (no alliances with major nations) to approximately 0.80 (alliance with most of the world’s economic weight). Alliance status after 2018 is forward-filled from the last observed year, reflecting the high persistence of formal treaties; this imputation affects only the final six years of the sample and is unlikely to drive the estimated medium-run impulse response pattern. Country identifiers are mapped from Correlates of War codes to ISO3 using standard concordances.

Results. We estimate the same univariate and joint local projection specifications as in the sanctions analysis (equations B4–B5), replacing p_{ct}^{Sanction} with p_{ct}^{Alliance} and including four lags of each measure.

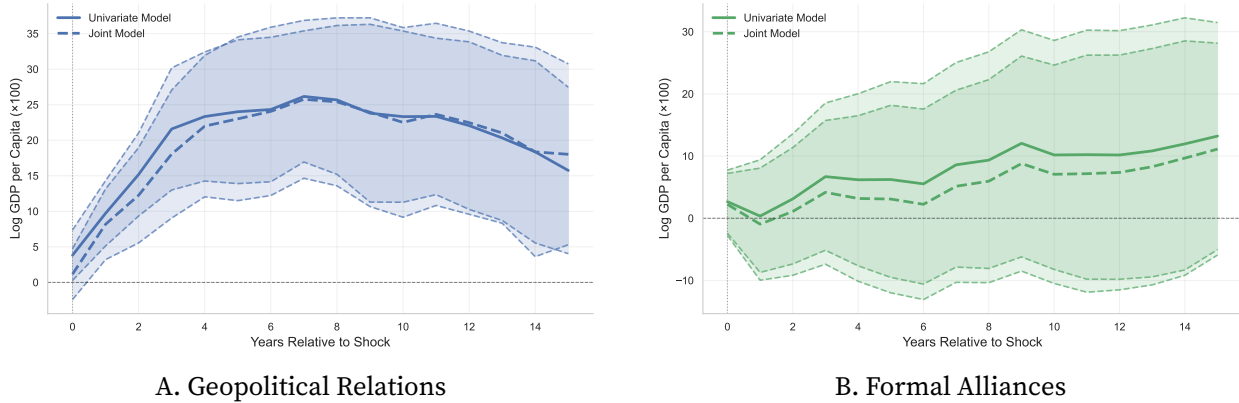


FIGURE B15. Horse-Race: Geopolitical Relations versus Formal Alliances

This figure compares univariate and joint specifications for geopolitical relations and formal alliance exposure. Panel (a) shows impulse responses for our geopolitical relations measure, comparing the univariate model (solid line) with the joint specification controlling for alliances (dashed line). Panel (b) presents analogous results for alliance exposure. Both panels include 95% confidence intervals based on Driscoll-Kraay standard errors. All specifications include country and region-year fixed effects.

Figure B15 displays the results. Panel (a) confirms that the geopolitical relations effect is stable: the impulse response peaks at approximately 26 log points around year 7 in both the univariate and joint specifications, with the two trajectories closely tracking each other throughout the 15-year horizon. Controlling for alliance exposure does not alter the estimated effect of geopolitical relations.

Panel (b) shows that alliance exposure has a positive univariate effect on GDP per capita, rising gradually to approximately 10–12 log points by year 10–15. In the joint specification, this effect attenuates to approximately 5–8 log points—a reduction of roughly 40–50%. The timing of the attenuation is informative: the short-run alliance coefficient largely disappears once geopolitical relations are controlled for, consistent with the event-based measure absorbing the diplomatic co-operation surrounding alliance formation and maintenance. At longer horizons, a modest positive residual persists, though with wide confidence intervals. This residual is not inconsistent with the broader measure’s dominance: treaty-based security commitments are durable institutional stocks that may contain information about long-run coordination beyond what annual event flow fully summarizes.

Defense Pacts. As a robustness check, we restrict the alliance measure to defense pacts—the strongest form of alliance commitment, excluding non-aggression pacts and consultation agreements. The geopolitical relations coefficient remains stable in joint specifications. The defense pact effect is noisier, consistent with the narrower scope of this measure, but displays a similar pattern of attenuation when the broader geopolitical score is included.

Appendix C. LLM Prompt: Event Category, CAMEO, and Goldstein Score

C.1. Prompt Structure

This subsection describes the LLM prompt structure for compiling and analyzing major political events shaping the bilateral relationship between two countries, {country_1} (code: {country_1_code}) and {country_2} (code: {country_2_code}), or their historical predecessors, during {target_year}. The analysis employs the Conflict and Mediation Event Observations (CAMEO) framework and the Goldstein scale, both detailed in Section C.3, to classify events and assess their intensity. If no major events are identified, a historical context-based relationship assessment is provided. The output is a single JSON object for computational integration.

Relationship Assessment Framework. The bilateral relationship for {target_year} is classified into one of the following mutually exclusive categories, reflecting interaction intensity and nature:

- **State of War / Active Conflict:** Sustained, large-scale armed conflict.
- **Crisis / Intense Confrontation:** High tension with disputes or limited clashes, short of war.
- **Hostile / Antagonistic Relationship:** Animosity marked by sanctions or diplomatic friction.
- **Competitive / Rivalrous Relationship:** Strategic competition with limited cooperation.
- **Limited Contact / Cool Relationship:** Minimal, neutral interaction.
- **Selective Cooperation / Transactional Relationship:** Cooperation on specific interests amid competition.
- **Broad Cooperation / Partnership:** Extensive sectoral cooperation with regular dialogue.
- **Strategic Partnership:** Deep coordination on strategic issues with high trust.
- **Alliance:** Formal treaty-based mutual support, often military.

Analytical Steps. The analysis follows five steps for rigorous event identification, classification, and assessment:

- a. **Verify Political Entities:** Confirm {country_1} (code: {country_1_code}) and {country_2} (code: {country_2_code}) existed in {target_year}. If not, identify the primary political entity controlling the relevant territory (e.g., Soviet Union for Russia before December 26, 1991; Russian Federation thereafter). Ambiguities are noted in the evaluation summary, with analysis using the best-identified entities, reflected in JSON fields `country1` and `country2`.
- b. **Data Collection:** Use search tools to compile interactions between verified entities during {target_year}.
- c. **Identify Major Political Events:** Identify events in {target_year} significantly influencing the bilateral relationship, verified by reliable sources. Events include economic diplomacy, diplomatic actions, high-level interactions, and security measures, with details in Section C.2.
- d. **Event Analysis:** For each event:
 - i. Assign `country1` and `country2` as initiator/target or participants.
 - ii. Provide `event_name` and `event_description`.

- iii. Classify using CAMEO (see Section C.3): CAMEO_quad_class (Verbal/Material Cooperation/Conflict), CAMEO_root_code (e.g., 04), CAMEO_event_code (e.g., 043), emphasizing economic actions (e.g., 163 for sanctions) and mediation (e.g., 045).
 - iv. Assign Goldstein_Scale (−10.0 to +10.0; see Section C.3), reflecting intensity, adjusted for bilateral context but consistent with CAMEO.
 - v. Classify economic_event: Tariffs, Economic Sanctions, Trade Agreements and Treaties, Other Economic Policies, or Not an economic event.
 - vi. Provide evaluation_summary justifying classifications and scores.
- e. **Overall Relationship Assessment:** Select one relationship category for {target_year}, integrating event patterns (via CAMEO/Goldstein) and historical context. If no events are found, assess based on interaction absence and historical trends.

JSON Output. The output is a JSON object with key `historical_political_events`:

- If events found: List of objects with fields:
 - year, country1, country2, event_name,
 - event_description, CAMEO_quad_class, CAMEO_root_code,
 - CAMEO_event_code, economic_event, Goldstein_Scale,
 - relationship, evaluation_summary.
- If no events found: Single object with event_name = “No Major Bilateral Events Found,” null CAMEO/Goldstein fields, and context-based relationship.

Implementation Notes.

- Entity verification ensures historical accuracy using country codes.
- “Major” events have significant political impact.
- The relationship assessment is uniform for {target_year} and historically contextualized.
- JSON output adheres to the specified structure for automation.

C.2. Event Identification Details

This subsection elaborates on the criteria for identifying major political events that significantly influence the bilateral relationship between {country_1} (code: {country_1_code}) and {country_2} (code: {country_2_code}), or their historical predecessors, during {target_year}, as referenced in Section C.1. Events are selected for their demonstrable impact on the relationship’s trajectory.

Event Verification. Potential events are critically evaluated using reliable sources to confirm authenticity. Unverified or fabricated events are excluded to ensure analytical rigor. When events span multiple categories, the primary classification reflects the dominant mechanism or domain of impact, with secondary dimensions noted in the event description.

Event Dimensions. Events are identified across six major dimensions of bilateral relations, ensuring comprehensive coverage of politically consequential interactions.

Economic Relations. This dimension encompasses the full spectrum of economically mediated bilateral interactions, including both cooperative arrangements and coercive measures.

- *Trade Policy and Market Access:* Imposition, adjustment, or removal of tariffs; non-tariff barriers such as technical standards, sanitary measures, and import licensing; trade agreement negotiations, signings, ratifications, withdrawals, or dispute settlement proceedings.
- *Financial and Monetary Relations:* Financial sanctions including asset freezes, transaction bans, and SWIFT exclusions; currency swap agreements and bilateral payment arrangements; foreign investment restrictions or liberalizations; bilateral investment treaty developments.
- *Economic Coercion and Inducements:* Comprehensive trade embargoes and sectoral sanctions; export controls and entity list designations; technology transfer restrictions; foreign aid packages, development finance, and debt relief programs.
- *Strategic Economic Sectors:* Energy supply agreements and pipeline projects; telecommunications and digital economy restrictions (e.g., 5G bans); critical resource arrangements covering rare earth elements, strategic minerals, and food security.
- *Economic Integration and Infrastructure:* Bilateral infrastructure initiatives; supply chain arrangements including friend-shoring and critical supply agreements; regional economic arrangement participation.

Diplomatic and Political Relations. This dimension captures formal state-to-state interactions and official communications that shape the bilateral relationship.

- *Formal Diplomatic Engagement:* Embassy or consulate openings and closures; ambassador appointments and recalls; diplomatic staff expulsions; formal protests, démarches, and official condemnations or commendations.
- *High-Level Political Interactions:* Presidential or prime ministerial visits and bilateral summits; ministerial meetings across foreign affairs, defense, and trade portfolios; joint commission sessions and strategic dialogues.
- *Public Diplomacy and Rhetoric:* Major policy speeches affecting bilateral relations; parliamentary resolutions; government white papers on the bilateral relationship; significant public campaigns or propaganda efforts.

Security and Defense. This dimension encompasses military cooperation, competition, and security-related incidents between the two countries.

- *Military Cooperation and Competition:* Defense cooperation agreements and military alliances; status of forces agreements; major weapons sales or cancellations; arms embargoes; joint military exercises and military-to-military exchanges.

- *Security Incidents*: Border skirmishes and territorial claim assertions; airspace or maritime violations; naval encounters and air intercepts; demarcation agreements.
- *Intelligence and Cyber Operations*: Publicly revealed espionage scandals; intelligence officer expulsions; intelligence sharing agreements or suspensions; state-sponsored cyber attacks and cyber security cooperation.

Legal, Territorial, and Movement. This dimension covers international legal proceedings, sovereignty disputes, and policies governing the movement of people.

- *International Legal Actions*: Bilateral disputes before international courts such as the International Court of Justice or International Tribunal for the Law of the Sea; WTO disputes with significant political dimensions; contentious extradition cases.
- *Territorial and Maritime Issues*: Exclusive economic zone disputes; continental shelf claims; fisheries agreements or conflicts; freedom of navigation operations; border demarcation and sovereignty recognition.
- *Movement of People*: Visa regime changes and travel bans; visa-free agreements; guest worker programs; readmission agreements; refugee and asylum policy changes.

Multilateral and Global Governance. This dimension captures bilateral dynamics manifested through international organizations and global issue areas.

- *International Organizations*: United Nations Security Council confrontations; General Assembly coalition building; specialized agency disputes; regional organization membership changes.
- *Global Issues*: Climate agreement positions and environmental cooperation; pandemic response coordination and vaccine diplomacy; human rights criticism or defense in international fora.

Other Significant Events. This residual category encompasses additional politically consequential interactions.

- *Historical and Symbolic*: Apologies for historical wrongs; memorial visits; monument disputes; anniversary commemorations with bilateral significance.
- *Humanitarian and Disaster Response*: Aid offers or rejections following natural disasters; joint rescue operations; humanitarian access disputes.
- *Technology and Space*: Joint space missions; satellite cooperation; technology theft accusations; research collaboration terminations.
- *Communications and Media*: Journalist expulsions; broadcasting restrictions; undersea cable disputes.

Selection Criteria. Events are prioritized based on their significant effect on, or strong indication of, the bilateral relationship's trajectory. When economic tools are employed for political purposes, events are classified under economic categories. Only key interactions meeting these

significance thresholds are included, ensuring analytical focus on politically consequential events that characterize the bilateral relationship.

C.3. Conflict and Mediation Event Observations and Goldstein Score

Our analysis employs the Conflict and Mediation Event Observations (CAMEO) framework (Schrodt and Yilmaz 2012) to systematically classify and quantify bilateral political events. CAMEO provides a comprehensive coding scheme that categorizes international political actions along two primary dimensions: the nature of the interaction (cooperation versus conflict) and the form of action (verbal versus material). This framework enables consistent, objective classification of diverse political events while preserving crucial information about their intensity and character.

CAMEO Classification Structure. The CAMEO framework organizes events into four quadrant classes based on the intersection of cooperation-conflict and verbal-material dimensions:

- **Verbal Cooperation:** Diplomatic statements, consultations, expressions of intent to cooperate, and formal diplomatic cooperation including treaty signing and public endorsements.
- **Material Cooperation:** Tangible cooperative actions such as economic aid provision, military cooperation, intelligence sharing, and policy concessions.
- **Verbal Conflict:** Critical statements, accusations, demands, rejections, threats, and public protests that express disagreement or hostility.
- **Material Conflict:** Concrete hostile actions including economic sanctions, military demonstrations, coercive measures, and various forms of violence.

Within each quadrant, CAMEO provides a hierarchical coding system with root codes (two-digit) representing general action categories and event codes (three-digit) specifying precise actions. For example, root code 16 (REDUCE RELATIONS) includes event codes 163 (impose economic sanctions) and 161 (reduce diplomatic relations), allowing for nuanced differentiation within broader conflict categories.

Implementation in Our Analysis. Our LLM-based analysis applies CAMEO classification through structured prompt engineering that guides the model through systematic event categorization. The process begins with identifying the core bilateral action in each event description, determining its cooperative or conflictual nature, and assessing whether the action is primarily verbal or material. The LLM then selects the most appropriate root code within the identified quadrant class and chooses the specific event code that best captures the action's essence.

We pay particular attention to economic diplomacy events, ensuring that economic tools of statecraft receive appropriate classification. For instance, we distinguish between broad economic sanctions (code 163) and targeted administrative sanctions (code 172), recognizing their different mechanisms and intensities. Similarly, we differentiate between various forms of diplomatic cooperation, from routine consultations (code 040) to formal agreement signing (code 057), capturing the spectrum of cooperative engagement.

Integration with Goldstein Scale Scoring. CAMEO classifications inform our Goldstein Scale scoring (Goldstein 1992), which assigns numerical values from -10.0 (maximum conflict) to $+10.0$ (maximum cooperation) to each event. The LLM uses the CAMEO event code as the primary reference point for determining baseline intensity, then applies contextual adjustments based on the specific circumstances described in the event. This approach ensures consistency with established conflict-cooperation measurement while allowing for nuanced assessment of event significance within particular bilateral contexts.

The combination of CAMEO’s systematic classification with Goldstein Scale quantification enables our methodology to capture both the categorical nature of political actions and their relative intensity, providing a foundation for empirical analysis of how different types of geopolitical events affect economic outcomes. This dual-coding approach addresses the limitations of purely categorical measures while maintaining the interpretability necessary for economic research applications.

Empirical Patterns in Event Classification. Figures C1 and C2 illustrate the distribution and evolution of our compiled geopolitical events across the CAMEO classification system and Goldstein Scale from the 1960s through the 2020s, revealing distinct patterns that align with major shifts in the international order.

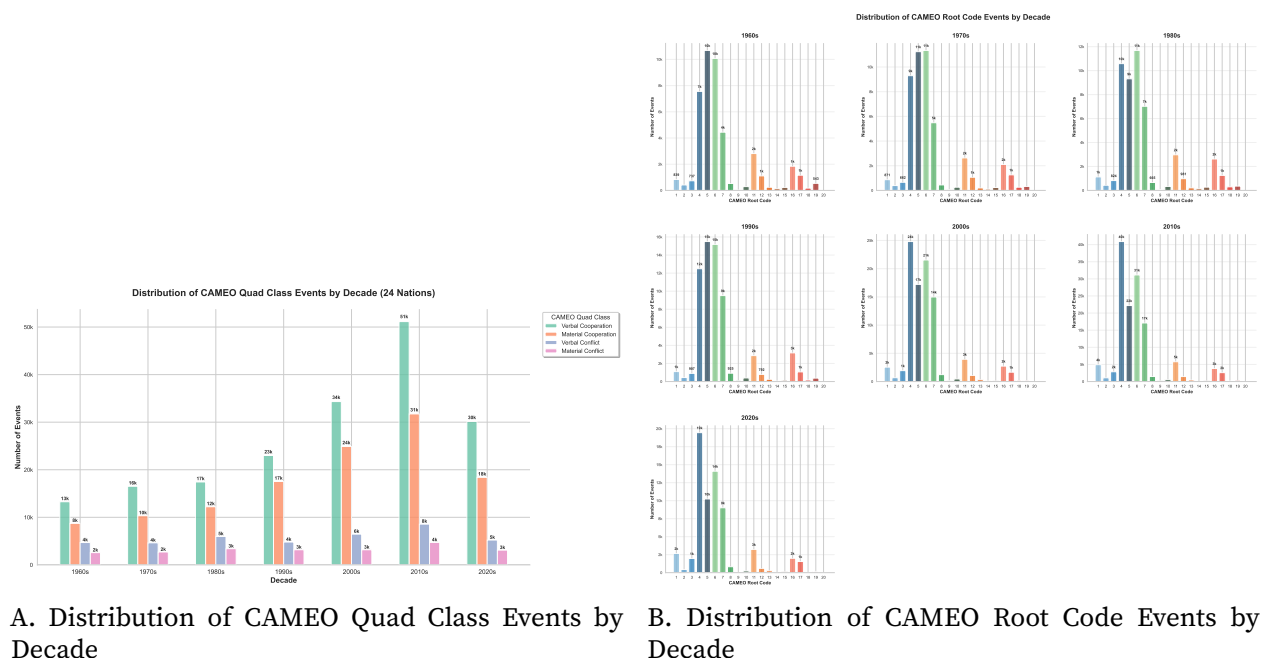


FIGURE C1. Evolution of CAMEO Event Classification Patterns

Figures C1A and C1B document the evolution of bilateral event patterns across six decades. The Cold War decades show balanced distributions across cooperation and conflict categories, with total events growing from approximately 29,000 in the 1960s to 39,000 in the 1980s. The post-Cold War globalization era (1990s–2000s) brought a sharp expansion of cooperative interactions—verbal

cooperation roughly doubled and material cooperation more than doubled—while conflict levels remained stable. The 2010s represent peak recorded activity at over 96,000 events (more than triple the 1960s level), with both cooperation and conflict expanding in absolute terms. The 2020s maintain elevated engagement (approximately 56,000 events in a partial decade) with an increased conflict share relative to the globalization era, consistent with the geopolitical fragmentation documented in Section 2.3.

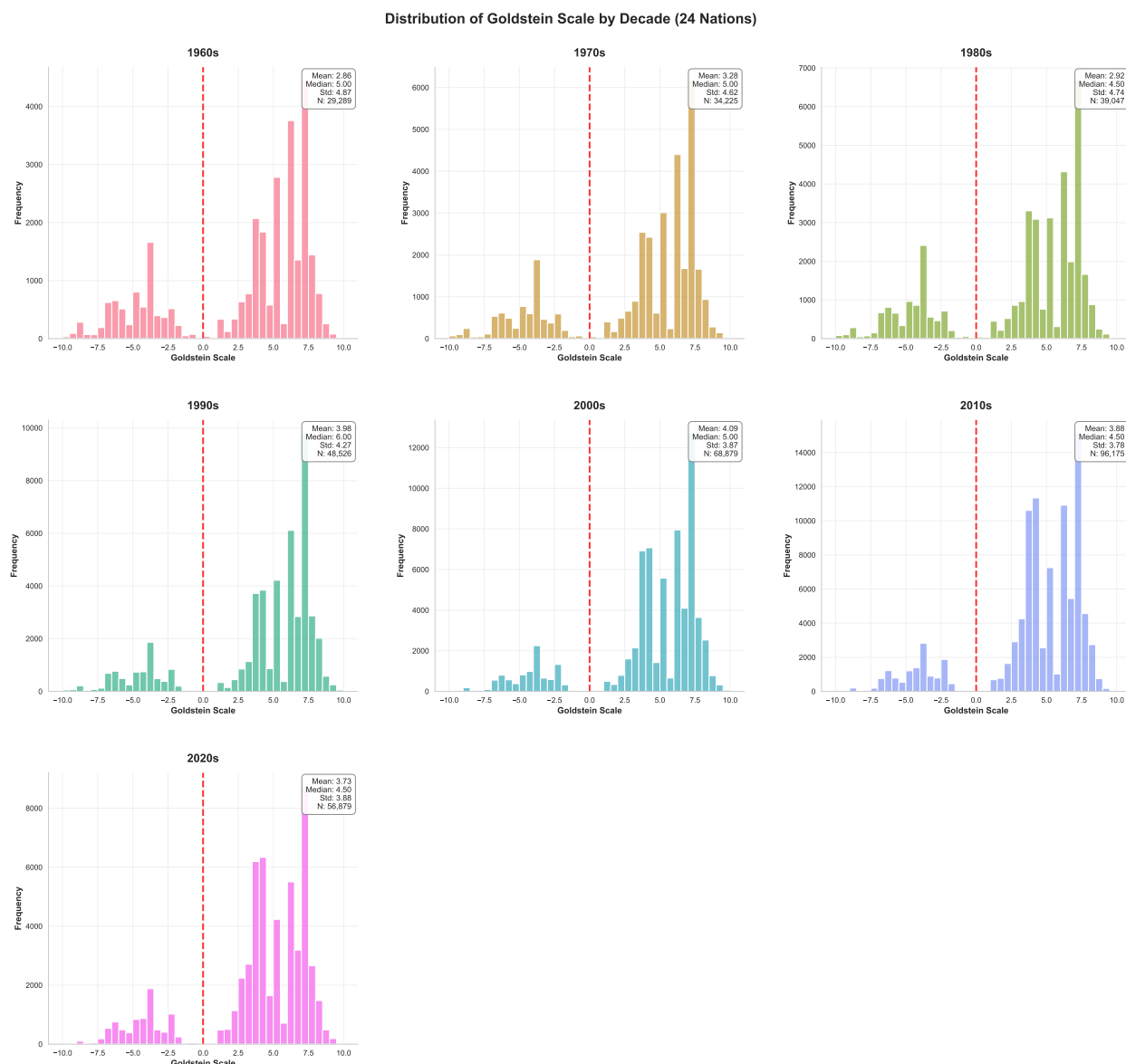


FIGURE C2. Distribution of Goldstein Scale Scores by Decade

Figure C2 shows the corresponding Goldstein Scale distributions. The Cold War decades display wide dispersions (standard deviations of 4.6–4.9) with mean cooperation scores of 2.9–3.3. The 1990s and 2000s exhibit the highest mean scores (4.0–4.1) and reduced dispersion (standard deviations of 3.9–4.3), reflecting the cooperative orientation of the post-Cold War period. The 2010s and

2020s show a partial retreat—mean scores decline to 3.7–3.9 and the negative tail thickens—though distributions remain right-skewed, indicating that cooperative interactions continue to dominate even as conflictual events become more frequent.